

H 560

Haematology analyser



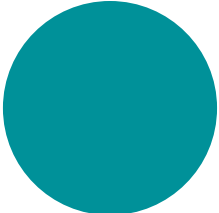
USER MANUAL

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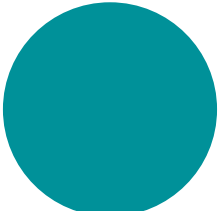
Preface

The H560 User Manual is published by Erba Lachema. Neither the whole nor any part of the information contained herein may be adapted or reproduced in any material for, except with the prior written consent of Erba Lachema.

All information, of a technical nature, and particulars of the H560 Analyser and its use are given by Erba Lachema in good faith, but may contain errors. This manual is intended only to assist the user in the use of the H560 Analyser and therefore Erba Lachema shall not be liable for any loss or damage whatever arising from the use or any information or particulars in, or any errors, or omission in this manual.

Users must respect the precautions and notes intended to protect them against injuries and/or instrument damage.

Misuse of the instrument and none respect of the prescribed use and the instrument maintenance procedures will void the warranty and may result in injuries.



Declaration

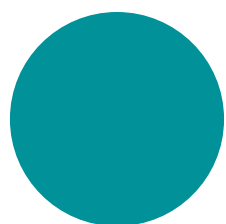
This user manual may be modified without notice.

Erba Lachema reserves the right of final interpretation of this user manual.

The pictures in this user manual are for reference only. If there is inconsistency between the pictures and the actual product, the actual product shall prevail. Do not use the pictures for other than intended use.

Erba Lachema shall be responsible for the safety, security, and performance of the product only when all of the following conditions are met:

- The assembly, re-commissioning, extension, modification, and repair of the product are performed by the authorized personnel of Erba Lachema.
- The product is operated based on this user manual.
- The electrical appliances in the relevant working room comply with applicable national and local requirements.



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Manual Overview

1.1 Introduction

This chapter explains how to use this operator's manual of H560 Haematology Analyser, which is shipped with the H560 Haematology Analyser and contains reference information about the analyser and procedures for operating, troubleshooting and maintaining the analyser. Read this manual carefully before operating the analyser and operate your analyser in strict accordance with this manual.

1.2 Who Should Read This Manual

This manual contains information written for clinical laboratory professionals to:

- Learn about the hardware and software of the analyser.
- Customize system settings.
- Perform daily operations.
- Perform system maintenance and troubleshooting.

1.3 How to Find Information

This operator's manual comprises 16 chapters. Find the information you need by referring to the Table 1-1.

Table 1-1 Description of chapters

See...	You can find...
1 Manual Overview	Instructions for using the H560 Haematology Analyser.
2 Installation	Installation requirements for the H560 Haematology Analyser.
3 System Overview	Applications, measurable parameters, instrument configuration, software interface and software operations of the H560 Haematology Analyser.
4 Working Principle	Measuring principle and procedures of the H560 Haematology Analyser.
5 Setup	Settings of the system parameters such as the software date format and parameter units.
6 Daily Operations	Daily operations such as sample collection and preparation, the analysis procedures, start-up and shutdown of the instrument.
7 Sample Analysis	Sample analysis procedure and handling of the analysis results.
8 Result Review	Review of the analysis results.
9 Quality Control	Basic requirements for quality control and the quality control methods provided by the H560 Haematology Analyser.
10 Calibration	Basic requirements for calibration and the calibration methods provided by the H560 Haematology Analyser.

11 Reagent Management	Settings and management of the reagents for the H560 Haematology Analyser.
12 Service	Methods for maintaining and testing the H560 Haematology Analyser.
13 Troubleshooting	Troubleshooting methods for the H560 Haematology Analyser.
14 Traceability and Reference Methods	Traceability and reference methods of hematology calibrator and controls.
15 Specifications	Specification indicators of the H560 Haematology Analyser.
16 Terms and Abbreviations	Terms and abbreviations for the H560 Haematology Analyser.
17 Packing List	Packing list for the H560 Haematology Analyser.

1.4 Conventions Used in This Manual

The texts with special meaning in the manual are highlighted by different fonts and formats.

Table 1-2 Meanings of format use in user manual

Format	Definition
[XX]	All uppercase characters enclosed in [] indicate the name of a key on the analyser or the (external) keyboard, such as [ENTER].
XX	Bold characters indicate text displayed on the screen, such as Report .
XX	XX indicates variables and the specific content depends on the actual situation.
<i>XX</i>	Bold and italic characters Indicate chapter titles, such as <i>1.1 Introduction</i> .

1.5 Symbol Conventions

When you see ...	Then...
	BIOHAZARD Follow the instruction below the symbol to avoid potential biocontamination.
	WARNING Follow the instruction below the symbol to avoid personnel injury.
	CAUTION Follow the instruction below the symbol to avoid analyser damage and failure, or unreliable analysis results.
	Follow the instruction below the symbol. The symbol highlights the important information in operating procedures that calls for special attention.
	Puncture Warning: The sampling probe is sharp and may contain biohazardous materials. Special care should be taken when working with it.
	Laser Warning: This sign serves as a reminder of laser radiation.

The analyser or the outer packaging may have the following labels or symbols.

NOTE

If the labels are damaged or missing, please contact Erba Lachema or Erba Lachema's agents for replacement.

All illustrations in this manual are provided as references only. They may not necessarily reflect actual analyser configuration or display.

When you see ...	Then...
	Caution
	Biohazard
	Exercise caution to prevent puncture
	Laser radiation warning: It is a Class 3R laser product with 5.0 mW of maximum power output at 635nm. Avoid direct eye exposure to the laser beam.
	Instruction for Moving
	Network interface
	Protective grounding
	Alternating current (AC)
	For <i>in vitro</i> diagnosis only
	Lot No.
	Expiry date
	Serial No.
	European CE declaration of conformity
	Date of manufacture
	Manufacturer

When you see ...	Then...
	Storage temperature
	Humidity level for storage
	Atmospheric pressure level for storage
	Consult the operator's manual
	Avoid sunlight
	Keep dry
	No rolling
	No Stacking
	Let this side face upward
	Fragile, handle with care
	Recyclable materials
	The symbol on the product indicates that this product may not be treated as a household waste. Instead it shall be handled to the applicable collection point for the recycling of electrical and electronic equipment. By ensuring this product is disposed of correctly, you will help prevent potential negative consequences for the environment and human health, which could otherwise be caused by inappropriate waste handling of this product. Please contact your local city office or your distributor of this product. Pursuant to the EU directive 2002/96/EC for medical devices sold from that time by Erba Lachema the corresponding costs are divided up as described below: 1. The concerned device delivery to Erba Lachema will be paid by the Customer 2. Device dismantling sorting of parts and elimination of wastes will be supported by Erba Lachema according to the existing local regulation 3. In case of a sale to a third-party the first Customer shall inform Erba

When you see ...	Then...
	Lachema of the name and address of the new owner to guarantee the device traceability and to allow it's further elimination, and shall inform of the new owner that he will pay for the delivery of the device to Erba Lachema for its elimination 4. Otherwise the first Customer will pay all of the costs and all penalties that the legal authorities should impose on the manufacturer for default of the device elimination traceability as requested by the regulation – For medical devices sold before that time, except in particular cases, the elimination of the device will be supported by the Customer. Upon request Erba Lachema could provide this elimination. Contact us for quotation – For medical devices sold and used in other countries, the Customer should contact the Erba Lachema Representative to be informed of his responsibilities

1.6 Safety Information



- All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.
- If leak happens to the analyser, the leak liquid is potentially biohazardous.



WARNING

- Please check the firmness of all the door/ covers/panels before running the analyser to prevent unexpected opening or loosening when the analyser is working.
- Make sure all the safety measures are taken. Do not disable any safety device or sensor.
- Please respond to any alarm and error message immediately.
- Do not touch the moving parts.
- Contact Erba Lachema or Erba Lachema-authorized agents upon the identification of any damaged part.
- Be careful when opening/closing and removing/installing the doors, covers and panels of the analyser.
- Dispose the analyser according to government regulations.



CAUTION

- Please use the analyser in strict accordance with this manual.
- Please take proper measures to prevent the reagents from being polluted.

Installation

2.1 Introduction

**WARNING**

Installation by personnel not authorized or trained by Erba Lachema may cause personal injury or damage to the analyser. Do not install the analyser without the presence of Erba Lachema-authorized personnel.

Your analyser has passed strict tests before it is shipped from the factory. Internationally-recognized symbols and instructions show the carrier how to properly handle this electronic instrument in transportation. When you receive your analyser, carefully inspect the packaging. If you see any sign of mishandling or damage, contact Erba Lachema customer service department or your local agent immediately.

2.2 Installation Personnel

The analyser should only be installed by Erba Lachema or its authorized agents. You need to provide the appropriate environment and space. When the analyser needs to be relocated, please contact Erba Lachema or your local agents.

When you receive the analyser, please notify Erba Lachema or your local agent immediately.

2.3 Installation Requirements

**WARNING**

- Connect only to a properly grounded outlet.
 - Before turning on the analyser, make sure the input voltage meets the requirements.
-

**CAUTION**

- Using a patch board may introduce electrical interference and generate incorrect analysis results. Please place the analyser near the electrical outlet to avoid using the patch board.
 - Please use the original electrical wires shipped with the analyser. Using other electrical wires may damage the analyser or generate incorrect analysis results.
-

Installation requirements for the analyser are in Table 2-1.

Table 2-1 Installation requirements

Installation Environment	Requirements
Site	• Level ground and stable workbench with load capacity $\geq 50\text{kg}$. • Free of dust, mechanical vibration, heat and wind sources, contamination, heavy-noise source or electrical interference. • Avoid direct sunlight and keep good ventilation. • It's recommended to evaluate the electromagnetic environment of the laboratory before operating the analyser. • Keep the analyser away from sources of strong electromagnetic interference, otherwise, its proper functioning may be affected.
Space (In addition to the space required for the analyser itself, set aside:)	• At least 50 cm from each side, which is the preferred access to perform service procedures. • At least 20 cm from the back for cabling and ventilation. • Enough room on and below the countertop to accommodate for the diluent and waste containers. • Place the analyser near the electrical outlet and avoid being blocked by any objects, so that you can disconnect the power plug easily as required.
Temperature	15°C~30°C
Relative humidity	20%~85%
Operating atmospheric pressure	70kPa~106kPa
Ventilation	Keep air exchange to ensure good air circulation. The wind should not blow directly at the analyser.
Power Requirements	AC100V~240V, Input Power $\leq 200\text{VA}$, 50/60HZ.
Electromagnetic Wave	Keep the analyser away from electric-brush motors, flashing fluorescent and electric-contact equipment which is switched on/off frequently.
Waste Disposal	Dispose of the waste as per the requirements of the local environment protection authorities.

2.4 Damage Inspection

Before packing and shipping, Erba Lachema has applied rigid inspection on all the analysers. Upon receiving the analyser, please check carefully before unpacking to see if there are any of the following damages.

- The outer packaging is placed upside down or distorted.
- The outer packaging shows obvious signs of having been exposed to humid conditions.
- The outer packaging shows obvious signs of having been crashed.
- The outer packaging shows signs of having been opened.

Once you find the above damages, please notify your local agent immediately.

If the packaging is intact, please open the packaging in the presence of personnel from Erba Lachema or its agents and apply the following inspections:

- Check if all the items listed in the packing list are in the packaging.
- Carefully inspect the appearance of all the items to check if they are damaged or distorted.

2.5 Unpacking

Please unpack the analyser by taking the following steps:

1. Open the outer packing box; take out the accessory pack; take out the analyser together with the protective and cushioning materials.
2. Remove the foam and the protective PE bag.
3. Open the right door (open the linear-shaped cam lock on the right door with a slotted screwdriver).
4. Remove the binder clips, which are used for fixating two conveyor belts.
 - To avoid the possible collision resulting from the slippage caused by shaking and slanting during transportation, the central position of those two belts is fixated with binder clips before they are shipped from the factory. The binder clips must be removed during unpacking.
5. Remove the binder clips, which are used for fixating sampling assembly.

NOTE	To avoid damage during the transportation, the sampling assembly of the analyser is fixated with clamps. Do remove the clamps before using the analyser.
-------------	--

2.6 Connecting the Analyser System

2.6.1 Electrical Connections

Please refer to Figure 2-1 for the electrical connections of the analyser.



Figure 2-1 Connecting the electrical devices

2.6.2 Reagent Connections

WARNING



Be sure to dispose of reagents, waste, samples, consumables, etc. according to local legislations and regulations.

Reagents can be irritating to the eyes, skin, and mucosa. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them in the laboratory.

If the reagent accidentally comes in contact with your skin, wash it off immediately with plenty of water and see a doctor if necessary. Do the same if you accidentally get any of the reagent in your eyes.

CAUTION



Please make sure the length of the diluent pipe and the waste pipe should be no longer than 1 500mm; the length of the lyse pipe and the cleanser pipe should be no longer than 850mm.

Tighten the panel connector of the fluidic line so that the overall fluidic line is closed to prevent leakage and seepage caused by siphonage, etc.

Refer to Figure 2-2 for the connection of the reagents placed outside the analyser.

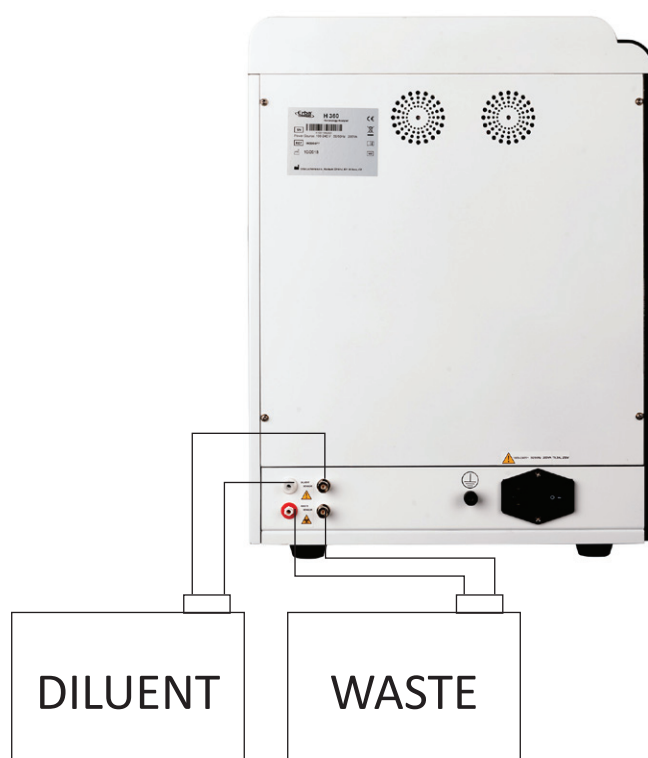


Figure 2-2 Connecting reagents placed outside the analyser

Refer to Figure 2-3 for the connection of the reagent placed inside the analyser.



Figure 2-3 Connecting reagents placed inside the analyser (left door opened)

2.6.3 Installing the Diluent Float Sensor and Replacing the Reagents

Please install the diluent float sensor and replace the diluent as per the approaches stated in this section.

2.6.3.1 Installing the Diluent Float Sensor

Install the diluent float sensor according to the following steps.

- Press down and remove the round cardboard with dotted cutting line on the top side of the diluent box so as to reveal a round hole.
- Pull out the cover of the container so that the cardboard around the round hole can seize the neck under the vial cap to prevent invagination.
- Turn and open the cap (keep the cap) and prevent any foreign objects from getting into the container.
- Install the diluent float sensor assembly in the accessory pack as shown in Figure 2-4. The float sensor shall be kept as vertical as possible during installation and the self-contained cap of the sensor shall be tightened.

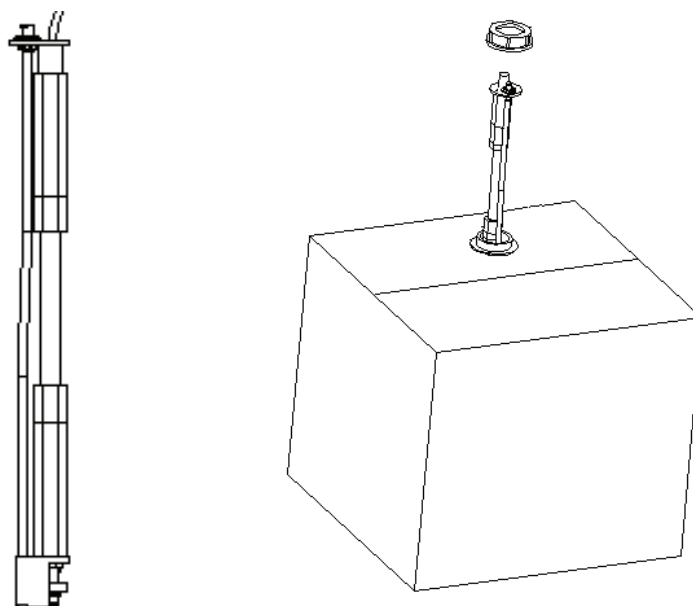


Figure 2-4 Installing the Diluent Float Sensor

2.6.3.2 Replacing Reagents

Steps for the replacing the diluent are the same as that for installing the sensor. Please keep the empty diluent container and the cap for future use.

2.6.4 Installing the Waste Float Sensor

NOTE

The float sensors used in the analyser are only applicable to Erba Lachema-supplied waste containers or the containers with the same specification and model (such as the vacant diluent container).

- Take a proper waste container (it can be a vacant diluent container, the opening of which is required to be pulled out of the hole of the box to expose the opening) and open the vial cap.
- Install the waste float sensor assembly in the accessory pack as shown in Figure 2-5. The float sensor shall be kept as vertical as possible during installation and the self-contained cap of the sensor shall be tightened at the same time to prevent the spilling of the waste.

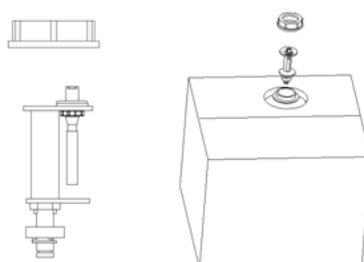


Figure 2-5 Installing the Waste Float Sensor

The waste container can be replaced according to the steps mentioned above. The replaced waste shall be properly disposed to avoid contamination.

**WARNING**

Be sure to dispose of reagents, waste, samples, consumables, etc. according to local legislations and regulations.

2.6.5 Connecting the LIS

If the analyser needs to be connected to laboratory information system (hereinafter referred to as LIS), you can complete the connection by following the steps in this section.

2.6.5.1 Installing LIS Workstation

- Install LIS workstation and set instrument type and model.
- Enter LIS workstation network setup interface after installation and set monitoring IP address and port number.

NOTE

Please contact the Erba Lachema customer engineer to get Description of LIS Communication Protocol for Haematology Analysers to complete the support of the LIS workstation to the LIS communication protocol.

2.6.5.2 Host Communication Settings

- Use a network cable to connect the analyser to LIS local area network.
- Please log on the H560 Haematology Analyser software as the administrator; if the analyser is turned on, skip this step.

For detailed, see **6.3 Start-up**.

The whole process lasts for 4 to 12 minutes. Please be patient.

- In the **Setup** interface, click **Host Communication** in the **Communication** selection to access the Laboratory Information System (LIS) communication setting interface. See *6.3 Start-up*.
- Set the IP address and other network information of the analyser according to the actual situation.

If the network is accessed through a router on the site, please select **Obtain an IP address automatically** and **Obtain DNS server address automatically**.

If the network is accessed through a network switch, or the analyser is directly connected to the LIS on the site, please select **Use the following address**, so as to manually set the IP address and subnet mask of the analyser. The IP addresses of the analyser and LIS must be in the same network segment. Furthermore, their subnet masks shall be the same, while other parameters can maintain null.

For detailed parameter descriptions, see **5.6.1 Host Network Settings**.

- Click **OK** to save the settings and close the dialog box.

Host Communication

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically
 ☒ Use the following address:

IP Address

.

.

.

Subnet mask

.

.

.

Default gateway

.

.

.

☐ Obtain DNS server address automatically
 ☒ Use the following DNS server addresses:

Preferred DNS server

.

.

.

Alternate DNS server

.

.

.

Details

Apply

OK

Cancel

Figure 2-6 Host Communication Settings

2.6.5.3 Connecting Analyser with LIS

- Please log on the H560 Haematology Analyser software as the administrator; if the analyser is turned on, skip this step.

For detailed, see 6.3 Start-up.

The whole process lasts for 4 to 12 minutes. Please be patient.

- In the **Setup** interface, click **LIS Communication** in the **Communication** selection to access the Laboratory Information System (LIS) communication setting interface. See Figure 2-7.
- Input the IP address and port of LIS workstation in **Network Settings** area.

LIS Communication

Network Settings

IP AddressPort

5600

Reconnect

Transmission Settings

☐ Auto-communication

☐ Transmit after result modified

☐ Bidirectional LIS/HIS Communication

Bidirectional LIS/HIS Communication Timeout

10

Sec.

Matched by

Sample ID

Protocol Settings

☐ Communication Acknowledgement

ACK timeout

10

Sec.

Graph Format

PNG

Histogram Transmission Method

Not transmit

Scattergram Transmission Method

Not transmit

DIFF Scattergram

☐ LS-MS

☐ LS-HS

☐ HS-MS

BASO Scattergram

☐ LS-MS

Apply

OK

Cancel

Figure 2-7 LIS Communication Settings

Find the IP address and port of LIS in the network setup interface in the LIS workstation; if IP address can't be found, try the method below:

- Enter the operating system of LIS workstation.
- Press combination key [Windows+R] to open the **Run** window.
- Input **cmd**, and then click **OK**.
- Input the **ipconfig** command into the cmd.exe window popped out.

The interface shows similar content as follows:

```

C:\Windows\system32\cmd.exe
Copyright (c) 2009 Microsoft Corporation. All rights reserved.
C:\Documents and Settings\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 10.0.0.102
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.0.1

Ethernet adapter Local Area Connection 2:

    Connection-specific DNS Suffix  . : 
    IPv4 Address. . . . . : 192.168.8.44
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.8.254

C:\Documents and Settings\Administrator>
  
```

Figure 2-8 IPv4 address

The IPv4 address in the red box is the IP address of LIS workstation

NOTE The IP address 192.168.8.44 of the LIS workstation shown as above is used as an example, real IP should be in the same network segment with LIS server.

- Click OK to save the settings.
- Check if the connection is successful.

The LIS icon in the upper right side on the analyser screen turns from white  to green , which indicates H560 Haematology Analyser software is connected to LIS successfully. If the icon stays grey, the connection fails. Please check if the IP address and port of LIS is correct and reconnect as the steps above; if the problem still exists, please contact the hospital network administrator or Erba Lachema customer service engineer to handle it.

2.6.6 Connecting the Printer Box (Optional)

If the analyser needs to be connected with the printer box, you can follow steps in this section to complete the connection.

2.6.6.1 Connecting the Analyser with the Printer Box

- Connect the analyser with the printer box physically.
- Access the software setting interface of the printer box, and set the IP address, port number, etc. of the printer box.

NOTE Please contact the Erba Lachema customer engineer to get Printer Box Operation Guide to complete the connection between the printer box and the analyser.

2.6.6.2 Setting Host Communication

You can skip this step if the analyser is connected with the printer box through a router on the site. If the analyser is connected with the printer box directly through a network cable, to make sure the successful connection, the administrator should set the IP address and other relative information on the Host Communication screen by following steps below.

1. Access **Setup > Communicate** interface, and click Host Communication. See Figure 2-9.

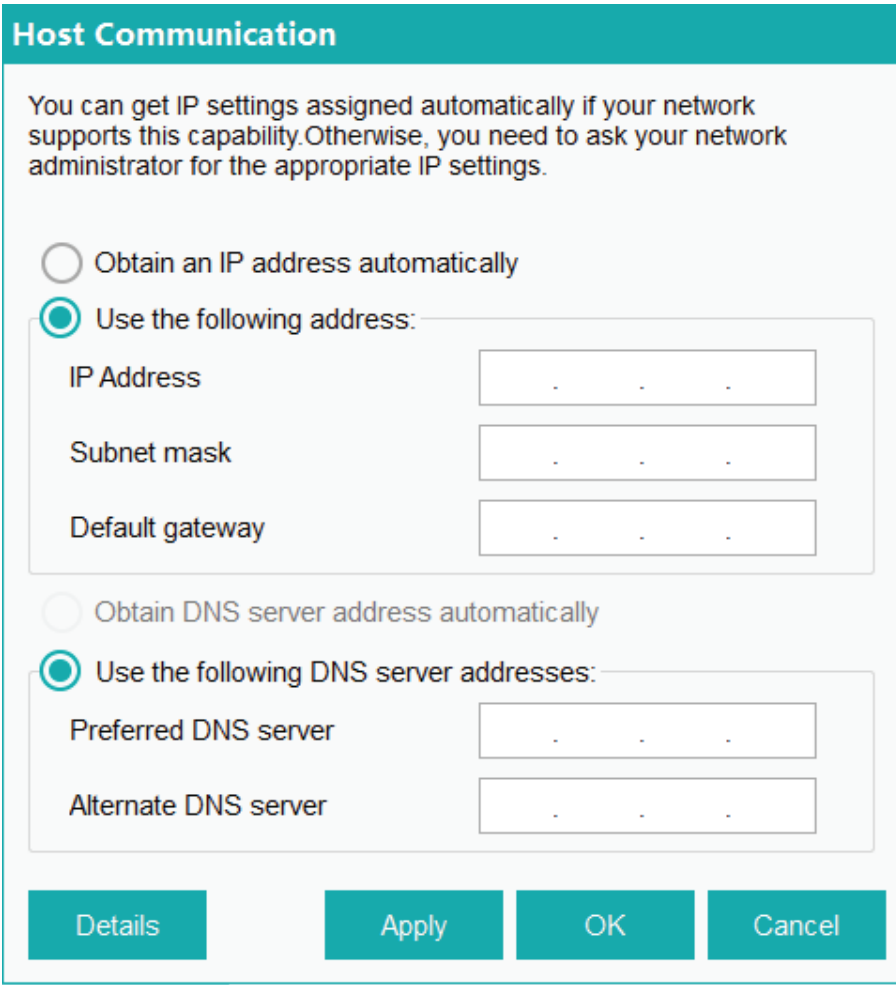
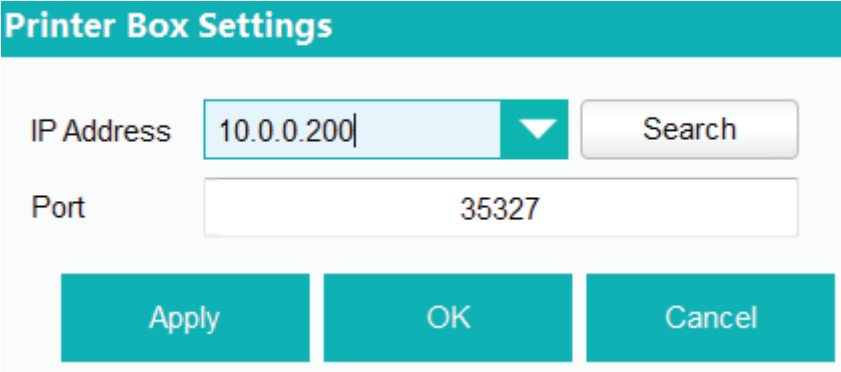
The image shows a 'Host Communication' configuration window. At the top, there is a teal header with the title 'Host Communication'. Below the header, a light gray box contains instructional text: 'You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.' Below this text are two radio button options. The first option is 'Obtain an IP address automatically' with an unselected radio button. The second option is 'Use the following address:' with a selected radio button. This second option is followed by a light gray box containing three input fields: 'IP Address', 'Subnet mask', and 'Default gateway'. Each field has a placeholder showing three dots separated by spaces. Below these fields are two more radio button options. The first is 'Obtain DNS server address automatically' with an unselected radio button. The second is 'Use the following DNS server addresses:' with a selected radio button. This is followed by two input fields: 'Preferred DNS server' and 'Alternate DNS server', each with a placeholder showing three dots separated by spaces. At the bottom of the window, there are four teal buttons: 'Details', 'Apply', 'OK', and 'Cancel'.

Figure 2-9 Host Communication

2. Click **Use the following address**, and set the IP address and subnet mask.
The IP address of the analyser and the printer box must be in the same network segment, and their subnet masks should be the same. Other parameters can be left blank.
3. Click **Apply**
4. Click **OK** to save the setting and close the interface.

2.6.6.3 Printer Box Settings

1. Click **Setup > Print Settings > Printer Box Settings**. See Figure 2-10.



The screenshot shows a 'Printer Box Settings' dialog box. It has a teal header bar with the title 'Printer Box Settings'. Below the header, there are two input fields. The first is labeled 'IP Address' and contains the text '10.0.0.200'. To its right is a small teal button with a white downward-pointing triangle. To the right of that is a light gray button labeled 'Search'. Below the IP Address field is a second input field labeled 'Port' which contains the text '35327'. At the bottom of the dialog, there are three teal buttons: 'Apply', 'OK', and 'Cancel'.

Figure 2-10 Printer Box Settings

2. Enter the IP address and port number of the printer box.

NOTE You need to enter the same IP address and port number set on the software of the printer box.

3. Click **Apply**.
4. Click **OK**.
5. Verify if the connection succeeds.

If the printer icon on the top right corner of the interface is changing from  to , then the connection succeeds. If not, the connection fails. Please check if IP address and port number are correct, and try to reconnect again as described in this section; if it still fails, please contact Erba Lachema customer engineer.

Click **Setup > Print Settings > Printer driver**, click the dropdown list and select Printer Box. See Figure 2-11.

Print Settings

Printer

Printer driverPrinter Box

Printer\\192.168.1.97

Printer ResolutionHigh

Report Settings

Report Titleogy Analysis Report

Copies1

Format Settings

Report TypeReport

Paper TypeA4

TemplateA4-Portrait-His

Paper size210*297 mm

RefreshImportDeleteExport

Auto Settings

AutoprintOnOff

Auto print after validation

Auto validate when printing

Print after validation

Printing Options

Print Flag

Print Ref. Range

Print Suspicious Flag

Print Ref. Range Flags

Print Microscopic Exam Para.

Print result edited flags

Two reports in one page (half of A4)

Update blank test time before be printed

Print as black and white(Report)

QC Graph SettingsPrinter Box Settings

Apply

OK

Cancel

Figure 2-11 Print settings

The drop-down list of **Printer** shows all printers that match with the current printer box. You can select one of them as the default printer.

NOTE

Known virtual printers are automatically blocked. If you select a virtual printer to print, you should restart the printer box.

Then you can perform the printing operation

3 System Overview

3.1 Introduction

This section describes in details the intended use, measurement parameters, structure, user interface and compatible reagents of the analyser.

3.2 Intended Use

The H560 is a fully automated 5-part differential hematology analyser for the *in vitro* diagnostic quantitative determination of cell counts in human blood. It is designed to report on the hematological status of patient samples, supporting the monitoring and diagnosis of pathological conditions. The H560 should be used by trained professionals in the hematology laboratory and only with Erba Mannheim reagents.

NOTE The analyser is intended for screening in the clinical examination. When making clinical judgment based on the analysis results, the doctors should also take into consideration the clinical examination results or other test results.

3.3 Measurement Parameters

The analyser performs sample analysis for different parameters according to different measurement modes (CBC or CBC+DIFF).

- In CBC+DIFF mode, the analyser provides quantitative analysis results for 34 parameters (including 26 haematology parameters and 8 research parameters), 3 histograms, and 4 DIFF scattergrams (including one BASO scattergram and three DIFF scattergrams).
- In CBC mode, the analyser provides quantitative analysis results for 16 haematology parameters, 3 histograms, and one BASO scattergram.

Refer to the Table 3-1 for the detailed parameters.

Table 3-1 Parameters

Type	Parameter Name	Abbreviation	CBC	CBC+DIFF
WBC (17 items)	White Blood Cell count	WBC	*	*
	Percentage of Neutrophils	Neu%	/	*
	Percentage of Lymphocytes	Lym%	/	*
	Percentage of Monocytes	Mon%	/	*
	Percentage of Eosinophils	Eos%	/	*
	Percentage of Basophils	Bas%	/	*

WBC (17 items)	Number of Neutrophils	Neu#	/	*
	Number of Lymphocytes	Lym#	/	*
	Number of Monocytes	Mon#	/	*
	Number of Eosinophils	Eos#	/	*
	Number of Basophils	Bas#	/	*
	Percentage of Abnormal Lymphocytes	ALY% (RUO)	/	*
	Percentage of Large Immature Cells	LIC% (RUO)	/	*
	Number of Abnormal Lymphocytes	ALY# (RUO)	/	*
	Number of Large Immature Cells	LIC# (RUO)	/	*
	Corrected White Blood Cells	cWBC# (RUO)	/	*
	Neutrophil-to-Lymphocyte Ratio	NLR (RUO)	/	*
RBC (10 items)	Red Blood Cell count	RBC	*	*
	Haemoglobin Concentration	HGB	*	*
	Mean Corpuscular Volume	MCV	*	*
	Mean Corpuscular Haemoglobin	MCH	*	*
	Mean Corpuscular Haemoglobin Concentration	MCHC	*	*
	Red Blood Cell Distribution Width - Coefficient of Variation	RDW-CV	*	*
	Red Blood Cell Distribution Width - Standard Deviation	RDW-SD	*	*
	Haematocrit	HCT	*	*
	Percentage of Nucleated red blood cells	NRBC% (RUO)	/	*
	Number of Nucleated red blood cells	NRBC# (RUO)	/	*
PLT (7 items)	Platelet count	PLT	*	*
	Mean Platelet Volume	MPV	*	*
	Platelet Distribution Width - Coefficient of Variation	PDW-CV	*	*
	Platelet Distribution Width - Standard Deviation	PDW-SD	*	*
	Plateletcrit	PCT	*	*
	Platelet-large cell ratio	P-LCR	*	*
	Platelet-large cell count	P-LCC	*	*
Histo-gram (3 items)	White Blood Cell Histogram	WBC Histogram	*	*
	Red Blood Cell Histogram	RBC Histogram	*	*
	Platelet Histogram	PLT Histogram	*	*
Scatter-gram	Differential Scattergram	DIFF Scattergram	/	*
	Basophils Scattergram	BASO Scattergram	*	*

NOTE “*” means the parameter is provided in the mode. “/” means the parameter is not provided. ALY%, LIC%, NRBC%, ALY#, LIC#, NRBC#, cWBC# and NLR are parameters for research use only (RUO), not for diagnostic use.

3.4 Structure of the Analyser



WARNING

Please check the firmness of all the doors, covers and boards before running the analyser. The analyser is heavy, so moving by one person alone may cause injury. It is advisable for two people to move it together when the transportation is necessary, and make sure you follow the instructions and use the proper tools.

Connect only to a properly grounded outlet.

To avoid electrical shocks, disconnect the power supply before opening the cover.

To prevent fire, use the fuses with specified model number and working current.



The sampling probe is sharp and may contain biohazardous materials. Special care should be taken when working with it.



This sign warns of laser radiation. Do not look directly at the laser beams or see through the optical instrument.

3.4.1 Main unit

The H560 Haematology Analyser consists of the main unit (analyser) and accessories. The main unit is the main part for analysis and data processing.

- Front of the analyser



Figure 3-1 Front of the analyser

1. Aspirate key
2. Sample probe
3. Power/Status indicator
4. Touch screen

- Back of the analyser

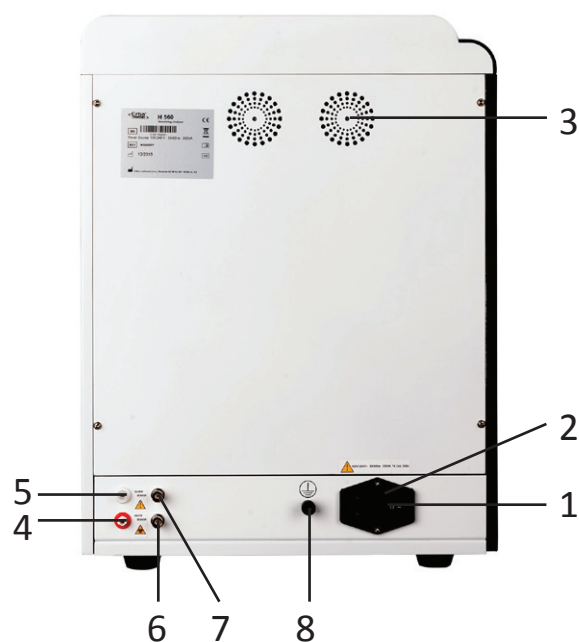


Figure 3-2 Back of the analyser

1. Power switch
2. AC input
3. Cooling fan
4. Waste outlet
5. Diluent inlet
6. Waste level detection connector
7. Diluent presence detection connector
8. Ground studs

- Side view of the analyser



Figure 3-3 Side view of the analyser

1. USB interface
2. Network interface
3. Right side door buckle
4. Left side small door buckle

3.4.2 Touch screen

The touch screen is located on the front side of the analyser for performing interface operations and displaying the information.

3.4.3 Aspirate key

The aspirate key is located in the middle of the front side (behind the sample probe) to start the sample analysis, to add diluent, or to cancel sleep.

3.4.4 Power/Status indicator

The status indicator is located in the middle section of the right part of the analyser (front side). It shows the status of the analyser including ready, running, error, sleep and on/off, etc. The indicators change with the status of the main unit. Details are shown in Table 3-2.

Table 3-2 Main Unit Status Indicators

Instrument Status	Indicator Status	Remarks
Shutdown	Off	The main unit has been shut down.
Stopped running with error conditions	Red light on	Stopped running with the occurrence of errors
Running with error conditions	Red light flickering	Running with the occurrence of errors
Time sequence deactivated	Yellow light on	Initialization or sleep status irrelevant to running
Running	Green light flickering	Execution of the sequence actions is in process.
Ready	Green light on	Execution of the sequence actions is allowed.

NOTE While the analyser is running, if the indicator turns dim or off, please contact Erba Lachema or Erba Lachema's agent for maintenance.

3.4.5 Power switch



CAUTION

To avoid damage, do not power on/off the analyser repetitively within a short time.

A power switch is located in the bottom back of the analyser. It turns on or shuts down the analyser.

3.4.6 USB interface

The USB interface is located on the right side of the main unit. There are 4 interfaces in total for external equipment (printer, barcode scanner, mouse or keyboard, and so on) connection or data transmission.

3.4.7 Network interface

The network interface is located on the right side of the main unit. There is 1 network interface in total for connecting with the Ethernet.

3.4.8 External Equipment (Optional)

The analyser can be connected with the following external equipment:

- **Keyboard**
The keyboard is connected with the USB interface on the right side of the analyser for controlling the analyser.
- **Mouse**
The mouse is connected with the USB interface on the right side of the analyser for operations on the analyser.
- **Printer**
The printer is connected with the USB interface on the right side of the analyser for printing reports and other information displayed on the screen.
- **Barcode Scanner**
The barcode scanner is connected with the USB interface on the right side of the analyser for entering barcode information in an easy and fast way.
- **USB flash disk**
The USB flash disk is connected with the USB interface on the right side of the analyser for exporting sample data.

3.5 User Interface

After the start-up procedure, you will enter the user interface. See Figure 3-4.

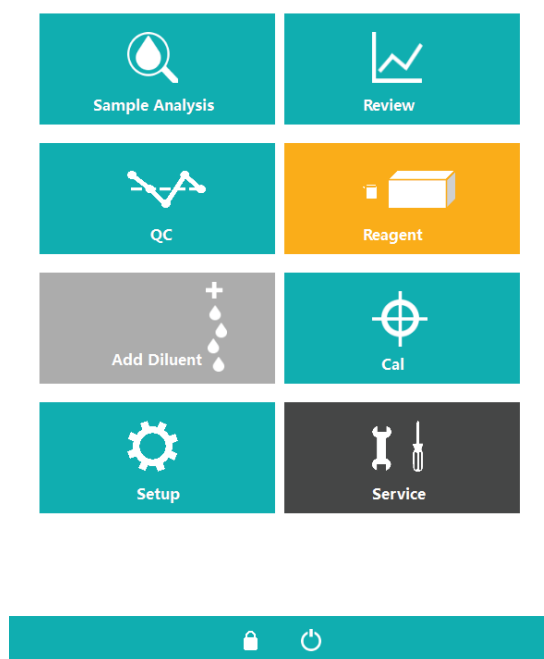


Figure 3-4 Default screen

3.5.1 Sample analysis screen

The interface can be divided into several areas as follows according to their functions:

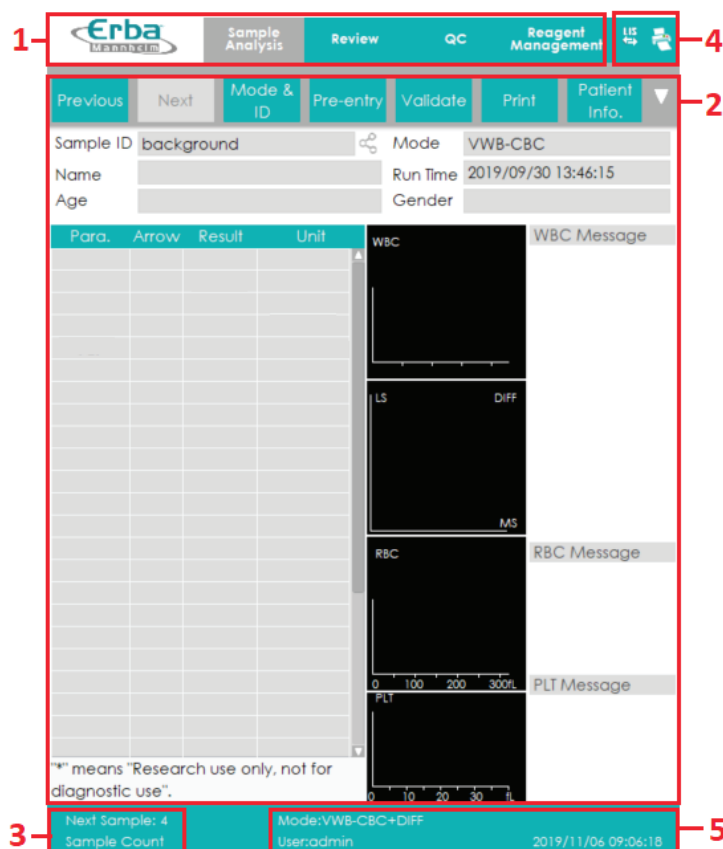


Figure 3-5 Sample analysis screen

- 1. Menu navigation area**
On the top of the screen is the menu navigation area. Once a menu button is pressed, the system goes immediately to the corresponding screen.
- 2. Menu content display area**
It displays the selected screen and the corresponding function buttons.
- 3. Error message area**
Upon the occurrence of a system failure, the corresponding error message will appear in this area. When there is more than one failure, the error message for the latest failure will appear in this area.
Click in this area, you can deal with the failures in the popup dialog box of troubleshooting help. For more information, see **13 Troubleshooting**.
- 4. Status display area**
On the top right of the screen is the status display area where the connection status between the analyser and the LIS system and printer status are displayed from left to right. The icons change with the status of the main unit, as shown in Table 3-3.

Table 3-3 Status Icon Description

Status	Icon	Remarks
LIS/HIS status	Grey icon 	The computer is not connected to the LIS/HIS.
	Black icon 	The computer is connected to the LIS/HIS.
Print status	Grey icon 	The external printer is not connected to the analyser yet.
	Colour icon 	The external printer is connected to the analyser.

5. Current user, date and time of the analyser.

3.6 Reagents, Controls and Calibrators

Because the analyser, reagents, controls, and calibrators are components of the system, system performance depends on the combined integrity of all the components. You should only use the Erba Lachema-specified reagents (see 15.2. **Reagents**), which are formulated specifically for the fluidic system of your analyser in order to achieve optimal system performance. Do not operate the analyser using reagents from multiple suppliers. Under such circumstances, the analyser may not achieve the performance specified in this manual and may generate unreliable results. All references to “reagents” in this manual refer to the reagents specifically formulated for this analyser. Each reagent package should be examined before use. Inspect the package for signs of leakage or moisture. If there is evidence of leakage or improper handling, do not use the reagent.

After long-distance transportation, the reagent must be allowed to settle for more than one day before use.

Store and use the reagents by following the instructions for use of the reagents.

NOTE When you have changed the diluents or lyses, run a background check to see if the results meet the requirement.

Pay attention to the expiration dates and open-container stability days of all the reagents. Be sure not to use expired reagents.

3.6.1 Reagents

The following reagents are intended to be used with the analyser H560.

- **Erba H560 Dil**, cat. no. HEM00030

Erba H560 Dil is an *in vitro* diagnostic reagent used for automated dilution of human blood for counting and sizing of blood cells on hematology analysers H560. It is designed to report on the hematological status of patient samples, supporting the monitoring and diagnosis of pathological conditions. The Erba H560 Dil should be used by trained professionals in the hematology laboratory and only with Erba Mannheim reagents.

- **Erba H560 Lyse1**, cat.no. HEM00031

Erba H560 Lyse1 is an *in vitro* diagnostic reagent used to lyse red blood cells for quantitative counting of white blood cells and measuring of hemoglobin in human blood on automated hematology analysers H560. It is designed to report on the hematological status of patient samples, supporting the monitoring and diagnosis of pathological conditions. The Erba H560

Lyse1 should be used by trained professionals in the hematology laboratory and only with Erba Mannheim reagents.

- **Erba H560 Lyse2**, cat.no. HEM00032

Erba H560 Lyse2 is an *in vitro* diagnostic reagent used to lyse red blood cells for quantitative 5-part differentiation of white blood cells in human blood on automated hematology analysers H560. It is designed to report on the hematological status of patient samples, supporting the monitoring and diagnosis of pathological conditions. The Erba H560 Lyse2 should be used by trained professionals in the hematology laboratory and only with Erba Mannheim reagents.

- **ELite H Clean**, cat.no. HEM00023

ELite H Clean is an *in vitro* diagnostic solution used for cleaning, rinsing and washing of automated hematology analysers ELite 580, H560 and H360. The ELite H Clean should be used by trained professionals in the hematology laboratory and only with Erba Mannheim reagents.

3.6.2 Controls and Calibrator

The following controls and calibrator are used for quality control and calibration of H560.

- **ELite H5 CON L**, cat.no. HEM00024

- **ELite H5 CON N**, cat.no. HEM00025

- **ELite H5 CON H**, cat.no. HEM00026

ELite H5 CON L/N/H is an *in vitro* diagnostic reagent used for control of the accuracy and precision of automated hematology analysers ELite 580 and H560. ELite H5 CON L/N/H should be used by trained professionals in the hematology laboratory and only with Erba Mannheim reagents.

- **ELite H CAL**, cat.no. HEM00027

ELite H CAL is an *in vitro* diagnostic reagent used for calibration of the parameters WBC, RBC, HGB, MCV and PLT on automated hematology analysers ELite 580, H560 and H360. Intended for screening, monitoring and diagnosis of risk various diseases according to the calibrated parameters. ELite H CAL should be used by trained professionals in the hematology laboratory and only with Erba Mannheim reagents.

Working Principle

4.1 Introduction

The measurement methods used in this analyser are: the electrical Impedance method for determining the RBC and PLT data; the colorimetric method for determining the HGB; laser-based flow cytometry for determining the WBC data. During each analysis cycle, the sample is aspirated, diluted and mixed before the determination for each parameter is performed.

4.2 Aspiration

The analyser supports Whole Blood mode (including **Venous Whole Blood** and **Capillary Whole Blood**) and Predilute mode.

In Whole Blood mode, the analyser will aspirate 15µL of whole blood.

In Predilute mode, the analyser will aspirate the prediluted sample (with the dilution ratio of 1:25) which is a mixture of 20µL of whole blood/capillary blood sample and 480µL of diluent the diluted sample thus prepared is then delivered to the analyser for sampling and aspiration.

4.3 Dilution

After being aspirated into the analyser, the sample is divided into two parts. After the reaction with reagents in parallel dilution procedures, each part forms the sample for red blood cell/platelet, white blood cell count/haemoglobin measurement and white blood cell differential measurement. To meet different needs, the analyser offers two working modes (Whole Blood and Predilute), and two measurement modes (CBC and CBC+DIFF).

Taking CBC+DIFF mode as an example, this section introduces the dilution procedures of the test sample in Whole Blood mode and Predilute mode separately. (The dilution procedure in CBC mode is not introduced here since it's the same as that in CBC+DIFF mode.)

NOTE	CBC mode, namely complete blood cell count, is intended for counting only, not for white blood cell classification. CBC+DIFF mode is intended for both counting and white blood cell classification.
-------------	--

4.3.1 Dilution Procedure in Whole-blood CBC+DIFF Mode

Dilution Procedures in Whole-Blood CBC+DIFF Mode are shown in Figure 4-1.

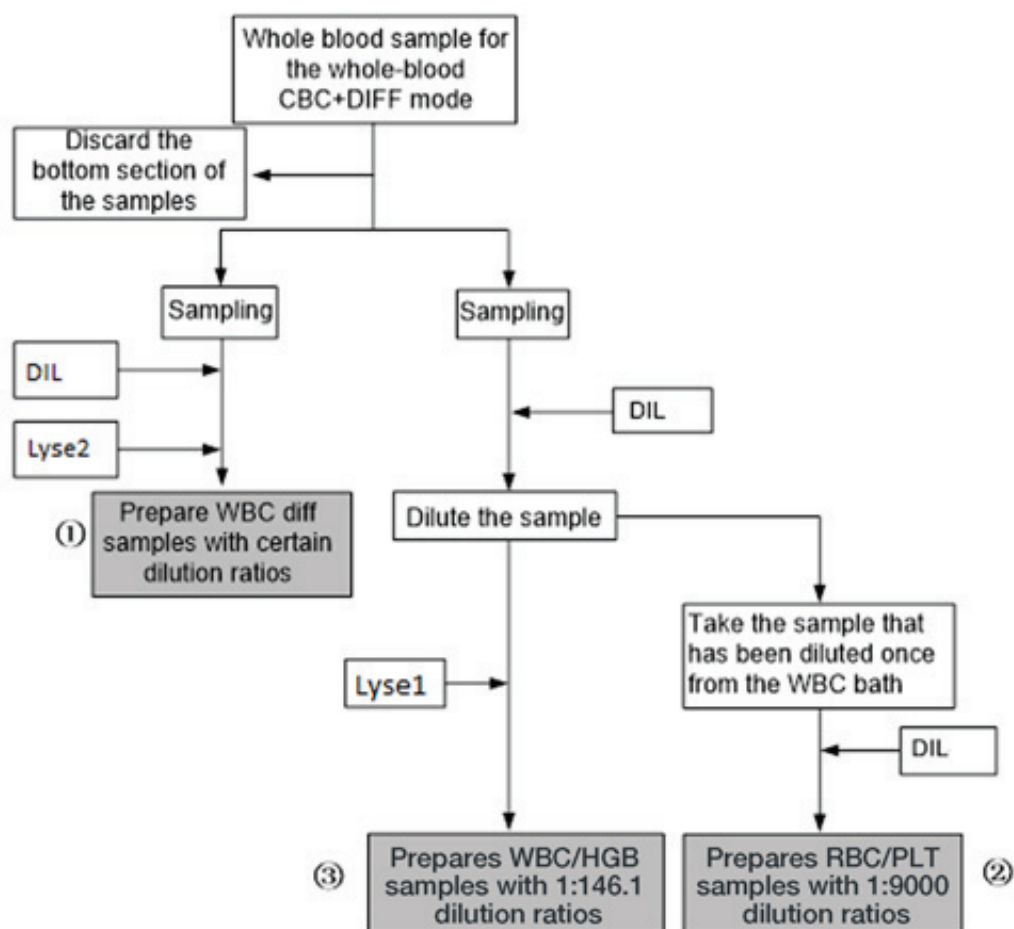


Figure 4-1 Dilution Procedure in Whole-blood CBC+DIFF Mode

Where,

1. is the dilution procedure for white blood cell diff, namely DIFF;
2. is the dilution procedure for red blood cell and platelet
3. is the dilution procedure for white blood cell count/haemoglobin; namely CBC.

4.3.2 Dilution Procedure in Predilute CBC+DIFF Mode

In CBC+DIFF mode, the dilution procedure for the prediluted sample is shown in Figure 4-2.

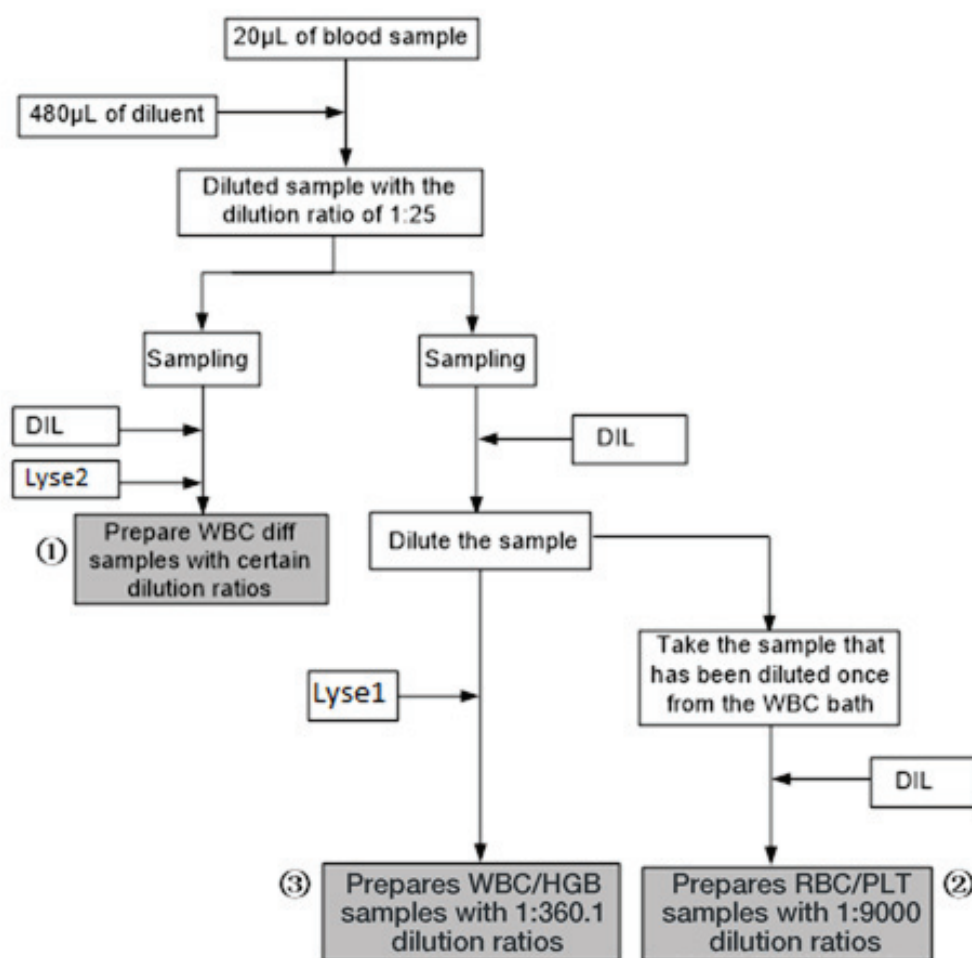


Figure 4-2 Dilution Procedure in Predilute CBC+DIFF Mode

Where,

1. is the dilution procedure for white blood cell diff, namely DIFF;
2. is the dilution procedure for red blood cell and platelet;
3. is the dilution procedure for white blood cell count/haemoglobin; namely CBC.

4.4 WBC Measurement

The analyser obtains the white blood cell 5-part classification results and white blood cell count/basophils count using a semiconductor-laser-based flow cytometry, and eventually calculates the parameters relevant to white blood cells.

4.4.1 Working Principle of Laser-based Flow Cytometry

The principle of laser-based flow cytometry is illustrated by Figure 4-3.

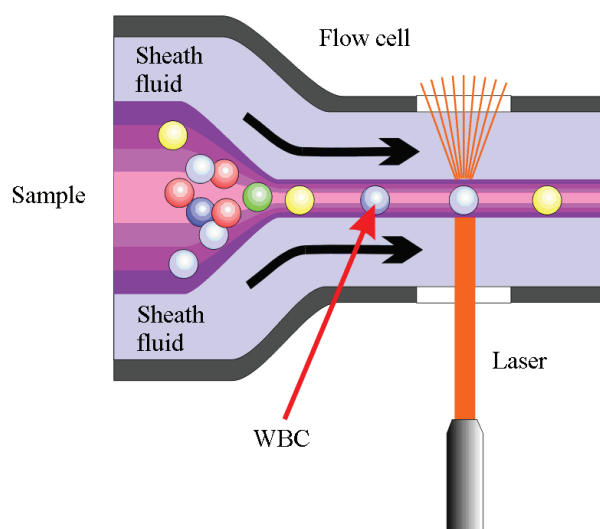


Figure 4-3 WBC Measurement

After a predetermined volume of blood is aspirated and diluted by a certain amount of reagent, it is injected into the flow chamber. Surrounded with sheath fluid (diluent), the blood cells pass through the center of the flow chamber in a single column at a faster speed. When the blood cells suspended in the diluent pass through the flow chamber, they are exposed to a laser beam. The intensity of scattered light reflects the blood cell size and intracellular density. The low-angle scattered light signal shows cell size, while the middle-angle and high-angle scattered light signal show intracellular information (nucleus and cytoplasm information). The optical detector receives this scattered signal and converts it into electrical pulses. Pulse data thus collected can be used to draw four 2-dimensional distributions (scattergrams) as shown in Figure 4-4.

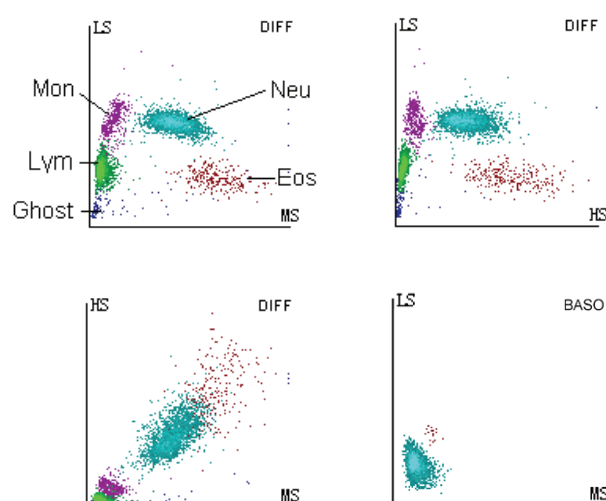


Figure 4-4 DIFF channel scattergram

Conduct dual channel detection to the white blood cells (WBCs). Use three-angle laser scattering and flow cytometry for the count and classification of various kinds of WBCs in dual channels. By analysing the DIFF channel scattergram, the analyser presents the Lym%, Mon%, Eos% and Neu%.

The independent WBC/Bas channel shall use a specific kind of haemolytic agent that can extract the Baso cell specificity, so as to reserve the complete information of Bas cells. Conduct precise and reliable WBC/Bas cell counting combined with three-angle laser scattering and flow cytometry.

4.4.2 Derivation of WBC-Related Parameters

Based on the DIFF scattergram and the analysis for the Lym zone, Neu zone, Mon zone and Eos zone, the analyser can get the percentage of lymphocytes (Lym%), the percentage of neutrophils (Neu%), the percentage of monocytes (Mon%) and the percentage of eosinophils (Eos%), and then get the number of basophils (Bas#), the number of lymphocytes (Lym#), the number of neutrophils (Neu#), the number of monocytes (Mon#) and the number of eosinophils (Eos#) based on the calculation with the white blood cell count obtained with the working principle of laser-based flow cytometry. The unit of the number of cells is $10^3/\mu\text{L}$.

- **White Blood Cell count**
WBC count is the number of leukocytes measured directly by counting the leukocytes passing through the flow chamber.
- **Number of Basophils (Bas#)**
Bas# is the number of Basophils measured directly by counting the basophils passing through the flow chamber.
- **Percentage of Basophils (BAS%)**

$$\text{Bas\%} = \frac{\text{Bas\#}}{\text{WBC}} \times 100\%$$
- **Percentage of Lymphocytes (Lym%)**

$$\text{Lym\%} = \frac{\text{Particles in Lym region of DIFF channel}}{\text{Sum of all particles in DIFF channel except those in Ghost region}} \times 100\%$$
- **Percentage of Neutrophils (Neu%)**

$$\text{Neu\%} = \frac{\text{Particles in Neu region of DIFF channel}}{\text{Sum of all particles in DIFF channel except those in Ghost region}} \times 100\%$$
- **Percentage of Monocytes (Mon%)**

$$\text{Mon\%} = \frac{\text{Particles in Mon region of DIFF channel}}{\text{Sum of all particles in DIFF channel except those in Ghost region}} \times 100\%$$
- **Percentage of Eosinophils (EOS%)**

$$\text{Eos\%} = \frac{\text{Particles in Eos region of DIFF channel}}{\text{Sum of all particles in DIFF channel except those in Ghost region}} \times 100\%$$
- **Number of lymphocytes (Lym#)**

$$\text{Lym\#} = \text{WBC} \times \text{Lym\%}$$
- **Number of Neutrophils (Neu#)**

$$\text{Neu\#} = \text{WBC} \times \text{Neu\%}$$
- **Number of Monocytes (Mon#)**

$$\text{Mon\#} = \text{WBC} \times \text{Mon\%}$$
- **Number of Eosinophils (Eos#)**

$$\text{Eos\#} = \text{WBC} \times \text{Eos\%}$$
- **Neutrophil-to-Lymphocyte Ratio (NLR)**

$$\text{NLR} = \text{Neu\#} / \text{Lym\#}$$

4.5 HGB Measurement

HGB is determined by the colorimetric method.

4.5.1 Colorimetric Method

The WBC/HGB diluent is delivered to the HGB bath where it is mixed with a certain amount of lyse, which converts haemoglobin to a haemoglobin complex that is measurable at 525 nm. An LED is mounted on one side of the bath and emits a beam of monochromatic light with a central wavelength of 525nm. The light passes through the sample and is then measured by an optical sensor mounted on the opposite side. The signal is then amplified and the voltage is measured and compared with the blank reference reading (readings taken when there is only diluent in the bath).

4.5.2 HGB

The HGB is calculated using the following equation and expressed in g/dL.

$$\text{HGB(g/dL)} = \text{Constant} \times \ln \left(\frac{\text{Blank Photocurrent}}{\text{Sample Photocurrent}} \right)$$

4.6 RBC/PLT Measurement

The analyser detects the red blood cell count and platelet count and their volume distribution by impedance method and eventually obtains the results of related parameters.

4.6.1 Electrical Impedance Method

RBCs/PLTs are counted and sized by the Electrical Impedance method. This method is based on the measurement of changes in electrical resistance produced by a particle, which in this case is a blood cell, suspended in a conductive diluent as it passes through an aperture of known dimensions. An electrode is submerged in the liquid on both sides of the aperture to create an electrical pathway. As each particle passes through the aperture, a transitory change in the resistance between the electrodes is produced. This change produces a measurable electrical pulse. The number of pulses thus generated is equal to the number of particles that passed through the aperture.

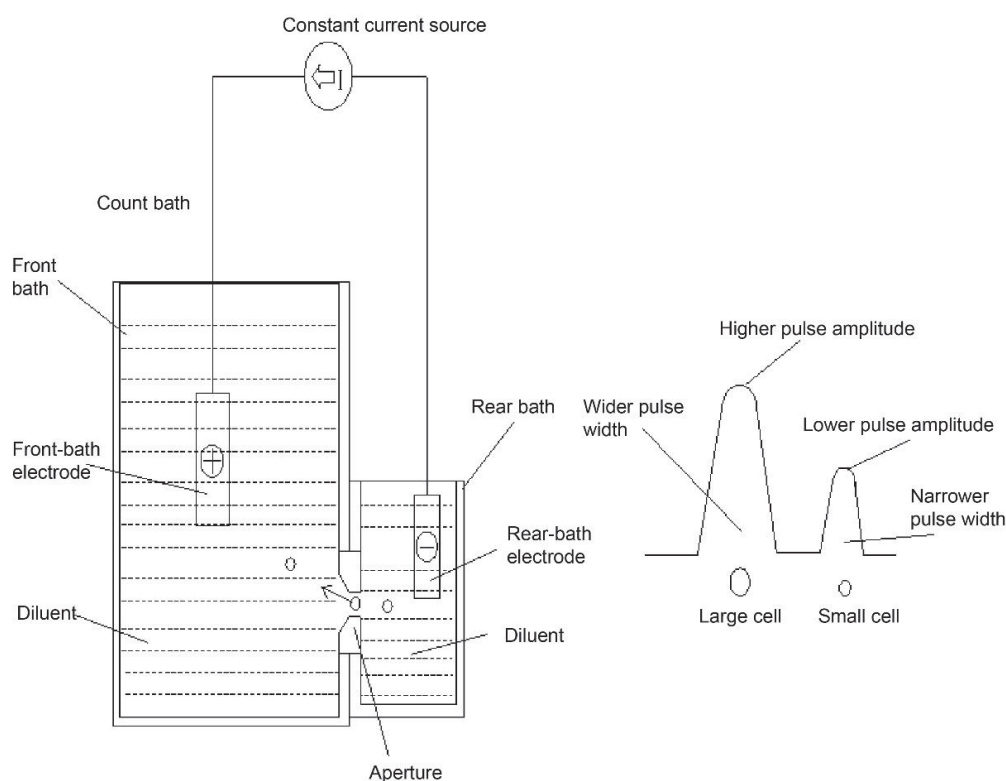


Figure 4-5 Electrical Impedance method

Each pulse is amplified and compared to the internal reference voltage channel, which only accepts the pulses of a certain amplitude. If the pulse generated is above the RBC/PLT lower threshold value, it is counted as a RBC/PLT. The analyser presents the RBC/PLT histogram, where the x-coordinate represents the cell volume (fL) and the y-coordinate represents the number of the cells.

4.6.2 RBC

- Red Blood Cell count
RBC ($10^6/\mu\text{L}$) is the number of erythrocytes measured directly by counting the erythrocytes passing through the aperture.
- Mean Corpuscular Volume (MCV)
Based on the RBC histogram, this analyser calculates the MCV and expresses the result in fL.
- Haematocrit (HCT), Mean Corpuscular Haemoglobin (MCH), Mean Corpuscular Haemoglobin Concentration (MCHC)
This analyser calculates the HCT (%), MCH (pg) and MCHC (g/dL) as follows, where the RBC is expressed in $10^6/\mu\text{L}$, MCV in fL and HGB in g/dL.

$$\text{HCT} = \frac{\text{RBC} \times \text{MCV}}{10}$$

$$\text{MCH} = \frac{\text{HGB}}{\text{RBC}}$$

$$\text{MCHC} = \frac{\text{HGB}}{\text{HCT}} \times 100$$

- Red Blood Cell Distribution Width - Coefficient of Variation (RDW-CV)
Based on the RBC histogram, this analyser calculates the CV (Coefficient of Variation, %) of the erythrocyte distribution width.
- Red Blood Cell Distribution Width - Standard Deviation (RDW-SD)
RDW-SD (RBC Distribution Width – Standard Deviation, fL) is obtained by calculating the standard deviation of the red blood cell size distribution.

4.6.3 PLT

- Platelet count
PLT ($10^3/\mu\text{L}$) is measured directly by counting the platelets passing through the aperture.
- Mean Platelet Volume (MPV, fL)
Based on the PLT histogram, this analyser calculates the MPV.
- Platelet Distribution Width - Coefficient of Variation (PDW-CV)
Based on the PLT histogram, this analyser calculates the CV (Coefficient of Variation, %) of the platelet distribution width.
- Platelet Distribution Width - Standard Deviation (PDW-SD)
PDW-SD (PDW Distribution Width – Standard Deviation, fL) is obtained by calculating the standard deviation of the platelet size distribution.
- Plateletcrit (PCT)
This analyser calculates the PCT as follows and expresses it in %, where the PLT is expressed in $10^3/\mu\text{L}$ and the MPV in fL.

$$\text{PCT} = \frac{\text{PLT} \times \text{MPV}}{10000}$$
- Platelet-Large Cell Count (P-LCC, $10^3/\mu\text{L}$) – P-LCC is measured directly by counting the large platelets passing through the aperture.
- Platelet-Large Cell Ratio (P-LCR)

$$\text{P-LCR} = \frac{\text{P-LCC}}{\text{PLT}} \times 100\%$$

4.7 Flushing

After each analysis cycle, each component of the analyser is flushed.

5 Setup

5.1 Introduction

The analyser has been initialized before delivery. The interfaces upon the initial start-up of the analyser are system settings by default. Some parameters of the analyser can be reset to meet various demands in practical applications.

The analyser divides the operators into two access levels, common user and administrator. Note that an administrator can access all the functions accessible to a common user. This chapter introduces how to customize your analyser as an administrator.

5.2 Interface Introduction

After logging in the system (see **6.3 Start-up**), click **Setup** to access the **Setup** interface. See Figure 5-1.

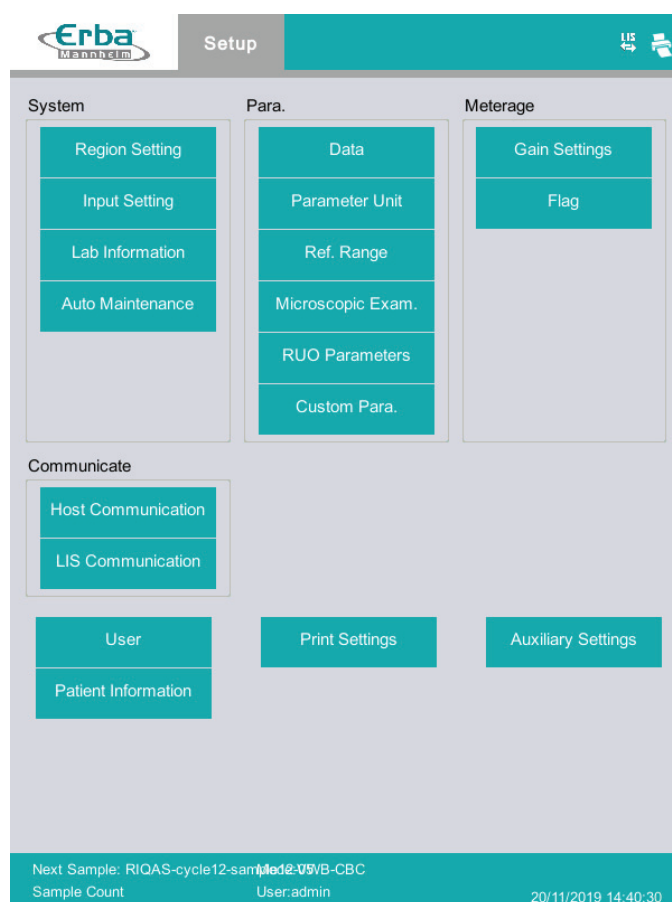


Figure 5-1 Setup

The administrator is allowed to set the following functions in the **Setup** interface:

- System settings
- Parameter settings
- Meterage settings
- LIS communication
- User management
- Print settings
- Auxiliary settings
- Patient information

5.3 System Settings

5.3.1 Region Setting

You can set the current date and time, as well as the date display format in the analyser system. Specific steps are shown below:

- Click **Region Setting** in the **System** area.

The date and time format setting interface pops up.

Figure 5-2 Setup of the Region Setting

- Click the Date and Time dropdown list and set the current date and time of the system in the popup dialog box.

Figure 5-3 Setup of the date and time

Related descriptions:

The input sequence of the controls is the same with the date format on the top right corner of the dialog box. For example, if the data format is **yyyy/MM/dd HH:mm:ss**, you should input the data in the sequence of year, month, date, hour, minute, and second.

Click  or  to select a date and time or enter the information in the textbox directly.

Click  to clear the current data and re-enter the information.

- Click **OK** to save and close the message box.
- Select the format setting from the dropdown list of the **Date Format**.

See Figure 5-4.

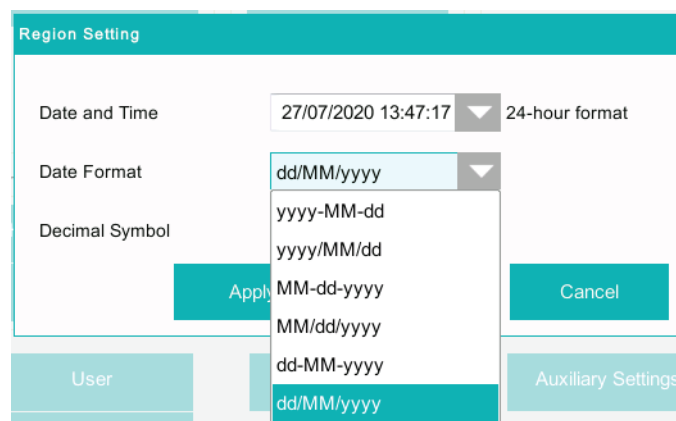


Figure 5-4 Setting the Date Format

- Click **Apply**.

The system message will pop up, indicating the successful setting. See Figure 5-5.

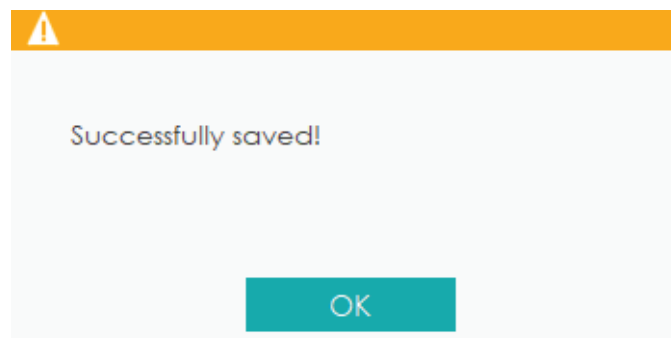


Figure 5-5 Successful Setting of the Date Format

The date and time at the bottom right corner will be displayed in the newly set format.

- Click **OK** to close the message box.
- Click **OK** to exit.

5.3.2 Input Settings

Click **Input Setting** in the **System** area, and then you can set the soft keyboard for screen input. As shown in Figure 5-6, you can set to turn the soft keyboard on or off.

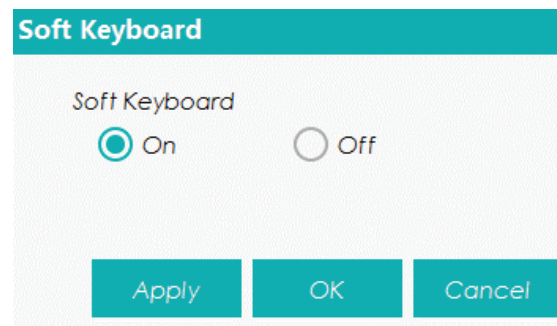


Figure 5-6 Input Settings

- Soft Keyboard

On (default)

You can enter content using the soft keyboard popped up on the screen. Functions and applications for the keys are shown in Figure 5-7.



Figure 5-7 Soft Keyboard

Off

You need to use an externally connected USB keyboard for entering content.

5.3.3 Lab Information

Click **Lab Information** in the **System** selection, then you can set the lab information. See Figure 5-8.

Lab Information

Hospital Name

Lab Name

Responsible Person

Responsible Person Contact Info

Customer Service Contact

Customer Service Contact Info

Analyzer SN

Installation Date

/ /

Remarks

Apply

OK

Cancel

Figure 5-8 Setting Lab Information

NOTE

Only the administrator has the access for setting the lab information. General users are only allowed to browse such information.

Refer to Table 5-1 for the detailed instructions of parameter setting.

Table 5-1 Setting Lab Information

Parameter	Setting Description
Hospital Name	Enter the name of the hospital where the lab is located.
Lab Name	Enter the lab name.
Responsible Person	Enter the responsible person of the lab.
Contact Info	Enter the contact information (telephone number or E-Mail) of the lab.
Customer Service Contact	Enter the name of the contact person in Service Department.
Customer Service Contact Info	Enter the contact information of the contact person in the Service Department.
Analyser SN	Display the serial number of the analyser. Read only.
Installation Date	Display the installation date of the analyser. Read only.
Remarks	Enter the remarks regarding the lab.

5.3.4 Auto Maintenance

Click **Auto Maintenance** in the **System** selection to access the **Auto Maintenance** setting interface. The system auto sleep waiting time and cleanser maintenance time can be set in the **Auto Maintenance** interface.

Figure 5-9 Auto Maintenance

Auto Sleep

In the **Wait** textbox, the administrators can set the waiting time for entering the sleep state after the main unit is halted. The range is between 15 and 120 minutes and the default value is 120 minutes.

Auto Cleanser Soak

The administrator is allowed to set the start time of the cleanser soak in the **Start Time** textbox. The acceptable value ranges from 0:00 to 23:59 and the default value is **17:00**.

In the **Wait** text box, the administrators can set the time interval to remind the user to perform the cleanser soak. When the system reminds the user to perform cleanser, if the user cancels the operation, the system will remind again after the set waiting time. The range is between 1 and 30 minutes and the default value is **10** minutes.

5.4 Parameter Settings

5.4.1 Data Dictionary

You can set shortcut codes for the relevant items of the patient information.

If a shortcut code is set, the shortcut code corresponding to the above mentioned item can be entered directly when the information is input or numbered, then the complete information can be displayed without entering (or selecting) complete information. It is a shortcut operation.

Different items can share one shortcut code.

5.4.1.1 Accessing the interface

Click **Data** in the **Para.** selection to access the data dictionary setting interface. See Figure 5-10. You can set the shortcut code for the relevant items of the patient information in this interface.

Data			
Department	Name	Shortcut Code	Remarks
Submitter	Internal Medicine	Nk	
Patient Type	Surgery	Wk	
Gender			
Area			
Bed No.			
Sample Type			

New Edit Delete Cancel

Figure 5-10 Shortcut Code

You can set the shortcut code for the following items: **Department**, **Submitter**, **Patient Type**, **Gender**, **Area**, **Bed No.** and **Sample Type**.

5.4.1.2 Adding a New Item

This section takes the adding of a new department as an example to introduce the method for adding a new item and its shortcut code. The method for adding other new items is similar and is not introduced in details herein.

Steps for adding a new department are shown as follows:

1. Click New in the Department interface.
A dialog box will pop up as shown in Figure 5-11.

New

Name

Shortcut Code

Remarks

OK

Cancel

Figure 5-11 Adding a New Item

2. Enter a new department name, shortcut code and remarks.

NOTE Newly added department name must be entered and it cannot be the same as existing ones.
The shortcut code is not necessary to be entered, but once set, every code must be unique.

3. Click **OK** to save the information about the new department.
Information about the newly added department will be displayed in the department interface.

5.4.1.3 Editing Items/Shortcut Code

This section takes the editing of a department as an example to introduce the method for editing items and its shortcut code. The method for editing other new items is similar and is not introduced in details herein.

Steps for editing a department are shown as follows:

1. Select the department to be modified in the **Department** interface (for example the Internal Medicine), then click **Edit**.
2. Modify the **Name**, **Shortcut Code** and **Remarks** in each textbox according to the actual demand.

NOTE	Newly added department name must be entered and it cannot be the same as existing ones. The shortcut code is not necessary to be entered, but once set, every code must be unique.
-------------	---

3. Click **OK** to save the information.

5.4.1.4 Deleting a Shortcut Code

This section takes the deleting of a department as an example to introduce the method for deleting items and its shortcut code. The method for deleting other new items is similar and is not introduced in details herein.

Steps for deleting a department are shown as follows:

1. Select the department to be deleted in the **Department** interface, and then click **Delete**.
2. Click **Yes** to delete the department.

5.4.2 Parameter Unit

Some of the parameters of the analyser can use different units which can be chosen as per user demand.

5.4.2.1 Accessing the Interface

Click Parameter Unit in the Para. selection to access the Parameter Unit setting interface. See Figure 5-12.

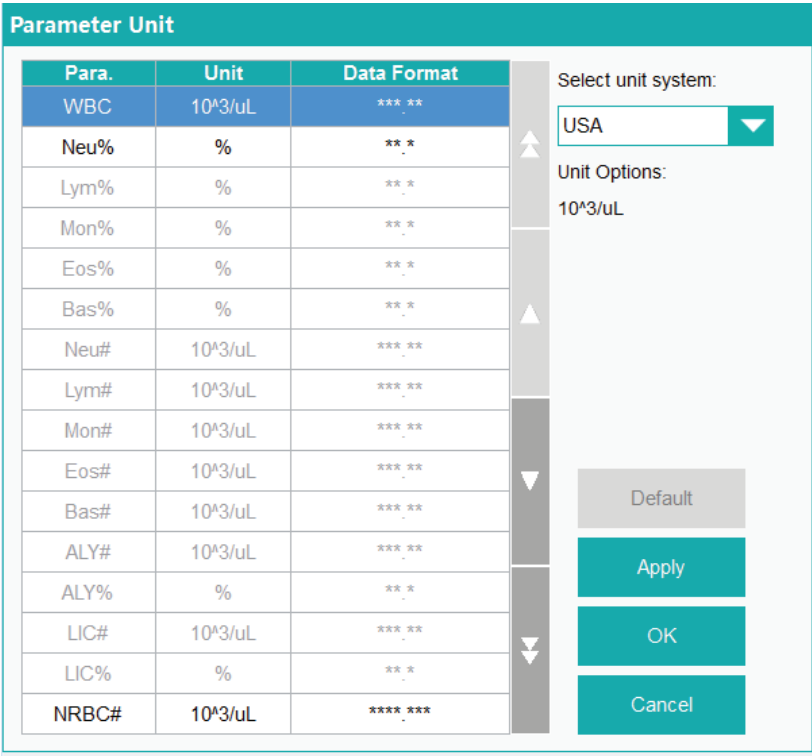


Figure 5-12 Setting Parameter Unit

5.4.2.2 Selecting Unit System

Click the **Select unit system** dropdown list and select a unit system for the parameters among the 6 unit systems (**Custom**, **International**, **Britain**, **Canada**, **USA** and **Netherlands**). The default unit system is **USA**.

NOTE When selecting different unit standards, the corresponding unit list and unit option will be displayed differently.
If another option is selected except the Custom, then the unit of each parameter can only be browsed.

5.4.2.3 Customizing Parameter Unit

- 1. Select **Custom** from the dropdown list of **Select unit system**.
- 2. Click the parameter, of which the unit is to be set, from the parameter list (such as WBC).
- 3. Select a new parameter unit from the **Unit Options** list.
- 4. Click **Apply** or **OK** to save the configuration.

NOTE

For parameters in the same group, if the unit of any parameter changes, the units of the other parameters change accordingly. (In the list, parameters will be sorted by group; the first parameter will be displayed in black and the other parameters in the same group will be displayed in grey.)

If the parameters units change, the display format of the list data will change accordingly.

5.4.2.4 Retrieving Defaults

When setting the **Custom** unit system, if you click **Default**, the unit of the parameters can be restored to the initial default values.

5.4.3 Reference Range

It is recommended that each laboratory establish its own expected reference intervals based upon the laboratory's patient population. The CLSI document* C28-A2 contains guidelines for determining reference intervals and intervals for quantitative laboratory tests. If the analysis result of a sample is beyond the reference range, it will be regarded as clinically abnormal. The **Ref. Range** interface is where you view and set the high and low limits for your patients. The analyser flags any parameter value above  or below  these limits.

This analyser divides the patients into 5 demographic groups: General, Man, Woman, Child and Neonate. You can also customize other groups. The recommended limits are for reference only. To avoid misleading parameter flags, be sure to set the patient limits according to the characteristics of local population.

** How to Define and Determine Reference Intervals in the Clinical Laboratory; Approved Guideline - Second Edition, CLSI document C28-A2, (ISBN P56238-269-1), CLSI (Clinical and Laboratory Standards Institute), 940 West Valley Road, Suite 1400, Wayne, PA 19087-1898, 2000.*

Table 5-2 Preset reference intervals of General reference group

Parameter	Lower Limit	Upper Limit	Unit
WBC	3.50	9.50	10 ⁹ /L
Neu%	40.0	75.0	%
Lym%	20.0	50.0	%
Mon%	3.0	10.0	%
Eos%	0.4	8.0	%
Bas%	0.0	1.0	%
Neu#	1.80	6.30	10 ⁹ /L
Lym#	1.10	3.20	10 ⁹ /L
Mon#	0.10	0.60	10 ⁹ /L
Eos#	0.02	0.52	10 ⁹ /L
Bas#	0.00	0.06	10 ⁹ /L

Parameter	Lower Limit	Upper Limit	Unit
RBC	3.80	5.80	10 ¹² /L
HGB	115	175	g/L
HCT	35.0	50.0	%
MCV	82.0	100.0	fL
MCH	27.0	34.0	pg
MCHC	316	354	g/L
RDW-CV	11.0	16.0	%
RDW-SD	35.0	56.0	fL
PLT	125	350	10 ⁹ /L
MPV	6.5	12.0	fL
PDW-CV	10.0	17.9	%
PDW-SD	9.0	17.0	fL
PCT	0.108	0.282	%
P-LCC	30	90	10 ⁹ /L
P-LCR	11.0	45.0	%

5.4.3.1 Accessing the Interface

Click **Ref. Group** in the **Para.** selection to access the reference group settings interface. See Figure 5-13.

Ref. Range

Ref. Group	Default	Lower Limit of Age	Upper Limit of Age	Gender
General	√			
Man		13 Years	999 Years	Male
Woman		13 Years	999 Years	Fem...
Child		28 Days	13 Years	
Neonatus		0 Hour	28 Days	

☒ Automatically match the customized reference group according to age and gender

Copy

New

Edit

Delete

Set as default

Close

Figure 5-13 Ref. Range

5.4.3.2 Copying a Ref. Group

Select a reference group and click **Copy**, and a new reference group with everything the same except the name of the reference group will be added to the system and a screen as shown in Figure 5-14 will pop up.

New

Para.	Lower Limit	Upper Limit	Unit	Para.	Lower Limit	Upper Limit	Unit
WBC	3.50	9.50	10 ³ /uL	RBC	3.80	5.80	10 ⁶ /uL
Neu%	40.0	75.0	%	HGB	11.5	17.5	g/dL
Lym%	20.0	50.0	%	HCT	35.0	50.0	%
Mon%	3.0	10.0	%	MCV	82.0	100.0	fL
Eos%	0.4	8.0	%	MCH	27.0	34.0	pg
Bas%	0.0	1.0	%	MCHC	31.6	35.4	g/dL
Neu#	1.80	6.30	10 ³ /uL	RDW-CV	11.0	16.0	%
Lym#	1.10	3.20	10 ³ /uL	RDW-SD	35.0	56.0	fL
Mon#	0.10	0.60	10 ³ /uL	PLT	125	350	10 ³ /uL
Eos#	0.02	0.52	10 ³ /uL	MPV	6.5	12.0	fL
Bas#	0.00	0.06	10 ³ /uL	PDW-SD	9.0	17.0	fL
ALY#	0.00	0.20	10 ³ /uL	PDW-CV	10.0	17.9	%
ALY%	0.0	2.0	%	PCT	0.108	0.282	%
LIC#	0.00	0.20	10 ³ /uL	P-LCR	11.0	45.0	%
LIC%	0.0	2.5	%	P-LCC	30	90	10 ³ /uL
NRBC#	0.000	9999.999	10 ³ /uL				
NRBC%	0.00	9999.99	%				
cWBC	3.50	9.50	10 ³ /uL				
NLR	/	/					

Ref. Group

Lower Limit of Age

Gender

Upper Limit of Age

Save

Close

Figure 5-14 Copying a Ref. Group

You can edit the new reference group. Save and close the screen, and then the copied reference group will be shown in the reference group list.

Ref. Range

Ref. Group	Default	Lower Limit of Age	Upper Limit of Age	Gender
General	il			
Man		13 Years	999 Years	Male
Woman		13 Years	999 Years	Female
Child		28 Days	13 Years	
Neonatus		0 Hour	28 Days	
new group				

☒ Automatically match the customized reference group according to age and gender

Copy

New

Edit

Delete

Set as default

Close

Figure 5-15 New reference group in the list

NOTE

The reference group name entered is not allowed to be empty nor the same as the existing ones.

5.4.3.3 Adding a New Ref. Group

If the built-in reference groups cannot meet the actual demand, you can add new ones and manually enter the information such as reference ranges for each parameter, names and genders. The procedures are shown as below:

1. Click **New**, and a screen for adding a new reference group will pop up. See Figure 5-16.

New

Para.	Lower Limit	Upper Limit	Unit	Para.	Lower Limit	Upper Limit	Unit
WBC			10 ³ /uL	RBC			10 ⁶ /uL
Neu%			%	HGB			g/dL
Lym%			%	HCT			%
Mon%			%	MCV			fL
Eos%			%	MCH			pg
Bas%			%	MCHC			g/dL
Neu#			10 ³ /uL	RDW-CV			%
Lym#			10 ³ /uL	RDW-SD			fL
Mon#			10 ³ /uL	PLT			10 ³ /uL
Eos#			10 ³ /uL	MPV			fL
Bas#			10 ³ /uL	PDW-SD			fL
ALY#			10 ³ /uL	PDW-CV			%
ALY%			%	PCT			%
LIC#			10 ³ /uL	P-LCR			%
LIC%			%	P-LCC			10 ³ /uL
NRBC#			10 ³ /uL				
NRBC%			%				
cWBC			10 ³ /uL				
NLR							

Ref. Group

Lower Limit of Age

Year

▼

Gender

▼

Upper Limit of Age

Year

▼

Save

Close

Figure 5-16 Adding a New Ref. Group

2. Complete the entries for each parameter with reference to the parameter description in Table 5-3.

Table 5-3 Description of Ref. Group parameters

Parameter	Meanings	Operation
Ref. Group	Name of the new reference group.	Click the edit box and enter the information using the soft keyboard. English characters and numbers are allowed to be entered, while special characters are not. NOTE The reference group name entered is not allowed to be empty nor the same as the existing ones.
Lower Limit of Age	Lower limit of age of the reference group.	Enter an integer value in the textbox and select the age unit (year, month, week, day, or hour) from the drop list on the right. NOTE The Lower Limit of Age must be smaller than the Upper Limit of Age .
Upper Limit of Age	Upper limit of age of the reference group.	Enter an integer value in the textbox and select the age unit (year, month, week, day, or hour) from the drop list on the right. NOTE The Upper Limit of Age must be greater than the Lower Limit of Age .
Gender	Gender of the reference group.	Select Man, Woman, Not defined from the dropdown list. The default setting is empty.
Lower Limit (of parameter)	Lower limit of parameters of the reference group. If the test result is lower than this value, it would be regarded as clinically abnormal.	Click the Lower Limit cell which corresponds to the parameter and enter a new value. NOTE The Lower Limit must be smaller than the Upper Limit .
Upper Limit (of parameter)	Upper limit of parameters of the reference group. If the test result is higher than this value, it would be regarded as clinically abnormal.	Click the Lower Limit cell which corresponds to the parameter and enter a new value. NOTE The Upper Limit must be greater than the Lower Limit .

3. Click **Save** to save the settings.
4. Click **Close** to exit the interface.

5.4.3.4 Editing a Ref. Group

You can modify the reference range of the parameters according to actual needs and set suitable reference intervals (age range, gender, etc.).

The procedures are shown as below:

1. Select the reference group to be set, and click **Edit** to enter the interface as shown in Figure 5-17.

Edit

Para.	Lower Limit	Upper Limit	Unit	Para.	Lower Limit	Upper Limit	Unit
WBC	3.50	9.50	10 ³ /uL	RBC	3.80	5.80	10 ⁶ /uL
Neu%	40.0	75.0	%	HGB	11.5	17.5	g/dL
Lym%	20.0	50.0	%	HCT	35.0	50.0	%
Mon%	3.0	10.0	%	MCV	82.0	100.0	fL
Eos%	0.4	8.0	%	MCH	27.0	34.0	pg
Bas%	0.0	1.0	%	MCHC	31.6	35.4	g/dL
Neu#	1.80	6.30	10 ³ /uL	RDW-CV	11.0	16.0	%
Lym#	1.10	3.20	10 ³ /uL	RDW-SD	35.0	56.0	fL
Mon#	0.10	0.60	10 ³ /uL	PLT	125	350	10 ³ /uL
Eos#	0.02	0.52	10 ³ /uL	MPV	6.5	12.0	fL
Bas#	0.00	0.06	10 ³ /uL	PDW-SD	9.0	17.0	fL
ALY#	0.00	0.20	10 ³ /uL	PDW-CV	10.0	17.9	%
ALY%	0.0	2.0	%	PCT	0.108	0.282	%
LIC#	0.00	0.20	10 ³ /uL	P-LCR	11.0	45.0	%
LIC%	0.0	2.5	%	P-LCC	30	90	10 ³ /uL
NRBC#	0.000	9999.999	10 ³ /uL				
NRBC%	0.00	9999.99	%				
cWBC	3.50	9.50	10 ³ /uL				
NLR	/	/					

Ref. Group

General

Gender

▼

Lower Limit of Age

▼

Upper Limit of Age

▼

Default

Save

Close

Figure 5-17 Editing a Ref. Group

- Refer to Table 5-3 for the description of the parameters to finish the editing.

For the built-in reference group, you can modify the upper limit and lower limit of the parameters, but not its name, the upper limit and lower limit of age as well as gender.

NOTE Click Set as default to restore the setting of the selected reference group to the default value.

Non-built-in reference group (which is added by user) cannot restore defaults.

- Click **Save** to save the modification.
- Click **Close** to exit.

5.4.3.5 Deleting a Ref. Group

Click **Delete**, and select **Yes** in the pop-up dialog box to delete the selected customized reference group.

NOTE Built-in reference group cannot be deleted.

5.4.3.6 Setting Default Ref. Group

When you pre-enter patient information in the **Sample Analysis** interface, the **Ref. Group** displayed by default is the default reference group.

The default setting is **General**. You can change it as required. Select a reference group and click **Set as default** to set the selected reference group as the default reference group.

As shown in Figure 5-18, the reference group with a check mark in its **Default** column is a default reference group.

Ref. Range				
Ref. Group	Default	Lower Limit of Age	Upper Limit of Age	Gender
General				
Man		13 Years	999 Years	Male
Woman		13 Years	999 Years	Female
Child	☒	28 Days	13 Years	
new group				

Figure 5-18 Setting Default Ref. Group

5.4.3.7 Automatically Match the Customized Reference Group According to Age and Gender

If **Automatically match the customized reference group according to age and gender** is checked, the customized reference group will be automatically assigned patients by the system according to their age and gender when the patient information is entered. If it fails to find a matching customized reference group for a patient, the patient will be assigned to the built-in reference group. When the system automatically matches the reference group according to age and gender, the rules listed in Table 5-4 shall be followed.

Table 5-4 Rules for Matching the Reference Group

Automatically match the customized reference group according to age and gender	Customized Ref. Group	Match the reference group
Unchecked	N/A	Built-in reference group
Checked	None	Built-in reference group
Checked	Created	Preferentially match the customized reference group

NOTE

When the customized ref. groups are used to match the reference group, the matching will be performed from top down according to the customized ref. groups displayed in the screen.

5.4.4 Microscopic Exam. Settings

You can perform the microscopic exam. settings, including adding, editing, deleting and adjusting the list order as per the actual demand.

NOTE

The operations of adding, editing, deleting and adjusting the list order do not affect the sample record in which the microscopic examination results have been entered and saved. Such operations are only valid for the record in which the microscopic examination results have not been saved, and the samples analysed after the setting operations.

5.4.4.1 Accessing the Interface

Click **Microscopic Exam.** in the **Para.** selection to access the microscopic examination setting interface. See Figure 5-19.

Microscopic Exam. Settings		
No.	Parameter Name	Code System
1	Neutrophilic segmented granuloc...	
2	Neutrophilic band granulocyte	
3	Lymphocyte	
4	Monocyte	
5	Eosinophil	
6	Basophil	
7	Plasmacyte	
8	Atypical Lymph	
9	Blast	
New Edit Delete Cancel		

Figure 5-19 Microscopic Exam. Settings

5.4.4.2 Adding a New Microscopic Exam. Parameter

Do as follows to add a new microscopic examination parameter.

1. Click New in the Microscopic Exam. Settings interface.

A dialog box will pop up as shown in Figure 5-20.

New

Parameter Name

Code System

OK

Cancel

Note: coding system is the code ID in LIS transmission.
You may not input the value if it is not needed.

Figure 5-20 Adding a New Microscopic Exam. Parameter

2. Input the parameter name and its coding system in the corresponding textboxes.
 - The **Parameter Name** cannot be empty and up to 32 characters can be entered.
 - The **Code System** is the code ID of the parameter. It is used for LIS transmission only when the parameter is transmitted to the LIS. You may not input the value if it is not needed. Up to 20 characters can be entered.
3. Click **OK**.
The name of the new parameter will be displayed in the microscopic exam. parameter list.

5.4.4.3 Editing a Microscopic Exam. Parameter

Select a parameter name from the list and click **Edit** to modify it. See Figure 5-21.

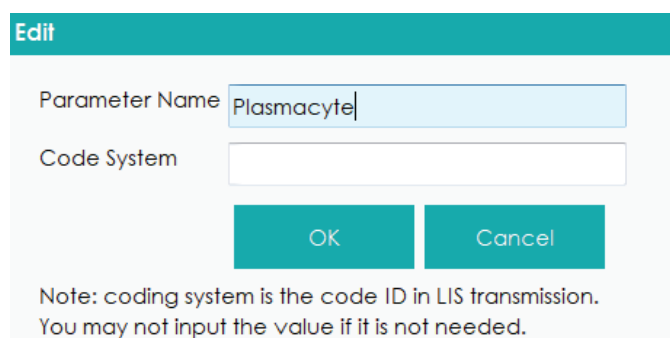


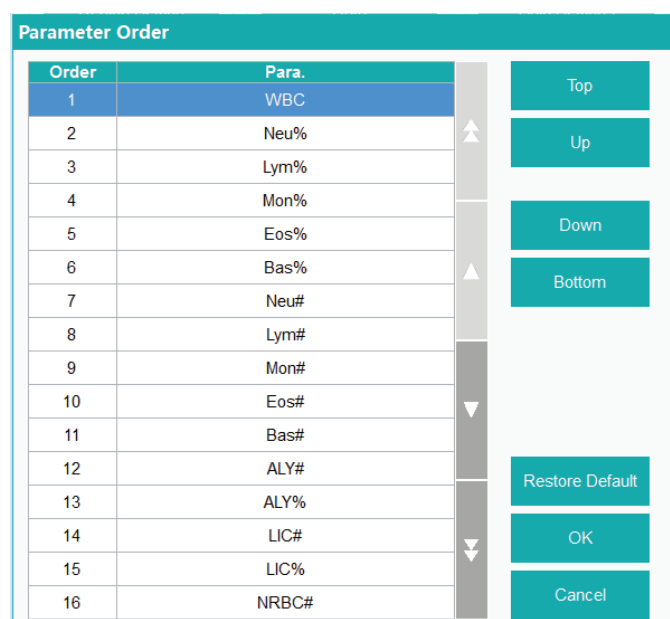
Figure 5-21 Editing a Microscopic Exam. Parameter

5.4.4.4 Deleting a Microscopic Exam. Parameter

Select a parameter name from the list, click the **Delete** button and then click **Yes** in the popup dialog box to delete this parameter.

5.4.4.5 Adjusting Parameter Order

Click **Adjust Order** to move the selected parameter to the top/bottom or move it upwards/downwards on the popup screen. Click **Restore Default**, and the parameter order will be restored to the default order. See Figure 5-22.



Order	Para.
1	WBC
2	Neu%
3	Lym%
4	Mon%
5	Eos%
6	Bas%
7	Neu#
8	Lym#
9	Mon#
10	Eos#
11	Bas#
12	ALY#
13	ALY%
14	LIC#
15	LIC%
16	NRBC#

Figure 5-22 Parameter Order

5.4.5 Research Use Only (RUO) Parameters

Click **RUO Parameters** in the **Setup > Parameter** interface to enter the **RUO Parameters** setting interface. See Figure 5-23.

RUO Parameters

RUO Parameters

☒ ALY%, LIC%, NRBC%, ALY#, LIC#, NRBC# ☒ cWBC

☒ NLR

Para.

☒ Display RUO parameters ☒ Display "*" mark

☒ Print RUO parameters ☒ Print "*" mark

☒ Display declaration ☒ Print declaration

Declaration

"*" means "Research use only, not for diagnostic use".

Apply OK Cancel

Figure 5-23 Setting RUO Parameters

The RUOs include ALY%, LIC%, NRBC% , ALY#, LIC#, NRBC#, cWBC# and NLR.

NOTE The RUO parameters are for research use only, not for diagnostic use.

Display RUO Parameters

It's checked by default, which means the information regarding the RUO parameters will be displayed in the counting results. If it's unchecked, the RUO parameters, the * mark and the declaration will not be displayed in the counting results.

Display "*" mark: It's checked by default, which means the "*" mark will be displayed in the counting results; If it's unchecked, the "*" mark and the declaration will not be displayed.

Display declaration: It's checked by default, which means the declaration will be displayed in the counting results; if it's unchecked, the declaration will not be displayed.

Print RUO parameters

It's checked by default, which means the RUO parameters will be printed in the report. If it's unchecked, the RUO parameters, the "*" mark and the declaration will not be printed in the report.

Print "*" mark: It's checked by default, which means the "*" mark will be printed in the report. If it's unchecked, the "*" mark and the declaration will not be printed in the report.

Print declaration: It's checked by default, which means the declaration will be printed in the report. If it's unchecked, the declaration will not be printed in the report.

Editing Declaration

The default declaration is: "*" means "research use only, not for diagnostic use". You can modify the declaration in the textbox as per the actual demand. Up to 50 characters can be entered, including all characters, numbers, letters and other special characters (except "/" and "\") on the keyboard.

NOTE

Any change made to the display settings or printing of the RUO parameters, the "*" mark and the declaration will be applied to all the RUO parameters (before and after the change).

5.4.6 Customized Parameters

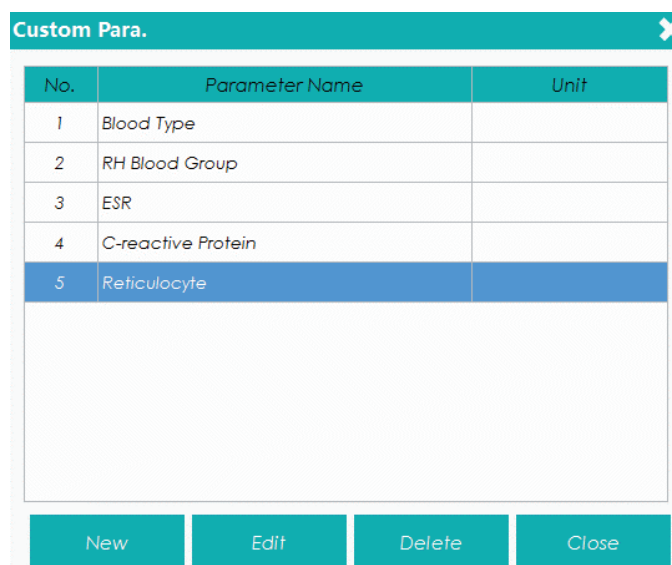
Except for this analyser's analysis parameters, parameters collected from other testing instruments or via manual testing by the user are customized parameters. You can set customized parameters so they can be printed together with this analyser's analysis parameter details on the Haematology Analysis Report.

This analyser's default customized parameters include: **Blood Type**, **RH Blood Group**, **ESR**, **C-reactive Protein** and **Reticulocyte**. You can set the unit and reference range of default customized parameters as well as add and set customized parameters.

5.4.6.1 Accessing the Interface

Click **Custom Para.** in the **Para.** selection.

The customized parameters setting interface as shown in Figure 5-24 will pop up on the screen.



The screenshot shows a window titled "Custom Para." with a close button (X) in the top right corner. Inside the window is a table with three columns: "No.", "Parameter Name", and "Unit". The table contains five rows of default parameters. The fifth row, "Reticulocyte", is highlighted in blue. Below the table is a large empty rectangular area for adding new parameters. At the bottom of the window are four buttons: "New", "Edit", "Delete", and "Close".

No.	Parameter Name	Unit
1	Blood Type	
2	RH Blood Group	
3	ESR	
4	C-reactive Protein	
5	Reticulocyte	

New Edit Delete Close

Figure 5-24 Customized Parameter Settings

5.4.6.2 Adding a Customized Parameter

1. Click New. The interface as shown in Figure 5-25 will pop up on the screen.

Ref. Group	Lower Limit	Upper Limit	Parameter Name
General			
Man			Unit
Woman			
Child			
Neonatus			

Apply

OK

Cancel

Figure 5-25 Adding a Customized Parameter

- Click the textboxes of **Parameter Name** and **Unit** respectively, and enter the name and unit of the customized parameter.
- Click corresponding cells of the **Upper Limit** and **Lower Limit** of the reference group, and input values.
- Click **OK**.

The added parameter will be displayed in the customized parameter list.

5.4.6.3 Editing a Customized Parameter

You can set the unit and reference range of customized parameters. Detailed steps are shown below:

- Select the customized parameter to be edited, and click **Edit**.
The interface as shown in Figure 5-26 will pop up on the screen.

Ref. Group	Lower Limit	Upper Limit	Parameter Name
General			Reticulocyte
Man			Unit
Woman			
Child			
Neonatus			

Apply

OK

Cancel

Figure 5-26 Editing a Customized Parameter

- 2. Click the textboxes of **Parameter Name** and **Unit** respectively, and modify the name and unit of the customized parameter.
- 3. Click corresponding cells of the **Upper Limit** and **Lower Limit** of the reference group, and modify the values.
- 4. You can also customize the reference group according to the actual situation. For details, see **5.4.3 Ref. Range**.
- 5. Click **Save**.

5.4.6.4 Deleting a Customized Parameter

Select a customized parameter, and click on **Delete**. Then, the parameter and its corresponding reference group will be deleted.

5.5. Meterage Settings

5.5.1 Gain Settings

You can adjust each digital pot at the **Gain Settings** interface. It is not recommended to adjust gains frequently.

Click **Gain Settings** in the **Meterage** selection to access the gain setting interface. Figure 5-27.

Gain Settings

Item	Current Value	Adjustment Rate
WBC		100%
RBC		100%
DIFF-LS		100%
DIFF-MS		100%
DIFF-HS		100%
BASO-LS		100%
BASO-MS		100%
BASO-HS		100%

HGB Current Value:

p

HGB Blank Voltage:

Apply

OK

Cancel

Figure 5-27 Gain Settings

NOTE New value of the gain adjustment = **Current Value** × **Adjustment Rate**.

- **Setting the WBC gain**

The WBC gain here is in whole blood mode.

Setting method I: click the **Current Value** of the WBC and enter the new value.

Setting method II: click the **Adjustment Rate** cell of the WBC and enter the adjustment rate of the new value relative to the current value.

- **Setting the WBC gain**

RBC channel gain.

Setting method I: click the **Current Value** of the RBC and enter the new value.

Setting method II: click the **Adjustment Rate** cell of the RBC and enter the adjustment rate of the new value relative to the current value.

- **DIFF-LS, DIFF-HS, DIFF-MS**

DIFF channel gain.

Setting method I: click the **Current Value** of the parameter and enter the new value.

Setting method II: click the **Adjustment Rate** cell of the parameter and enter the adjustment rate of the new value relative to the current value.

- **BASO-LS, BASO-HS, BASO-MS**

BASO channel gain.

Setting method I: click the **Current Value** of the parameter and enter the new value.

Setting method II: click the **Adjustment Rate** cell of the parameter and enter the adjustment rate of the new value relative to the current value.

- **Setting the HGB gain**

Current digital circuit gain. The purpose for adjusting the HGB channel gain is to change the HGB background voltage.

You can enter the value directly in the **HGB Current Value** textbox or click the adjusting button to adjust the HGB gain.

- **Setting the HGB Blank Voltage**

The background voltage derived from HGB gain cannot be modified. HGB Background Voltage can be adjusted within the specified range (4.2~4.8V) by modifying **HGB Current Value**.

5.5.2 Flag

When the test result meets the requirement of the flag rules, the corresponding flag will be displayed on the screen. You can edit the flag rules as per the actual demand and relevant lab procedures.

5.5.2.1 Accessing the Interface

Click **Flag** in the **Meterage** selection to access the flag rules setting interface. See Figure 5-28.

Flag

Flag	Flag Rules
Leucopenia	WBC < 2.50 (10 ³ /uL)
Leucocytosis	WBC > 18.00 (10 ³ /uL)
Lymphopenia	Lym# < 0.80 (10 ³ /uL)
Lymphocytosis	Lym# > 4.00 (10 ³ /uL)
Neutropenia	Neu# < 1.00 (10 ³ /uL)
Neutrophilia	Neu# > 11.00 (10 ³ /uL)
Monocytosis	Mon# > 1.50 (10 ³ /uL)
Eosinophilia	Eos# > 0.70 (10 ³ /uL)
Basophilia	Bas# > 0.20 (10 ³ /uL)
Erythrocytosis	RBC > 6.50 (10 ⁶ /uL)
Anisocytosis	RDW-CV > 22.0 (%) and RDW-SD > 64.0 (fL)
Macrocytosis	MCV > 113.0 (fL)
Microcytosis	MCV < 70.0 (fL)
Anemia	HGB < 9.0 (g/dL)
Hypochromia	MCHC < 29.0 (g/dL)
Thrombocytosis	PLT > 600 (10 ³ /uL)
Thrombopenia	PLT < 60 (10 ³ /uL)

Edit

Set as default

Close

Figure 5-28 Flag

5.5.2.2 Setting Flag Rules

You can select the name of the **Flag** in the **Flag** interface, then click **Edit** to modify the rules in the popup dialog box. See Figure 5-29.

Edit

Leucopenia

WBC <

2.50

10³/uL

OK

Close

Figure 5-29 Setting Flag Rules

5.5.2.3 Restoring Defaults

Click **Set as default** to restore the parameter to the default value.

5.6 Communication Settings

5.6.1 Host Network Settings

On the host communication screen, you can set the network information of the analyser to enable its network connection.

Click **Host Communication** in the **Communicate** selection to access the host network setting interface. See Figure 5-30.

Host Communication

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically
☒ Use the following address:

IP Address: . . .
 Subnet mask: . . .
 Default gateway: . . .

☐ Obtain DNS server address automatically
☒ Use the following DNS server addresses:

Preferred DNS server: . . .
 Alternate DNS server: . . .

Details Apply OK Cancel

Figure 5-30 Host Network Settings

Refer to Table 5-5 for the description of relevant parameters.

Table 5-5 Description of Host Communication Setting Parameters

Parameter	It means	Operation
Obtain an IP address automatically	The host gets the IP address dynamically from a DHCP server or a PPP dial-up network access server. This option is not applicable for the dial-up connection of SLIP server.	Please choose according to the actual situation.

Use the following address:	Specify the host to use the manually set IP address. If this option is selected, you need to set: • IP address The IP address obtained from the network administrator or Internet service provider. • Subnet mask The subnet mask obtained from the network administrator or Internet service provider. • Default gateway The IP address of the default gateway; the router's IP address for connecting the independent IP network segment.	Obtain the IP address, subnet mask and default gateway of the host from the network administrator or Internet service provider.
Obtain DNS server address automatically	Automatically obtain the IP address of the Domain Name Server (DNS).	Please choose according to the actual situation.
Use the following DNS server addresses:	Specify the IP address of the DNS server of the host. • Preferred DNS server The IP address of preferred or primary DNS servers. • Alternate DNS server (Optional) The IP address of alternative or secondary DNS servers of the host. This server will be used if the specified IP address of the Preferred DNS server is not available or if the DNS name cannot be resolved as the IP address of the DNS server which the host has inquired.	Obtain the IP address of DNS server from the network administrator or Internet service provider.

NOTE You can click **Details** to check the network information of the analyser, including physical address, IP address, subnet mask, default gateway, DNS server, etc.

5.6.2 LIS Communication

In the LIS Communication interface, You can set the communication between the system and the LIS, including network settings, protocol settings and transmission mode.

Click **LIS Communication** in the **Communication** selection to access the Laboratory Information System (LIS) communication setting interface. See Figure 5-31.

The screenshot shows the 'LIS Communication' settings window. It is organized into several sections:

- Network Settings:** Includes an 'IP Address' field with a dotted pattern, a 'Port' field set to '5600', and a 'Reconnect' button.
- Transmission Settings:** Contains checkboxes for 'Auto-communication' and 'Transmit after result modified'. There is also a checkbox for 'Bidirectional LIS/HIS Communication'. Below this is a 'Bidirectional LIS/HIS Communication Timeout' field set to '10 Sec.' and a 'Matched by' dropdown menu currently showing 'Sample ID'.
- Protocol Settings:** Includes a checkbox for 'Communication Acknowledgement' and an 'ACK timeout' field set to '10 Sec.'.
- Graph Format:** A dropdown menu set to 'PNG'.
- Histogram Transmission Method:** A dropdown menu set to 'Not transmit'.
- Scattergram Transmission Method:** A dropdown menu set to 'Not transmit'.
- DIFF Scattergram:** Three checkboxes labeled 'LS-MS', 'LS-HS', and 'HS-MS', all of which are currently unchecked.
- BASO Scattergram:** One checkbox labeled 'LS-MS', which is currently unchecked.

At the bottom of the window are three buttons: 'Apply', 'OK', and 'Cancel'.

Figure 5-31 Setting LIS Communication

Refer to Table 5-6 for the description of relevant parameters.

Table 5-6 Description of LIS Communication Setting Parameters

Parameter	It means		Operation
Network Settings	IP address	The IP Address of the LIS.	Please set it according to the actual situation.
	Port	The port of the LIS. The default value is 5600.	Please set it according to the actual situation. An integer between 1025 and 65535 can be entered. NOTE If the analyser is disconnected with the LIS, click the Reconnect button to connect the LIS again.

Parameter		It means	Operation
Transmission Mode	Auto-communication	Whether to upload the sample results automatically. - If checked, the system will automatically upload the result to the LIS upon the completion of the analysis. - If unchecked, the result of analysis will not be automatically uploaded. NOTE If the Bidirectional LIS/HIS Communication is checked, this parameter will be checked automatically.	Please choose according to the actual situation.
	Bidirectional LIS/HIS Communication	Whether to enable the bidirectional communication between the software and the LIS/HIS. - If checked, the system will automatically obtain the sample/patient information from LIS/HIS after the sample analysis is started or the patient information is edited, and automatically upload the result to the LIS upon the completion of the analysis. NOTE If the information is matched by sample ID, you only need to enter the sample ID; if the information is matched by Med Rec.No., you only need to enter the medical record number. - If unchecked, the software system will not obtain the sample/patient information, and decide whether to upload result based on the setting of the Auto-communication parameter.	Please choose according to the actual situation.
	Matched by	The matching method of the analyser with the LIS/HIS sample information. - Sample ID The sample information of LIS/HIS is matched by the sample ID when running samples. - Med Rec. No. The sample information of LIS/HIS is matched by the Med Rec.No. when running samples. If the Med Rec.No. is empty, the sample information will not be matched. NOTE The parameter is only valid when the Bidirectional LIS/HIS Communication is checked.	Please choose according to the actual situation.

Parameter		It means	Operation
Transmission Mode	Transmit after result modified	Whether to upload the sample results automatically after the sample results is edited. - If checked, the sample results will be uploaded automatically after the sample results is edited. - If unchecked, the system will not execute any operations.	Please choose according to the actual situation.
Protocol Settings	Communication Acknowledgement	If checked, the communication between the system and the LIS is successful when the ACK response from the LIS is received within the duration of ACK timeout; no response received indicates communication failure. If unchecked, the communication between the system and the LIS shall be considered successful no matter the ACK response from the LIS is received or not. NOTE The system will send the next message continuously no matter the communication is successful or not.	Please choose according to the actual situation.
	ACK timeout	Timeout duration of the ACK response. The default value is 10 seconds, that is, the communication will be considered failed if the system receives no ACK response within 10 seconds.	Click ↑ or ↓ or directly enter in the textbox. An integer between 1 and 120 can be entered. Unit: Second (sec.) NOTE The parameter is valid only when the Communication Acknowledgement is checked.
Graph Format		Graph transmission format, including PNG and BMP.	Please choose according to the actual situation.
Histogram Transmission Method		The methods for transmitting the histogram to the LIS when the result is transmitted by the system, including: - Not transmit Do not transmit the histogram to the LIS. - Bitmap Transmit the histogram to the LIS in the format of screen display. - Transmitting bitmap for printing The histogram is transmitted by the system to the LIS in the format of a printed report.	Please choose according to the actual situation.

Parameter	It means	Operation
Scattergram Transmission Method	The methods for transmitting the scattergram to the LIS when the result is transmitted by the system, including: • Not transmit Do not transmit the scattergram to the LIS. • Bitmap Transmit the scattergram to the LIS in the format of screen display. • Transmitting bitmap for printing The scattergram is transmitted by the system to the LIS in the format of a printed report.	Please choose according to the actual situation.
DIFF Scattergram	The DIFF scattergrams transmitted to the LIS, including LS-MS, LS-HS and HS-MS.	Please choose according to the actual situation. NOTE The parameter is invalid when Not transmit is set as the scattergram transmission method.
BASO Scattergram	The BASO scattergram transmitted to the LIS, namely LS-MS.	Please choose according to the actual situation. NOTE The parameter is invalid when Not transmit is set as the scattergram transmission method.

5.7 User management

After logging in the system, the administrator has the access to set the account information of general users and other administrators; common users can only browse the user list and change their own passwords.

5.7.1 Accessing the Interface

Click **User** in the **Setup** interface to access the user management interface as shown in Figure 5-32.

User Name	Name	User Group	Default User	Remarks
admin	admin	Administrator		

New

Edit

Delete

Set as default user

Reset Password

Change Password

Close

Figure 5-32 User management

5.7.2 Creating a User

Click **New** to set the account information of a new user in the popup interface, including username, first and last name, password, user group and remarks, etc. See Figure 5-33.

New

User Name

(Login Account)

Name

Password

Confirm Password

User Group

Common User

▼

Remarks

OK

Cancel

Figure 5-33 Creating a user

NOTE **User Group** includes **Common User** and **Administrator**. Users are assigned different access levels according to the user group they belong to.

Click **OK** after the setting is complete. The information of the new user will be shown in the user list.

5.7.3 Editing a User

Select the user to be edited and click **Edit** to modify the name and user group. See Figure 5-34.

Edit

User Name

admin

Name

admin

User Group

Administrator

Remarks

OK

Cancel

Figure 5-34 Editing a User

5.7.4 Deleting a User

Select the user to be deleted and click **Delete**, and then select **OK** in the pop-up dialog box to delete the user.

NOTE

The administrator cannot delete his/her own information.

5.7.5 Setting the Default User

Select a user and click **Set as default user** to set this user as the default user. After it is set successfully, the default user name will be displayed in the login box next time and you only needs to enter the corresponding password. See Figure 5-35.

Erba[®]

Mannheim

H 560

User Name:

PassWord:

LOG IN

Figure 5-35 Login after Setting the Default User

5.7.6 Changing Password

Click **Change Password**, enter the old password and new password of the user and confirm the new password in the popup dialog box, then click **OK**.

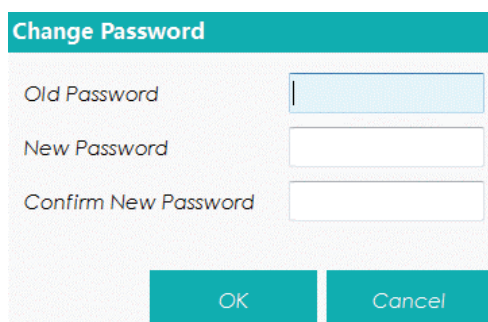
A screenshot of a 'Change Password' dialog box. The dialog has a teal header bar with the title 'Change Password'. Below the header, there are three text input fields labeled 'Old Password', 'New Password', and 'Confirm New Password'. The 'Old Password' field is currently active, showing a cursor. At the bottom of the dialog, there are two buttons: 'OK' and 'Cancel', both with teal backgrounds and white text.

Figure 5-36 Changing Password

NOTE You can only change his/her own password and cannot change the password of other users.

5.7.7 Resetting Password

If the user forgets the password or the password is required to be reset due to other reasons, please click **Reset Password** to reset the password of the selected user to the initial password. The reset password is the same as the user name.

NOTE The administrator is allowed to reset the password of all administrators and general users; general users do not have the access to reset the password.

5.8 Print Settings

Click **Print Settings** in the **Setup** interface for relevant print settings, including the default printer, template, report, copies and margins, etc. See Figure 5-37.

Figure 5-37 Print Settings

5.8.1 Printer Settings

You can set the printer and driver of the system in the Printer selection.

- **Printer Driver**
The system automatically detects the printer driver by default.
- **Printer**
Select a printer to be used from the dropdown list. If the dropdown list is blank, it indicates that no printer has been installed for the operating system. In this case, install a printer, and then perform the relevant settings and printing operations.
- **Printer Resolution**
Select a proper resolution from the dropdown list. The higher the resolution of the printer, the better the print quality.

5.8.2 Report Settings

You can set relevant parameters of the report in the **Report Settings** combo box.

- **Report Title**
Enter the title of the report in the **Report Title** textbox. The default setting is **Haematology Analysis Report**.
- **Copies**
You can enter the number of copies to be printed for a report in the **Copies** textbox according to the actual demand. Click to increase the number of copies and click to decrease the number of copies or enter the number of copies in the edit box directly. Range of the copies is between 1 and 100 and the default value is 1.

5.8.3 Format Settings

Report type and template of prints can be set in the **Format Settings** combo box. See Figure 5-38.

Format Settings	
Report Type	Report ▼
Paper Type	A4 ▼
Template	A4-Portrc ▼
Paper size	210*297 mm
Refresh	Import
Delete	Export

Figure 5-38 Format Settings

- **Selecting Report Type**
Select the format type to be set from the dropdown list of the **Report Type**. The default setting is **Report**.
- **Selecting Paper Type**
Select the paper type (size) from the dropdown list of the **Paper Type**, such as A4. After the selection is completed, the corresponding paper size will be shown at the bottom of the list, such as **210*148 mm**.
- **Selecting Template**
Select the template to be set from the dropdown list of the **Template**.
- **Refresh**
Click **Refresh** to refresh the format list after the customization by the administrator.
- **Importing/Exporting template**
You can export the existing template to a USB flash disk, and edit the template. After editing, import the template to the system to complete the customization of the template.

NOTE

Before importing/exporting template, insert a USB flash disk in the USB interface on the analyser.

- **Exporting template**
Select the template to be exported from the dropdown list of **Template** and click **Export**. Select the export path in the popup dialog box, and click **Save**.
- **Importing template**
Click **Import** and select the required template in the pop-up dialog box, then click **Open**.
- **Deleting template**
Select the template to be deleted from the dropdown list of the **Template**.

NOTE

Only customized templates can be deleted, the built-in templates cannot be deleted.

5.8.4 Auto Settings

- **Autoprint**

The default setting is **Off**, which means the report should be printed manually after the results are obtained.

If it is set to **On**, the system will automatically print the report of the sample as per the current report template once the counting results are obtained.

NOTE	If Print after validation is checked, the autoprint function becomes invalid. Auto print is not applicable for the background results.
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- **Auto print after validation**

It's unchecked by default, which means the system can print the report automatically without validation.

If it's checked, the report will be printed automatically after it's been validated instead of being printed right after the results are obtained each time.

NOTE	The parameter is valid only when the Autoprint is set to On .
-------------	---

- **Auto validate when printing**

It's unchecked by default, which means the report will not be automatically validated by the system at the time of printing.

If it's checked, the report will be automatically validated and printed by the system at the time of printing.

- **Print after validation**

It's unchecked by default, which means the report can be printed without validation.

If it's checked, the report can be printed only after validation and autoprint is inexecutable.

5.8.5 Printing Options

- **Print Flag**

It's checked by default, which means the flag information will be printed in the report. If it's not checked, it will not be printed.

- **Print Ref. Range**

It's checked by default, which means the reference range of the parameter will be shown in the printed report; If it's unchecked, the results alone, rather than reference range, will be shown in the printed report and the reference range will not.

- **Print Suspicious Flag**

It's unchecked by default, which means the suspicious flag "?" will not be shown in the printed report; if it's checked, such flag can be shown.

- **Print Ref. Range Flags**

It's checked by default, which means the printed report can show the ref. range flag (↑ or ↓); If it's unchecked, such a flag will not be shown.

- **Print Microscopic Exam. Para.**

It's checked by default, which means the result of **Microscopic Exam. Parameters** will be printed in the report. If it's not checked, it will not be printed.

- **Print result edited flags**

It's unchecked by default, which means the mark for the edited results will not be shown in the printed report.

If checked, the mark (**M** or **m**) for the edited results will be shown in the printed report if the parameters have been modified.

- **Two reports in one page (half of A4)**

It's unchecked by default. If this is checked, the default template size in Format Settings is half an A4 page (e.g., A4_Half-Portrait-Parameters), so two reports can be printed in one piece of A4 paper.

NOTE When **Autoprint** is **On**, a page remains to be printed with one report.

- **Update blank test time before be printed**

It's unchecked by default, which means the blank test time will not be processed by the system. If it's checked, the **Delivery Time** will be automatically updated as the **Run Time** by the system at the time of printing.

- **Print as black and white (Report)**

It's unchecked by default, which means the report will be printed according to the default settings of the printer.

If it's checked, the report will be printed as black and white.

NOTE The parameter is valid only when the **Report Type** is set to **Report**.

- **Printer Box Settings**

Figure 5-39 Printer Box Settings

IP Address

The default value is 10.0.0.200. You can input the address or select from the drop-down list. The IP address must be the same with the one on the software of the printer box.

Search

Search the IP address of the printer box in the current LAN. But you cannot search when the analyser is connected to the printer box with a network cable directly.

Port

The default port number is 35327. You can input if needed, and the input range is 1025-65535.

The port number must be the same with the one on the software of the printer box.

5.9 Auxiliary Settings

Click **Auxiliary Settings** in the Setup interface to access the Auxiliary Settings interface. See Figure 5-40.

Figure 5-40 Auxiliary Settings

The administrator is allowed to set the following functions in the **Auxiliary Settings** interface:

- Sample Numbering Rules
- Start-up sample IP and mode
- Predilute
- Other

5.9.1 Sample Numbering Rules

Set the sample ID entry rules.

- **Sample ID Entry Method**

Click the dropdown list of the **Sample ID Entry Method** and select the entry method of the sample ID from the following options.

1. Auto increment (default setting)
2. Manual entry

- **Prefix Length**

When **Auto Increment** is selected as the Sample ID entry method, you can add a prefix to a certain batch of samples for identification.

Enter the prefix length ranging from 0 to 24 (e.g. 2) of the sample ID in the **Prefix Length** textbox. The prefix length will applied to all sample IDs after the setting is saved.

5.9.2 Start-up sample ID and mode

Set the sample ID and measurement mode for the next sample after start-up.

- **Next Sample ID and mode after start-up**

The sample ID and mode set by the user will be used by the system after the next start-up when the specified sample ID is entered into the textbox and the measurement mode (CBC or CBC+DIFF) is selected from the dropdown list.

NOTE If the **Effective tomorrow** is checked, the modification of the next sample ID and mode after start-up will become effective on the next day.

- **Continue using the sample ID and mode before the last shutdown**

If checked, the system will by default add 1 to the last sample ID analysed before shutdown as the next sample ID after start-up.

5.9.3 Predilute

Set if you wish to see a popup dialog box when you perform the Predilute counting.

- Ask for confirmation (default setting): in the **Predilute** mode, when you press the aspirate key to start the analysis, a dialog box will pop up to remind you that the ongoing analysis is for **Predilute** counting.
- Do not ask for confirmation: the dialog box for confirming the Predilute counting will not pop up.

5.9.4 Quick Save

- **“Sample ID” in “Mode” interface**

Set whether the Quick Save function is enabled for the **Mode** interface.

If checked (default setting), when you enter the sample ID and press [Enter] (with the keyboard) or scan the sample ID (with barcode scanner) in the **Mode** interface, the sample information will be saved automatically.

If unchecked, the sample information should be saved manually.

- **“Med Rec. No.” in “Pre-entry” interface**

Set whether the Quick Save function is enabled for the **Pre-entry** interface.

If checked, when you enter the Med. Rec. No. in the **Pre-entry** interface and press [Enter] (with the keyboard), the sample information will be saved automatically.

If unchecked (default setting), the sample information should be saved manually.

5.9.5 Other

- **Show Result Edited Flags**

It's unchecked by default, which means the edited results are marked with an **M** at the end, while the corresponding results with manual modifications are marked with an **m** at the end. M or m is displayed between the result data and the parameter unit by default.

If unchecked, the edited result will not be marked with an **M** or **m**.

- **Automatically generate the delivery date**

It is checked by default, which means you don't need to manually enter the **Delivery Time** when you modify patient information after running a sample. The operating date will be displayed in the date textbox.

If unchecked, the **Delivery Time** shall be manually entered when patient information is modi-

fied in **Sample Analysis** interface.

- **Automatically generate the sampling date**

It is checked by default, which means you don't need to manually enter the **Sampling Time** when you modify patient information after running a sample. The operating date will be displayed in the date textbox.

If unchecked, the **Sampling Time** shall be manually entered when patient information is modified in **Sample Analysis** interface.

- **Suspicious Flag**

A single character (an English letter only) can be re-entered in the textbox as a suspicious flag. The default value is ?.

- **Ref. Range Flags**

You can select the **Ref. Range Flags** from the dropdown list. The default high flag is ↑ (or H) and the default low flag is ↓ (or L).

6 Daily Operations

6.1 Introduction

This chapter introduces the daily operations from the start-up to the shutdown of the analyser. A flow chart indicating the common daily operation process is presented below in Figure 6-1.

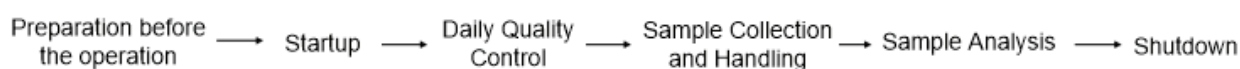


Figure 6-1 Daily Operations Procedure

6.2 Pre-operation Preparation



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.



WARNING

Be sure to dispose of reagents, waste, samples, consumables, etc. according to local legislations and regulations.

Reagents can be irritating to the eyes, skin and mucosa. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them in the laboratory.

If the reagent accidentally comes in contact with your skin, wash it off immediately with plenty of water and see a doctor if necessary. Do the same if you accidentally get any of the reagent in your eyes.

Keep your clothes, hairs and hands away from the moving parts to avoid injury.

The sample probe tip is sharp and may contain biohazardous materials. Exercise caution to avoid contact with the probe when working around it.

You should only use the Erba Lachema-specified reagents. Store and use the reagents as specified in instructions for use of the reagents.

Check if the reagents are connected correctly before using the analyser.

NOTE

After long-distance transportation, the reagent must be allowed to settle for more than one day before use.

Be sure to use clean K₂EDTA vacutainer blood collection tubes with anticoagulant, fused silica glass/plastic test tubes, centrifugal tubes and borosilicate glass capillary tubes.

Be sure to use the Erba Lachema-specified disposable products including vacutainer blood collection tube, vacutainer blood collection tubes with anticoagulant and capillary tubes etc.

Perform the following checks before turning on the analyser.

- Waste container

Check and make sure the waste container is empty.

- Fluidic tubing and power connections

Check and make sure the reagents and waste tubing are properly connected and not bent.

Check and make sure the power cord of the analyser is properly plugged into the power outlet.

- Printer (Optional)

Check and make sure enough paper is installed.

Check and make sure the power cord of the printer is properly plugged into power outlet, and the printer is properly connected to the peripheral computer.

- Network Cable (Optional)

Check and make sure the network cable is properly connected to the analyser.

6.3 Start-up

This section introduces the operations related to the start-up of the analyser.

NOTE

If you failed to start the analyser continuously, please contact Erba Lachema customer service department or your local agent immediately.

After start-up, please make sure the data/time displayed on the screen is correct.

1. Place the power switch at the back of the analyser in the [I] position.
The power indicator light will be on.
2. Check the indicator light on the analyser.
If the indicator light is on, it indicates the analyser has been started up. The analyser will perform self-test and initialization in sequence. The whole process will last for 4 to 10 minutes. (Time needed for initializing the fluidic systems depends on how the analyser was previously shut down.)
3. Enter the correct user name and password in the Login message box. See Figure 6-2.

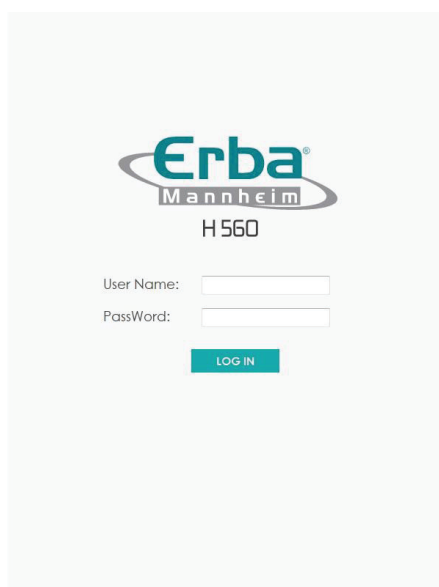


Figure 6-2 Login

The initial user name and password of administrator are **admin**, which was set by service engineer.

1 to 12 digits of numeric characters can be entered for the user name and the password. No Chinese character is allowed.

4. Click **Log in** to enter the user interface.

The system will display the **Sample Analysis** screen by default and display the test result of the background when the analyser is started.

The background test is designed for detecting particle interference and electrical interference.

For the background reference range of each parameter, please see **15.4.2 Normal Background**.

NOTE

The sample ID for the background test is **background**.

If the background results exceed the Ref. Range for the first time during fluidics initialization, then the analyser will run the background test one more time.

Running a test when there is a Background abnormal, you would obtain an unreliable testing result.

If any error is detected during initialization (e.g. the background results exceed the Ref. Range), the analyser will activate the alarm. For details, see **13 Troubleshooting**.

6.4 Daily Quality Control

To ensure reliable analysis results, conduct daily QC analysis on the analyser before running samples. For details, see **9 Quality Control**.

6.5 Sample Collection and Handling



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.



WARNING

Do not touch the patients' blood sample directly.



CAUTION

Do not re-use such disposable products as collection tubes, test tubes, capillary tubes, etc. Prepare the samples as per the procedures recommended by the reagent manufacturer.

NOTE

Be sure to use clean K₂EDTA vacutainer blood collection tubes with anticoagulant, fused silica glass/plastic test tubes, centrifugal tubes and borosilicate glass capillary tubes. Be sure to use the Erba Lachema-specified disposable products including vacutainer blood collection tube, vacutainer blood collection tubes with anticoagulant and capillary tubes etc.

For the cell blood count, store the whole blood at +15 to +25 °C and run them within 5 hours after collection.

6.5.1 Venous Whole Blood Samples

The procedure for preparing venous whole blood sample is as follows:

Use clean K₂EDTA (1.5 – 2.2mg/mL) vacutainer blood collection tubes with anticoagulant to collect venous blood samples.

Mix the venous blood with the anticoagulant well in the tube immediately.



CAUTION

For vacutainer blood collection tube (12 x 75 mm, cap excluded), please make sure the volume of the whole blood sample is not less than 0.5 mL.

6.5.2 Capillary Whole Blood Samples

Collect the capillary whole blood sample with a vacuum blood collection tube specified by the manufacturer.



CAUTION

To ensure the accuracy of the analysis, make sure the volume of the capillary whole blood sample is not less than 100 µL.

NOTE

Run the capillary whole blood sample within 3 minutes to 2 hours after its collection. The tube shall be placed vertically upward, not tilted or upside down. Otherwise, the inner wall of the tube may be stained with excessive sample, resulting in waste. Moreover, it may cause unevenly mixed sample and unreliable analysis results.

6.5.3 Prediluted Samples

**CAUTION**

Do not use anticoagulant during the prediluted sample analysis procedure; otherwise, the analysis result will be affected.

NOTE

Be sure to evaluate predilute stability based on your laboratory's sample population and sample collection techniques or methods.

The procedure for preparing prediluted sample is as follows:

1. Click the  on the top left corner and enter the menu screen as shown in Figure 6-3.

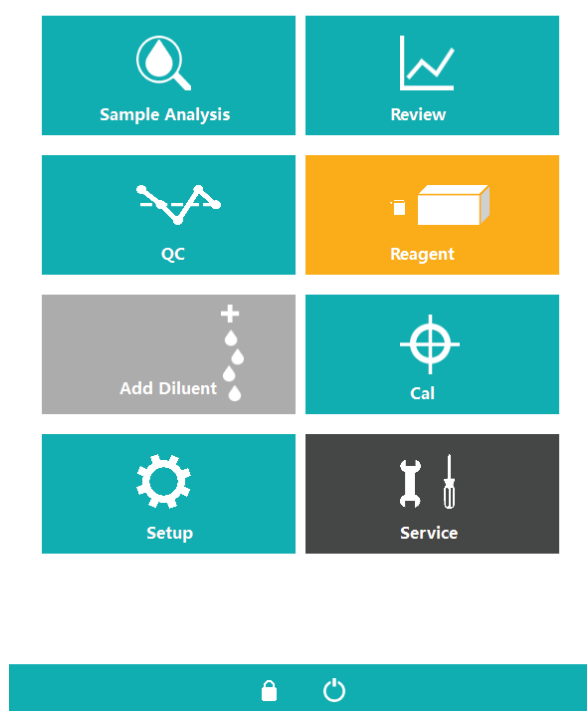


Figure 6-3 Menu Screen

2. Click the **Add Diluent** icon.
3. Take a clean centrifugal tube, uncap it and present it to the sample probe in a manner as shown in the following Figure 6-4 in which the probe tip is vertically in contact with the bottom of the tube so as to avoid bubbles, liquid attached to the inner wall or spatter.

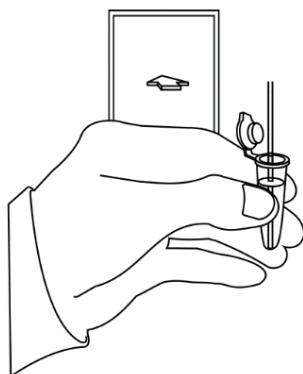


Figure 6-4 Sample under sample probe

4. Press the aspirate key and add the diluent (480 μ L at a time). After the diluent is added and you hear a beep, you can remove the centrifugal tube.

NOTE	You can also dispense 480 μ L of diluent by pipette into the tube.
-------------	--

5. If more portions of diluent are needed, repeat steps Take a clean centrifugal tube, uncap it and present it to the sample probe in a manner as shown in the following 4 in which the probe tip is vertically in contact with the bottom of the tube so as to avoid bubbles, liquid attached to the inner wall or spatter.~Press the aspirate key and add the diluent (480 μ L at a time)..
6. Add 20 μ L of blood to the diluent, close the tube cap and shake the tube to mix the sample.
7. After the prediluted sample is prepared, click **Cancel** to exit dispensing the diluent.

The prediluted sample prepared after single blood collection can be counted twice.
Be sure to keep dust from the prepared diluent.

Be sure to run the prediluted samples within 30 minutes after the mixing.

NOTE	Be sure to mix any sample that has been prepared for a while before running it. The centrifugal tube shall be placed vertically upward, not tilted or upside down. Otherwise, the inner wall of the tube would be stained with excessive sample, resulting in waste. Moreover, it may cause unevenly mixed sample and unreliable analysis results.
-------------	--

6.6 Sample Analysis

After the sample is prepared, you can perform the operations for sample analysis. For details, see **7 Sample Analysis**.

6.7 Shutdown



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.



The sample probe is sharp and potentially biohazardous. Exercise caution to avoid contact with the probe when working around it.



CAUTION

Do not turn on the analyser immediately after its shutdown. Wait at least 10 seconds before power-on to avoid damage to the machine.

To ensure stable analyser performance and accurate analysis results, be sure to perform the Shutdown procedure to shut down the analyser after it has been running continuously for 24 hours.

NOTE

When the analyser is running or performing other fluidics sequence, do not force shutdown the analyser.

If any error is detected during shutdown procedure, the analyser will return to the status before the shutdown procedure is performed, and then activate the alarm.

See **13 Troubleshooting**.

for details of removing the error.

Be sure to shut down the analyser in strict accordance with the instruction below.

Procedures for shutting down the analyser are as follows:

1. Click the  button on the menu screen. The interface pops up a dialog box.
2. Click **Yes**.
The system starts to execute the shutdown sequence and a message box pops up showing the procedures for cleanser maintenance.
3. Follow the instructions and set the cleanser under the sample probe, and press the aspirate key on the analyser or click **Aspirate** to run the cleanser aspiration.
Place the [O/I] switch at the back of the main unit in the [O] position.
4. After shutdown, empty the waste in the waste container, and dispose of it.



WARNING

Be sure to dispose of reagents, waste, samples, consumables, etc. according to local legislations and regulations.

7 Sample Analysis

7.1 Introduction

Sample analysis is the most important function of the H560 Haematology Analyser. You can get the blood cell count, HGB concentration and the 5-part classification counting results of the white blood cells by performing the sample analysis.

The summary of sample analysis procedures are as follows:

1. Entering the sample information.
2. Running the samples.
3. Processing the analysis results.

7.2 Interface Introduction

The **Sample Analysis** interface is the main interface of the analyser (Figure 7-1). You can complete the operations such as entering the sample information, performing sample analysis, reviewing/printing analysis results in the **Sample Analysis** interface.

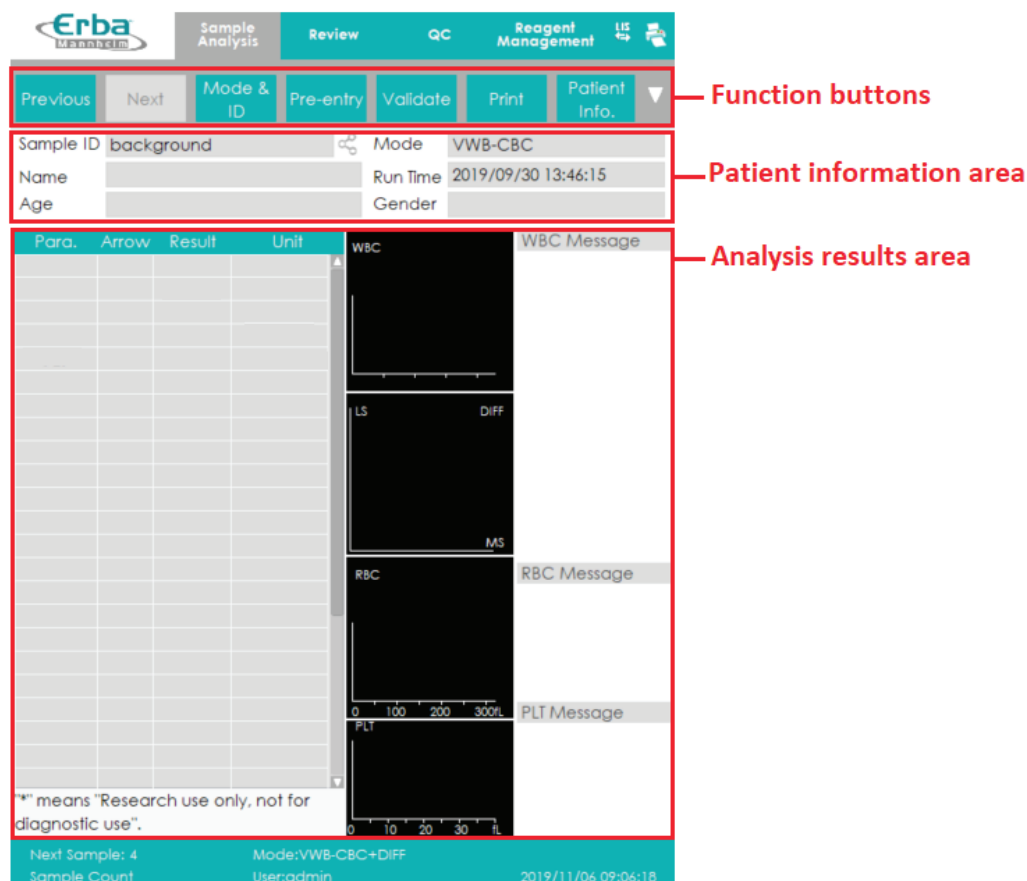


Figure 7-1 Sample analysis interface

Related descriptions:

- **Function buttons**
You can perform operations such as setting the mode for the samples, pre-entering information, reviewing previous/next records and printing. Click  and view all function buttons. See section **7.6 Functions of the Buttons**.
- **Patient information area**
It displays the patient information corresponding to the current sample.
- **Analysis results area**
It displays the analysis results of the sample, including the parameter results, Flags, DIFF scattergrams, BASO scattergram and histograms (including WBC, RBC and PLT). The system displays the analysis results of the most recent run by default.

Parameter Results

- This list displays the analysis results of all the parameters of the samples.
You can compare the values in the Result column with the corresponding Ref. Range. If the values are within the reference range, it means that they are normal. If not, it indicates that the sample may be abnormal and the corresponding symbols will be displayed in the Flag column.

WBC Message

- Displays the alert message regarding the WBC.

RBC Message

- Displays the alert message regarding the RBC.

PLT Message

- Displays the alert message regarding the platelet.

DIFF

- WBC DIFF scattergram in the CBC+DIFF mode. Click the scattergram, three WBC DIFF scattergrams including LS-MS, LS-HS and HS-MS and one BASO scattergram will be displayed.

WBC

- WBC distribution histogram.

RBC

- RBC distribution histogram.

PLT

- Platelet distribution histogram.

7.3 Entering Sample Information

You can enter the worklist information of the samples to be tested before the analysis.

NOTE

If the **Bidirectional LIS/HIS Communication** is checked and the sample information is Matched by Sample ID in the **Setup > Communicate > LIS Communication** interface, you don't need to pre-enter the sample information. The analyser automatically obtains the patient information from HIS/HIS by the sample ID entered in the **Mode** or **Worklist** interface. For details, see **5.6 Communication**.

If the **Bidirectional LIS/HIS Communication** is checked and the sample information is **Matched by** Med Rec.No. in the **Setup > Communicate > LIS Communication** interface, you only need to enter the **Med. Rec. No.**. The analyser obtains the other

information of sample from LIS/HIS.

You can also enter sample/patient information after the sample analysis is completed. For details, please refer to **Result 8 Result Review**.

Detailed steps are shown below:

1. Click the **Pre-entry** button in the function button area.

The interface as shown in Figure 7-2 will pop up on the screen.

Figure 7-2 Pre-entering Patient Information

2. Enter patient information with reference to the parameter description in Table 7-1.

Table 7-1 Parameter Description

Parameter	It means	Operation
First Name	First name of patient.	Input in the textbox directly.
Last Name	Last name of patient.	Input in the textbox directly.
Patient Type	Type of patient. Values: • (Null) • Inpatient • Physical Examination • STAT • Outpatient	Select from the dropdown list.

Parameter	It means	Operation
Sample Type	Type of sample for microscopic examination. Values: • Venous blood • Capillary • Cord blood • Blood	Click the Sample Type dropdown list box and select the type of sample for microscopic examination.
Med Rec. No.	Medical record number of patient.	Input in the textbox directly.
Gender	Gender of patient. Values: • (Null) • Male • Female • Not defined	Select from the dropdown list.
Birthday	Birthday of a patient.	Select from the date control. • The input sequence of the controls is the same with the date format on the top right corner of the dialog box. For example, if the data format is yyyy/MM/dd, you should input the data in the sequence of year, month, and date. • Click  or  to select the date or click the textbox to enter them directly. • Click  to clear the current data and re-enter the information.
Age	Age of a patient.	Select the unit of age from the dropdown list (Year, Month, Day or Hour) and enter the age of the patient in the textbox before the age unit. NOTE If the Birthday is set, the age will be displayed automatically.
Ref. Group	Reference group of the sample under analysis. The result is judged according to the reference range of the reference group and the result beyond the normal range will be flagged.	Select from the dropdown list. NOTE If the Automatically match the customized reference group according to age and gender is set, gender and age of a patient will automatically match the reference group according to the corresponding relationship (No matter the reference group is selected or not). Refer to 5.4.3 Ref. Range for the setting of the reference group and range.
Department	Department receiving the patient.	Select from the dropdown list.
Area	Ward area of patient.	Input in the textbox directly.

Parameter	It means	Operation
Bed No.	Bed No. of inpatient.	Select from the dropdown list or input directly. NOTE The bed No. is required to be filled only for inpatients.
Sampling Time	Date and time when the sample is collected.	Click the date control for the settings. - The input sequence of the controls is the same with the date format on the top right corner of the dialog box. For example, if the data format is yyyy/MM/dd HH:mm, you should input the data in the sequence of year, month, date, hour, and minute. - Click  or  to select the date or click the textbox to enter them directly. - Click  to clear the current data and re-enter the information. NOTE The system automatically displays the current time as sampling time. The sampling time can be no later than the current system time.
Submitter	Personnel submitting the sample.	Select from the dropdown list or input directly.
Delivery Time	Date and time when the sample is delivered.	Click the date control for the settings. - The input sequence of the controls is the same with the date format on the top right corner of the dialog box. For example, if the data format is yyyy/MM/dd HH:mm, you should input the data in the sequence of year, month, date, hour, and minute. - Click  or  select the date or click the textbox to enter them directly. - Click  to clear the current data and re-enter the information. NOTE The system automatically displays the current time as sample delivery time. The delivery time can be no later than the current system time and cannot be earlier than the sampling time.
Remarks	Clarifications or notes.	Input in the textbox directly.

3. Click **Apply** or **OK** to save the configuration.

7.4 Running Samples



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.



WARNING

The sample probe tip is sharp and may contain biohazardous materials. Exercise caution to avoid contact with the probe when working around it.



CAUTION

Do not re-use such disposable products as collection tubes, test tubes, capillary tubes, etc.

Make sure that the entered sample ID and mode exactly match those of the samples to be run.

The tube (or centrifugal tube) shall be placed vertically upward, not tilted or upside down. Otherwise, the inner wall of the tube may be stained with excessive sample, resulting in waste. Moreover, it may cause unevenly mixed sample and unreliable analysis results.

During aspiration, the tip of the probe should be kept at a certain distance from the bottom of the sample container, otherwise the accuracy of aspiration volume will be affected.

NOTE

Keep the tip of the probe from contacting with the wall of the test tube to avoid blood splashing.

Proper reference range shall be selected on the **Setup** interface before analysis. Otherwise, the results may be flagged erroneously.

The default system setting for counting mode is **Venous Whole Blood (VWB)-CBC+DIFF**. When the analyser is running the samples, you can switch to Review interface to perform operations including browsing and exporting, etc., and you can also switch to other interfaces. But all the functions related to the fluidics sequence are not available.

Take the following steps to perform sample analysis.

1. Prepare samples as instructed by **6.5 Sample Collection and Handling**.
For details about the preparation of venous whole blood samples, see **6.5.1 Venous Whole Blood Samples**.
For details about the preparation of capillary whole blood samples, see **6.5.2 Capillary Whole Blood Samples**.
For details about the preparation of prediluted samples, see **6.5.3 Prediluted Samples**.
2. Shake the capped tube of sample for a homogeneous specimen.
3. When the green indicator light is steady-on, click **Mode & ID** in **Sample Analysis** interface.
A dialog box will pop up as shown in Figure 7-3. The analyser supports six counting modes: venous whole blood (VWB)-CBC+DIFF, venous whole blood (VWB)-CBC, capillary whole blood (CWB)-CBC+DIFF, capillary whole blood (CWB)-CBC, Predilute (PD)-CBC+DIFF, and Predilute (PD)-CBC.

Mode & ID

☒ Venous Whole Blood (VWB)

☐ Capillary Whole Blood (CWB)

☐ Predilute (PD)

☐ CBC

☒ CBC+DIFF

Sample ID

1

☐ Bidirectional LIS/HIS Communication

OK

Cancel

Figure 7-3 Mode & ID Settings

4.

Select the blood sample mode Venous **Whole Blood (VWB)**, **Capillary Whole Blood (CWB)** or **Predilute (PD)** of the sample.
5.

Select the measurement mode **CBC** or **CBC+DIFF** according to the actual test case, and enter the **Sample ID**.

Refer to Table 7-2 for the description of relevant parameters.

Table 7-2 Sample Analysis Parameter Descriptions

Parameter	It means	Operation
CBC	Complete Blood Count with no differential count for white blood cells. The counting results comprise 16 parameters, 3 histograms (including WBC, RBC and PLT), and one BASO scattergram	Selected from the radio box.
CBC+DIFF	Complete Blood Count plus differential count for white blood cells. The counting results comprise 26 measurement parameters, 7 RUO parameters, one DIFF scattergram, one BASO scattergram, and three histograms (including WBC, RBC and PLT).	Selected from the radio box.

Parameter	It means	Operation
Sample ID	Identification number for the samples to be run.	Enter in the textbox directly. NOTE - Letters, numbers and all characters that can be entered through the keyboard (including special characters) are allowed for the Sample ID. Chinese and other languages (such as Japanese, Korean, etc.) are not supported. - The length of the entries ranges from 1 to 25 and the entries shall not be empty. The last character of a sample ID must be numeric, but a string of "0" only is not an acceptable sample ID.

6. Click **OK**.
7. Remove the tube cap carefully and place the sample under the probe so that the probe can aspirate the well-mixed sample.
8. Press the aspirate key on the analyser to start running the sample.
The sample will be automatically aspirated by the sample probe.
9. When you hear a beep, remove the sample tube.
10. The analyser will automatically run the sample and the analysis status icon and analyser indicator is flickering in green. When the analysis is complete, the analyser indicator returns to constantly-on green.
11. Repeat steps Prepare samples as instructed by **6.5 Sample Collection and Handling**. When you hear a beep, remove the sample tube. to run the remaining samples.

7.5 Dealing with the Analysis Results

7.5.1 Automatic saving of analysis results

This analyser automatically saves sample results. When the maximum number has been reached, the newest result will overwrite the oldest (already backed up).

7.5.2 Parameter Flags

- If parameter is followed by a "↑" or "↓", it means the analysis result has exceeded the upper or lower limit of the reference range but still within the display range.
- If the parameter is followed by a "?", it means the analysis result is suspicious.
- If you see "***" instead of a result, it means the result is either invalid or beyond the display range.

NOTE For the background test, the flags for parameters or abnormal blood cell differential and morphology are not available.

7.5.3 Flags of Abnormal Blood Cell Differential or Morphology

The analyser will flag abnormal or suspicious WBC, RBC and PLT according to the scattergrams and histograms. The flag information is defined in the Table 7-3.

Table 7-3 Flags of abnormal blood cell differential or morphology

Flag Type		Flag information
WBC	Abnormal	Leucocytosis Leucopenia Neutrophilia Neutropenia Lymphocytosis Lymphopenia Monocytosis Eosinophilia Basophilia
	Suspicious	WBC abnormal Abnor. WBC scattergram Abnor. WBC histogram Left Shift? Immature Cell? RBC Lyse Resistant? Abn./Atypical Lym? Abnormal WBC Channel Abnormal DIFF Channel
RBC/HGB	Abnormal	Erythrocytosis Anisocytosis Macrocytosis Microcytosis Anaemia Hypochromia
	Suspicious	Abnor. RBC Distr. Dimorphologic Iron Deficiency? HGB Abnor./Interfere? RBC Clump? Abnormal RBC Channel Abnormal HGB Channel
PLT	Abnormal	Thrombocytosis Thrombopenia
	Suspicious	Abnor. PLT Distr. PLT Clump?

The system shows flags for abnormal or suspicious items in different samples and measurement modes in accordance with the impact of the abnormal or suspicious WBC, RBC or PLT items on the results of the parameters. The correlation is shown in the Table 7-4.

Table 7-4 Flags for abnormal or suspicious items in different samples and measurement modes

Type	Flag	Whole Blood		Predilute (PD)	
		CBC	CBC+DIFF	CBC	CBC+DIFF
WBC	WBC abnormal?	√	√	√	√
	RBC Lyse Resistant?	×	√	×	√
	Abnor. WBC scattergram	×	√	×	√
	Abnor. WBC histogram	√	√	√	√
	Left Shift?	×	√	×	√
	Immature Cell?	×	√	×	√
	Abn./Atypical Lym?	×	√	×	√
	Leucocytosis	√	√	√	√
	Leucopenia	√	√	√	√
	Neutrophilia	×	√	×	√
	Neutropenia	×	√	×	√
	Lymphocytosis	×	√	×	√
	Lymphopenia	×	√	×	√
	Monocytosis	×	√	×	√
	Eosinophilia	×	√	×	√
	Basophilia	×	√	×	√
	Abnormal WBC Channel	×	√	×	√
	Abnormal DIFF Channel	×	√	×	√
RBC/HGB	Dimorphologic	√	√	√	√
	HGB Abnor./Interfere?	√	√	√	√
	Anisocytosis	√	√	√	√
	Microcytosis	√	√	√	√
	Macrocytosis	√	√	√	√
	Erythrocytosis	√	√	√	√
	Anaemia	√	√	√	√
	Hypochromia	√	√	√	√
	Abnor. RBC Distr.	√	√	√	√
	Iron Deficiency?	√	√	√	√
	RBC Clump?	√	√	√	√
	Abnormal RBC Channel	√	√	√	√
	Abnormal HGB Channel	√	√	√	√
PLT	PLT Clump?	√	√	√	√
	Thrombocytosis	√	√	√	√
	Thrombopenia	√	√	√	√
	Abnor. PLT Distr.	√	√	√	√

NOTE “√” indicates that flags will be displayed in the mode.”×” indicates that flags will not be displayed in the mode.

When the PLT value is less than $100 \times 10^3/\mu\text{L}$, a manual count by the microscope is recommended.

7.6 Functions of the Buttons

7.6.1 Previous/Next

Click **Previous**, and the screen will display the sample analysis results prior to the current one.

Click **Next**, and the screen will display the sample analysis results after the current one.

7.6.2 Mode & ID

Click this button to set the sample mode and measurement mode during the sample analysis. See section **7.4 Running Samples**.

7.6.3 Pre-entry

Click this button, and you can pre-enter the information of the sample to be tested before performing the sample analysis. See section **7.3 Entering Sample Information**.

7.6.4 Validate/Cancel Validation

After running sample, you can click Validate to validate the sample. After validating, the button will be replaced by **Cancel Validation**. After validating, you cannot edit the sample/patient information and the result.

If the current sample has been validated, the sample validation can be cancelled by clicking **Cancel Validation**. After cancelling the validation, you can edit the sample/patient information and the result.

7.6.5 Print

You can click **Print** to print the report of the sample result.

7.6.6 Patient Information

You can browse and edit the patient information of the selected sample in the **Sample Analysis** interface. The operation procedures are as shown below:

1. Click **Patient Info.** to enter the patient information setting interface as shown in Figure 7-4.

Patient Info.

First Name	Last Name	Sample ID
		50_1104_163
Patient Type	Sample Type	Department
Med Rec. No.	Area	Bed No.
Gender	Birthday	Age
	--	Year
Ref. Group	Sampling Time	Delivery Time
General	2017-03-27 17:36	2017-03-27 17:36
Submitter	Operator	Run Time
	admin	2017-03-27 17:36
Mode	Approver	Report Time
Venous Whole Blood		-- :
Diagnosis		
Remarks		
Apply OK Cancel		

Figure 7-4 Patient Information

- Enter patient information with reference to the parameter description in Table 7-5.

Table 7-5 Parameter Description of Patient Information

Parameter	Meaning	Operation
Sample ID	Number of the selected sample.	It will be displayed automatically, and you can modify it manually.
First Name	First name of patient.	Input in the textbox directly.
Last Name	Last name of patient.	Input in the textbox directly.
Patient Type	Type of patient. Values: - Inpatient - Physical Examination - STAT - Outpatient	Select from the dropdown list.
Sample Type	Type of selected sample. - Venous blood - Capillary - Cord blood - Blood	Select from the dropdown list.
Med Rec. No.	Med Rec. No. of patient.	Input in the textbox directly.
Gender	Gender of patient. Values: (Null) - Male - Female - Not defined	Select from the dropdown list.

Parameter	Meaning	Operation
Birthday	Birthday of a patient.	Select from the date control. - The input sequence of the controls is the same with the date format on the top right corner of the dialog box. For example, if the data format is yyyy/MM/dd, you should input the data in the sequence of year, month, and date. - Click  or  to select a date and time or enter the information in the textbox directly. - Click  to clear the current data and re-enter the information.
Age	Age of a patient.	Select the unit of age from the dropdown list (Year, Month, Day or Hour) and enter the age of the patient in the textbox before the age unit.
Ref. Group	Reference group of the sample under analysis. The result is judged according to the reference range of the reference group and the result beyond the normal range will be flagged.	Select from the dropdown list. NOTE If the Automatically match the customized reference group according to age and gender is set, gender and age of a patient will automatically match the reference group according to the corresponding relationship (No matter the reference group is selected or not). Refer to 5.4.3 Ref. Range for the setting of the reference group and range.
Department	Department receiving the patient.	Select from the dropdown list.
Area	Ward area of patient.	Input in the textbox directly.
Bed No.	Bed No. of inpatient.	Select from the dropdown list or input directly. NOTE The bed No. is required to be filled only for inpatients.
Sampling Time	Date and time when the sample is collected.	Click the date control for the settings. - The input sequence of the controls is the same with the date format on the top right corner of the dialog box. For example, if the data format is yyyy/MM/dd HH:mm, you should input the data in the sequence of year, month, date, hour, and minute. Click  or  to select a date and time or enter the information in the textbox directly. - Click  to clear the current data and re-enter the information. NOTE The sampling time can be no later than the current system time.
Submitter	Personnel submitting the sample.	Select from the dropdown list or input directly.

Parameter	Meaning	Operation
Mode & ID	Counting mode of the selected sample. The format is <i>blood sample mode-measurement mode</i> .	You do not need to enter it and it will be displayed automatically.
Delivery Time	Date and time when the sample is delivered.	Click the date control for the settings. - The input sequence of the controls is the same with the date format on the top right corner of the dialog box. For example, if the data format is yyyy/MM/dd HH:mm, you should input the data in the sequence of year, month, date, hour, and minute. - Click  or  to select a date and time or enter the information in the textbox directly. - Click  to clear the current data and re-enter the information. NOTE The delivery time can be no later than the current system time and cannot be earlier than the sampling time.
Operator	Personnel running the sample.	You do not need to enter it and it will be displayed automatically.
Run Time	Time when the sample is run.	You do not need to enter it and it will be displayed automatically.
Approver	Personnel validating the sample.	This parameter will be automatically displayed after the sample is validated.
Report Time	The date and time when the report is printed for the first time.	This parameter will be automatically displayed after the report is printed.
Diagnosis	Suspected diagnosis information.	Input in the textbox directly.
Remarks	Clarifications or notes.	Input in the textbox directly.

3. Click **Apply** or **OK** to save the settings.

7.6.7 Customized Parameters

You can browse and edit the customized parameters results of the selected sample in the **Sample Analysis** interface. The procedures are shown as below:

1. Click **Custom Para.** to enter the customized parameters setting interface as shown in Figure 7-5.

Custom Para. 

No.	Parameter Name	Unit
1	Blood Type	
2	RH Blood Group	
3	ESR	
4	C-reactive Protein	
5	Reticulocyte	

New

Edit

Delete

Close

Figure 7-5 Customized Parameters

2. Click the cell corresponding to its **Value** column of the parameter, and enter the value.
If the unit and reference range of parameters have been set in the **Setup > Parameter > Custom Para.** interface, the corresponding unit and range (lower limit~upper limit) will be displayed in this tab. When both the value and range of parameters are numbers, and the number is out of the reference range, the relevant mark ↑ or ↓ will be displayed in the **Flag** column.
Please refer to **5.4.6 Customized Parameters** for customized parameters settings.

7.6.8 Microscopic Exam. Parameters

You can perform the microscopic exam. settings as per the following steps.

Click **Microscopic Exam. Para.**

The microscopic examination parameters interface as shown in Figure 7-6 will pop up on the screen.

Figure 7-6 Adding a New Microscopic Exam. Parameter

Refer to Table 7-6 for parameter description and operation methods regarding the microscopic examination.

Table 7-6 Microscopic Exam. Parameters

Parameter	It means	Operation
Sample Type	Type of sample for microscopic examination. • Venous blood • Capillary • Cord blood • Blood	Click the Sample Type dropdown list box and select the type of sample for microscopic examination.
Exam. Time	Time of microscopic examination.	Click the Exam. Time combo box and select the time and date for the microscopic examination. NOTE The Microscopic exam. time can be no later than the current system time.
Microscopic Description	Description of cells morphology.	Enter the morphology information for WBC, RBC and PLT respectively into the multi-line textbox.

7.6.9 Communication

You can transmit the current sample data (except the background sample) to the LIS/HIS system in the **Sample Analysis** interface.

- 1. Click  to unfold all function buttons.
- 2. Click **Comm..**

7.6.10 Edit Result

NOTE

You cannot edit the results of validated samples.

You cannot edit the results of the background.

In the CBC mode, only the results of the test parameters are available, the results concerning the percentage of the WBC diff parameters are not available.

You can edit the parameter result of the selected sample as per the following steps.

- 1. Click  to unfold all function buttons.
- 2. Click **Edit Result**.

The **Edit Result** dialog box will pop up on the screen as shown in Figure 7-7.

Edit Result

WBC	17.28	10 ³ /uL	RBC	5.51	10 ⁶ /uL	PLT	421	10 ³ /uL
Neu%	64.2	%	HGB	16.9	g/dL	MPV	9.0	fL
Lym%	20.5	%	HCT	55.0	%	PDW-SD	16.1	fL
Mon%	6.1	%	RDW-CV	13.5	%	PDW-CV	13.5	%
Eos%	9.2	%	RDW-SD	54.9	fL	P-LCR	30.8	%
Bas%	7.9	%						

Apply

OK

Cancel

Figure 7-7 Editing Parameter Result

- 3. Modify the counting results of the corresponding sample parameters.
 - 4. Click **Apply** or **OK** to save the changes.
- If the sum of the percentage of the diff parameters is not equal to 100.00% or the WBC value is invalid after modification, the system will prompt in a message box that the entered value is invalid. Please re-enter after confirmation.
- If the result of one parameter is modified, then the result of other related parameter(s) will be changed accordingly and the high or low/suspicious flags will also be updated.

NOTE

The result of the parameter that you modified manually will be flagged with an **M**.

If any parameter result is then changed due to the one that you modified manually, it will be flagged with an **m**.

7.6.11 Delete

Validated samples are not allowed to be deleted.

The common user has no access to delete the sample records.

1. Click  to unfold all function buttons.
2. Click **Delete**, and then click **Yes** in the pop-up dialog box to delete the sample.

8

Result Review

8.1 Introduction

Upon the completion of each sample analysis, the analyser will automatically save the sample information, result data, flag messages, histograms and scattergrams to the Review Database. In the **Review** Interface, you can browse the saved sample information, result data, flag messages, histograms and scattergrams, and can search, compare or export the saved sample information.

8.2 Interface Introduction

You can browse, search, compare, print, and export the existing results in the **Review** interface. Click **Review** to enter the sample review interface. See Figure 8-1.



Figure 8-1 Review

Interface Description:

- Result list: you can browse detailed sample records.
- Function buttons: you can perform the operations such as comparing or searching the sample results, deleting and viewing the Run Charts, exporting and printing reports.

8.3 Sample List

The review interface shows a list of the analysed samples, which contains the sample number, status, mode and results of various parameters and other information.

Click a sample or multiple samples in the list area, then you perform operations such as exporting in batch for the selected samples. To cancel the selection, click the selected samples again.

8.4 Functions of the Buttons

8.4.1 Validate

NOTE After validating, you cannot edit the sample/patient information and the result.

After running samples, you can validate the samples as per the following steps.

1. Click **Validate**. A dialog box will pop up as shown in Figure 8-2.
2. Select the sample which needs to be validated.

Selected Records: The selected sample results with blue background.

Samples on current page: Results of all the samples shown on the current page.

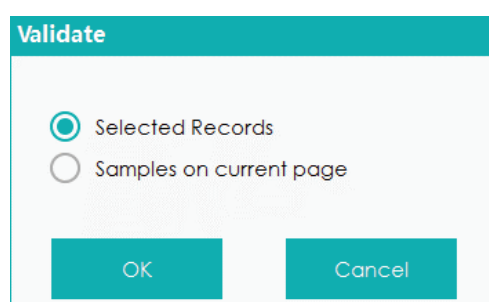


Figure 8-2 Options for validation of results

3. Click **OK**.
4. Click **OK** to close the message box.

8.4.2 Cancel Validation

NOTE After cancelling the validation, you can edit the sample/patient information and the result.

You can cancel the validation of validated samples. Detailed steps are shown below:

1. Click Cancel Validation.
2. Select the sample which needs to be validated.
 - Select **Selected Records**, and the system will cancel the validation for the selected sample results with blue background.
 - Select **Samples on current page**, and the system will cancel the validation for all the samples on the current page.
3. Click **OK**.
4. Click **OK** to close the message box.

8.4.3 Print

Click Print to **print** the result report of the selected sample.

8.4.5 Delete

NOTE	Validated samples are not allowed to be deleted. The common user has no access to delete the sample records.
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1. Select one or several sample records to be deleted.
 2. Click Delete.
- A prompt box will pop up on the screen as shown in Figure 8-3.

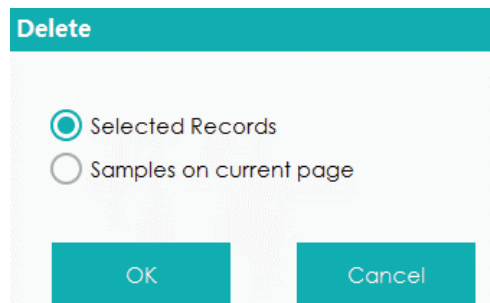


Figure 8-3 Delete Sample Records

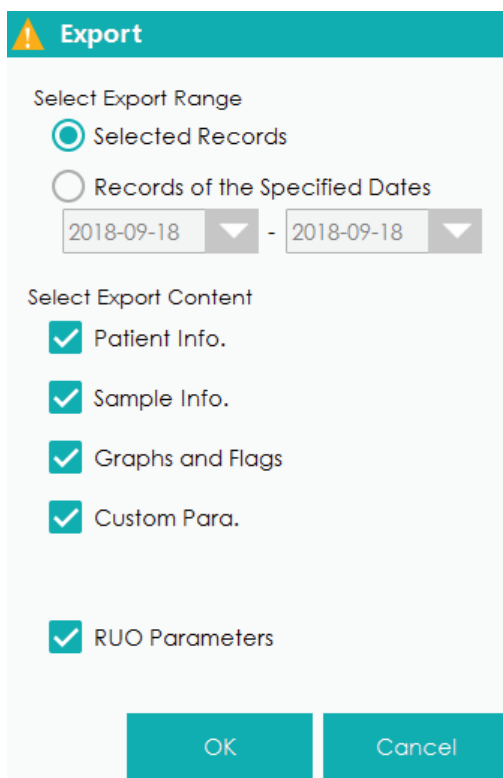
3. Select one or several sample records to be deleted according to the actual situation.
 - Selected Records: The selected sample results with blue background.
 - Samples on current page: Results of all the samples shown on the current page.
4. Click OK to delete the selected record(s).

8.4.5 Export

The operator can export the sample data to the USB flash disk for backup. There are two ways of exporting the sample data: exporting selected records and exporting records of specified dates.
Export Selected Records

1. Insert a USB flash disk in the USB interface on the analyser.
2. Select records to be backed up, and click **Export**.

As shown in the Figure 8-4, the export range of the system is **Selected Records** by default.



Export

Select Export Range

☒ Selected Records

☐ Records of the Specified Dates

2018-09-18 - 2018-09-18

Select Export Content

☒ Patient Info.

☒ Sample Info.

☒ Graphs and Flags

☒ Custom Para.

☒ RUO Parameters

OK Cancel

Figure 8-4 Export Selected Records

3. Select the content to be exported according to the actual demand.
Content available for export includes: **Patient Info.**, **Sample Info.**, **Graphs and Flags**, **Custom Para.**, **RUO Parameters**.
4. Click **OK**.
5. Select the data export path in the popup dialog box, enter the backup file name, and click **Save**.

The file will be exported to the root directory of the USB flash disk (/udisk/sda1) and named in the format of **SampleInfo_yyyyMMdd_hhmmss.csv**. Among which, yyyyMMdd_hhmmss means data export year, month, date, hour, minute, and second.

6. Click **Save**.

The system pops up a dialog box to indicate that the data export is successful.

Export Records of the Specified Dates

1. Insert a USB flash disk in the USB interface on the analyser.
2. Click **Export**.
3. Select **Records of the Specified Dates** and set the run date range of sample in the two date textboxes. See Figure 8-5.

Export

Select Export Range

☐ Selected Records

☒ Records of the Specified Dates

2018-09-18 - 2018-09-18

Select Export Content

☒ Patient Info.

☒ Sample Info.

☒ Graphs and Flags

☒ Custom Para.

☒ RUO Parameters

OK Cancel

Figure 8-5 Export Records of the Specified Dates

4. Select the content to be exported according to the actual demand.
Content available for export includes: **Patient Info.**, **Sample Info.**, **Graphs and Flags**, **Custom Para.**, **RUO Parameters**.
5. Click **OK**.
6. Select the data export path in the popup dialog box, enter the backup file name, and click Save.
The file will be exported to the root directory of the USB flash disk (/udisk/sda1) and named in the format of **SampleInfo_YYYYMMdd_hhmmss.csv**. Among which, **YYYYMMdd_hhmmss** means data export year, month, date, hour, minute, and second.
7. Click **Export**.
The system pops up a dialog box to indicate that the data export is successful.

8.4.6 Edit Result

You cannot edit the results of validated samples.

NOTE

Background result cannot be edited!

In the CBC mode, only the results of the test parameters are available, the results concerning the percentage of the WBC diff parameters are not available.

You can edit the parameter result of the selected sample as per the following steps:

1. Select a row of record from the result list and click the Edit Result button.
The Edit Result dialog box will pop up on the screen as shown in Figure 8-6.

The 'Edit Result' dialog box contains the following parameters and values:

Parameter	Value	Unit
WBC	17.28	10 ³ /uL
RBC	5.51	10 ⁶ /uL
PLT	421	10 ³ /uL
Neu%	64.2	%
HGB	16.9	g/dL
MPV	9.0	fL
Lym%	20.5	%
HCT	55.0	%
PDW-SD	16.1	fL
Mon%	6.1	%
RDW-CV	13.5	%
PDW-CV	13.5	%
Eos%	9.2	%
RDW-SD	54.9	fL
P-LCR	30.8	%
Bas%	7.9	%

Buttons: Apply, OK, Cancel

Figure 8-6 Editing Parameter Result

2. Modify the counting results of the corresponding sample parameters.
3. Click **Apply** or **OK** to save the changes.

If the sum of the percentage of the diff parameters is not equal to 100.00% or the WBC value is invalid after modification, the system will prompt in a message box that the entered value is invalid. Please re-enter after confirmation.

If the result of one parameter is modified, then the result of other related parameter(s) will be changed accordingly and the high or low/suspicious flags will also be updated.

8.4.7 Patient Info.

You can browse and edit sample/patient information after the sample analysis is completed. Detailed steps are shown below:

1. Click **Patient Info..**

The interface as shown in Figure 8-7 will pop up on the screen.

Patient Info.

First Name 	Last Name 	Sample ID 50_1104_163
Patient Type 	Sample Type 	Department
Med Rec. No. 	Area 	Bed No.
Gender 	Birthday - -	Age Year
Ref. Group General	Sampling Time 2017-03-27 17:36	Delivery Time 2017-03-27 17:36
Submitter 	Operator admin	Run Time 2017-03-27 17:36
Mode Venous Whole Blood	Approver 	Report Time - - :
Diagnosis 		
Remarks 		
Apply OK Cancel		

Figure 8-7 Patient Info

- Enter patient information with reference to the parameter description in Table 8-1.

Table 8-1 Parameter Description

Parameter	It means	Operation
Sample ID	Number of the selected sample.	It will be displayed automatically, and you can modify it manually.
First Name	First name of patient.	Input in the textbox directly.
Last Name	Last name of patient.	Input in the textbox directly.
Patient Type	Type of patient. Values: - Inpatient - Physical Examination - STAT - Outpatient	Select from the dropdown list.
Sample Type	Type of selected sample. - Venous blood - Capillary - Cord blood - Blood	Select from the dropdown list.
Med Rec. No.	Med Rec. No. of patient.	Input in the textbox directly.

Parameter	It means	Operation
Gender	Gender of patient. Values: • (Null) • Male • Female • Not defined	Select from the dropdown list.
Birthday	Birthday of a patient.	Select from the date control. • The input sequence of the controls is the same with the date format on the top right corner of the dialog box. For example, if the data format is yyyy/MM/dd, you should input the data in the sequence of year, month, and date. • Click  or  to select a date and time or enter the information in the textbox directly. • Click  to clear the current data and re-enter the information.
Age	Age of a patient.	Select the unit of age from the dropdown list (Year, Month, Day or Hour) and enter the age of the patient in the textbox before the age unit.
Ref. Group	Reference group of the sample under analysis. The result is judged according to the reference range of the reference group and the result beyond the normal range will be flagged.	Select from the dropdown list. NOTE If the Automatically match the customized reference group according to age and gender is set, gender and age of a patient will automatically match the reference group according to the corresponding relationship (No matter the reference group is selected or not). Refer to Ref. RangeRef. Range for the setting of the reference group and range.
Department	Department receiving the patient.	Select from the dropdown list.
Area	Ward area of patient.	Input in the textbox directly.
Bed No.	Bed No. of inpatient.	Input in the textbox directly. NOTE The bed No. is required to be filled only for inpatients.

Parameter	It means	Operation
Sampling Time	Date and time when the sample is collected.	Click the date control for the settings. - The input sequence of the controls is the same with the date format on the top right corner of the dialog box. For example, if the data format is yyyy/MM/dd HH:mm, you should input the data in the sequence of year, month, date, hour, and minute. - Click  or  to select a date and time or enter the information in the textbox directly. - Click  to clear the current data and re-enter the information. NOTE The sampling time can be no later than the current system time.
Submitter	Personnel submitting the sample.	Select from the dropdown list or input directly.
Mode & ID	Counting mode of the selected sample. The format is blood sample mode-measurement mode.	You do not need to enter it and it will be displayed automatically.
Delivery Time	Date and time when the sample is delivered.	Click the date control for the settings. - The input sequence of the controls is the same with the date format on the top right corner of the dialog box. For example, if the data format is yyyy/MM/dd HH:mm, you should input the data in the sequence of year, month, date, hour, and minute. - Click  or  to select a date and time or enter the information in the textbox directly. - Click  to clear the current data and re-enter the information. NOTE The delivery time can be no later than the current system time and cannot be earlier than the sampling time.
Operator	Personnel running the sample.	You do not need to enter it and it will be displayed automatically.
Run Time	Time when the sample is run.	You do not need to enter it and it will be displayed automatically.
Approver	Personnel validating the sample.	This parameter will be automatically displayed after the sample is validated.
Report Time	The date and time when the report is printed for the first time.	This parameter will be automatically displayed after the report is printed.
Diagnosis	Suspected diagnosis information.	Input in the textbox directly.
Remarks	Clarifications or notes.	Input in the textbox directly.

3. Click **Apply** or **OK** to save the configuration.

8.4.8 Query

You can view the test results of a patient within a certain test date range by entering the query conditions. The procedures are shown as below:

1. Click the **Query** button to enter the multi-conditional query dialog box as shown below.

Figure 8-8 Query Conditions

2. Determine the query conditions as needed.

For the specific parameter description, see Table 8-2.

Table 8-2 Parameter Description of Query Conditions

Parameter	It means	Operation Description
Sample ID	Sample ID to be queried.	Input in the textbox directly.
Name	Name of patient.	Input in the textbox directly.
Med Rec. No.	Med Rec. No. of patient.	Input in the textbox directly.
Para.	Parameter and its range to be queried.	Select a parameter from the first dropdown list, and a comparison symbol ($\geq$, $>$, $\leq$, $<$, $=$) from the second dropdown list, then input a value in the textbox. For example, if you select WBC and $>$, then input 3 in the textbox. The sample results which RBC value is greater than $3.0 \times 10^6/\mu\text{L}$ will be queried and displayed.
Run Date	Test date range of sample.	Select the starting and ending dates of the sample test in the two data controls successively.
Sample status	Status of validation, printing or communication of the sample. • Not Validated • Not Printed • Not Transmitted	Please choose according to the actual situation. The default value is Not Validated .

Auto select checked by default indicates that the query result is being selected (with a blue background colour). If it's unchecked, the query result will remain on a white background colour.

NOTE

Click All Samples to close the current window, display all the samples again and restore all the filter conditions to the default values.

3. Click Query.

The system will display all the query results which meet the conditions.

8.4.9 Graph

In the Review interface, you can click Graph to browse the selected sample graph results, parameter results and flag messages. The procedures are shown as below:

1. Select a result to review in graph interface.
2. Click  to unfold all function buttons.
3. Click **Graph** to enter the graph interface of the selected sample.

In the Graph interface, you can view sample information such as parameter results, graph results and flag messages. In addition, you can also print the analysis report as. See Figure 8-9.

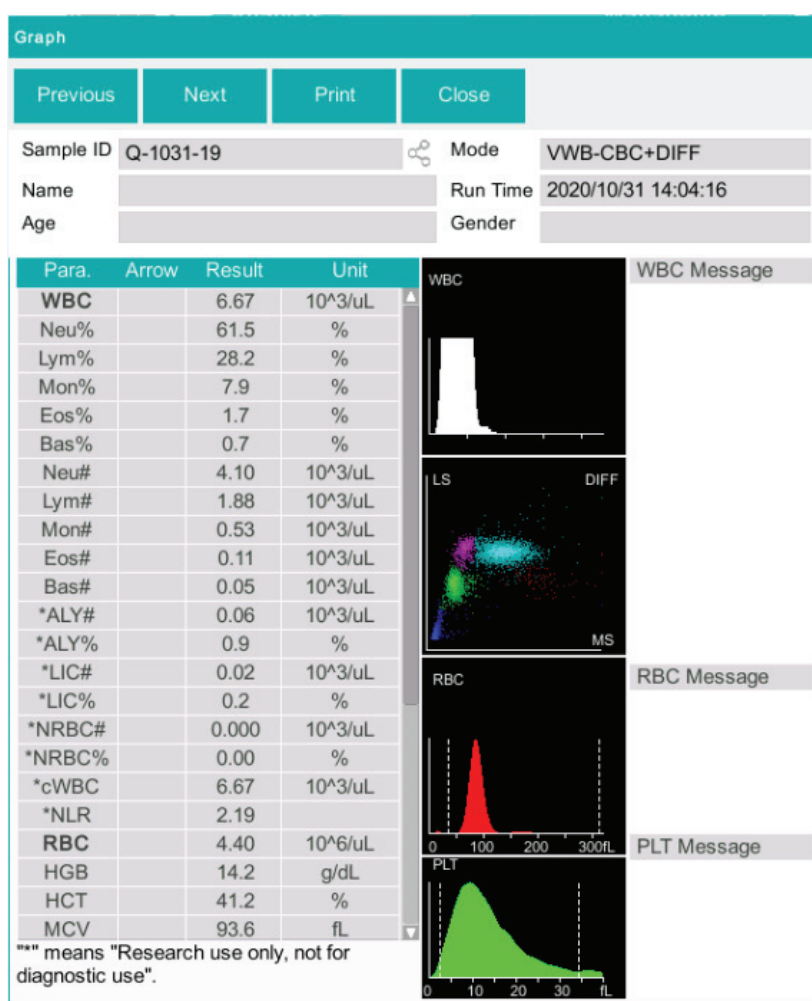


Figure 8-9 Graphs Review

8.4.10 Microscopic Exam. Parameters

You can perform the microscopic examination settings as per the following steps:

1. Click **Microscopic Exam. Para.**

The microscopic examination parameters interface as shown in Figure 8-10 will pop up on the screen.

Parameter Name	Value
Neutrophilic segmented gran...	
Neutrophilic band granulocyte	
Lymphocyte	
Monocyte	
Eosinophil	
Basophil	
Plasmacyte	
Atypical Lymph	
Blast	
Promyelocyte	
Neutrophilic myelocyte	
Eosinophilic myelocyte	
Basophilic myelocyte	
Neutrophilic metamyelocyte	
Eosinophilic metamyelocyte	
Basophilic metamyelocyte	
Preliminary lymphocyte	

Sample Type: Capillary

Exam. Time: / /

Microscopic Description

Apply OK Cancel

Figure 8-10 Adding a New Microscopic Exam. Parameter

2. Set the microscopic examination parameters by referring to Table 8-3.

Table 8-3 Microscopic Exam. Parameters

Parameter	It means	Operation
Sample Type	Type of sample for microscopic examination. - Venous blood - Capillary - Cord blood - Blood	Click the Sample Type dropdown list box and select the type of sample for microscopic examination.
Exam. Time	Time of microscopic examination.	- Click the date control for the settings. - The input sequence of the controls is year, month, date, hour, and minute. - Click or to select the date or click the textbox to enter them directly. - Click to clear the current data and re-enter the information. NOTE The Microscopic exam. time can be no later than the current system time.
Microscopic Description	Description of cells morphology.	Enter the morphology information for cells into the multi-line textbox.

8.4.11 Customized Parameters

You can browse and edit the customized parameters results of the selected sample in the **Review** interface. The procedures are shown as below:

1. Select one sample.
2. Click  to unfold all function buttons.
3. Click **Custom Para.** to enter the customized parameters setting interface as shown in Figure 8-11.

Custom Para.

Para.	Flag	Value	Unit	Range
Blood Type				
RH Blood Group				
ESR				
C-reactive Protein				
Reticulocyte				

Apply

OK

Close

Figure 8-11 Customized Parameters

4. Click the cell corresponding to its **Value** column of the parameter, and enter the value.
If the unit and reference range of parameters have been set in the **Setup > Parameter > Custom Para.** interface, the corresponding unit and range (lower limit~upper limit) will be displayed in this tab. When both the value and range of parameters are numbers, and the number is out of the reference range, the relevant mark ↑ or ↓ will be displayed in the **Flag** column.
Please refer to **5.4.6 Customized Parameters** for customized parameters settings.

8.4.12 Communication

You can transmit the selected sample data, the data in the current page or the data within the specified date range to the LIS/HIS system in the **Review** interface.

- Selected Records
 1. Select one or several sample data to be communicated in the result list.
 2. Click  to unfold all function buttons.
 3. Click **Comm.**

A dialog box will pop up as shown in Figure 8-12. The default option is **Selected Records**.

Figure 8-12 Communication for Selected Data

4. Click **OK**. After the data is transmitted to LIS/HIS, a message box will pop up.
 5. Click **OK** to close the message box.
- Samples on current page
 1. Click  to unfold all function buttons.
 2. Click **Comm.**
 3. Select **Samples on current** page. See Figure 8-13.

Figure 8-13 Communication for Data on Current Page

4. Click **OK**. After the data is transmitted to LIS/HIS, a message box will pop up.
 5. Click **OK** to close the message box.
- Records of the Specified Dates
 1. Click  to unfold all function buttons.
 2. Click **Comm.**
 3. Select **Specified Data**, and set the starting and ending dates of data to be communicated. See Figure 8-14.

A dialog box titled "Comm." with a teal header. It contains three radio button options: "Selected Records", "Samples on current page", and "Records of the Specified Dates". The "Records of the Specified Dates" option is selected. Below the options is a date range selector showing "2018-09-18" followed by a dropdown arrow, a hyphen, another "2018-09-18" followed by a dropdown arrow. At the bottom are two teal buttons: "OK" and "Cancel".

Figure 8-14 Communication for Data on Specified Dates

4. Click OK. After the data is transmitted to LIS/HIS, a message box will pop up.
5. Click OK to close the message box.

8.4.13 Run Chart

Operators can check and review run charts of sample parameter results in the database. There are three view modes: selected samples, samples on current page and samples on specified run dates.

- View the run chart of the selected sample (default)
 1. Check no fewer than three sample records.
 2. Click  to unfold all function buttons.
 3. Click **Run Chart**. The system pops up a dialog box as in Figure 8-15.

A dialog box titled "Run Chart" with a teal header. It contains three radio button options: "Selected Records", "Samples on current page", and "Run Date". The "Selected Records" option is selected. Below the options is a date range selector showing "2018-09-18" followed by a dropdown arrow, a hyphen, another "2018-09-18" followed by a dropdown arrow. At the bottom are two teal buttons: "OK" and "Cancel".

Figure 8-15 Viewing the Run Chart of the Selected Sample

4. Click **OK**.

The screen will show the parameter result run chart of the selected sample. Figure 8-16.

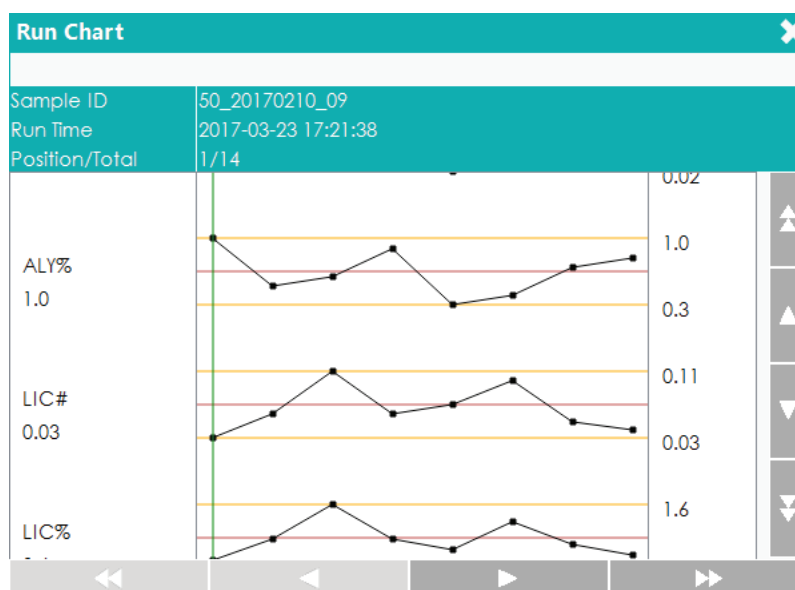


Figure 8-16 Run Chart

- View the run chart of samples on current page
 - Click on the current page to unfold all function buttons.
 - Click the **Run Chart** button and select Samples on current page in the pop-up dialog box. See Figure 8-17.

Run Chart

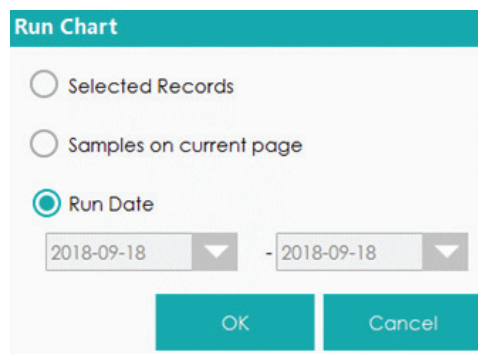
☐ Selected Records
☒ Samples on current page
☐ Run Date

2018-09-18 ▼ - 2018-09-18 ▼

OK Cancel

Figure 8-17 Viewing the Run Chart of Samples on the Current Page

- Click **OK**. The screen will show the parameter result run chart of the selected sample.
- View the run chart of samples on specified run dates
 - Click to unfold all function buttons.
 - Click the Run Chart button, and select Run Date in the pop-up dialog box. See Figure 8-18.

A dialog box titled "Run Chart" with a teal header. It contains three radio button options: "Selected Records", "Samples on current page", and "Run Date". The "Run Date" option is selected. Below the options are two date input fields, each with a dropdown arrow, separated by a hyphen. Both fields contain the date "2018-09-18". At the bottom are two teal buttons labeled "OK" and "Cancel".

Run Chart

☐ Selected Records

☐ Samples on current page

☒ Run Date

2018-09-18 - 2018-09-18

OK Cancel

Figure 8-18 Viewing the Run Chart of Samples on Specified Run Dates

3. Click the date edit box, set a date range in the pop-up dialog box, then click **OK**.

The input sequence of the controls is the same with the date format on the top right corner of the dialog box. For example, if the data format is **yyyy/MM/dd**, you should input the data in the sequence of year, month, and date.

Click  or  to select a date and time or enter the information in the textbox directly.

Click  to clear the current data and re-enter the information.

4. Click **OK**. The screen will show the parameter result run chart of the selected sample.

9 Quality Control

9.1 Introduction

Quality Control (QC) consists of strategies and procedures that measure the precision and stability of the analyser. The results imply the reliability of the sample results. QC involves measuring materials with known, stable characteristics at frequent intervals.

Analysis of the results with statistical methods allows the inference that sample results are reliable. Erba Lachema recommends running the QC program on a daily basis with low, normal and high level controls. A new lot of controls should be analysed in parallel with the current lot prior to their Exp. dates. This may be accomplished by running the new lot of controls twice a day for five days using any empty QC file.

You should only use the Erba Lachema-specified controls and reagents. Store and use the controls and reagents by following the instructions for use of the controls and reagents.

NOTE Controls beyond their Exp. date shall not be used. Controls (similar to standard blood samples) must be well mixed before use.
General users only have the access for browsing and executing the QC analysis other than editing.

9.2 L-J Quality Control

9.2.1 QC Principle

In the L-J quality control, quality control can be applied to 26 parameters. You can perform QC on relevant parameters according to the configured QC mode. After the QC settings, you can perform QC analysis on the corresponding parameters according to the set QC mode. Each QC file can be assigned 1 Lot number for high, normal and low level controls. Each QC file can store up to 500 QC results. When there are more than 500 QC results, the new QC results will overwrite the oldest results in sequence.

9.2.2 QC Settings



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.

Only users with administrator-level access can edit the L-J settings.

NOTE Before running a new batch of controls, you need to assign a QC file to each batch of controls. You can complete the QC settings by setting QC information in the QC files.

9.2.2.1 Entering QC Information

The administrator can set the QC files by operations such as Copy, New, and Edit. Detailed steps are shown below:

- 1. Click **QC** to access the **QC** interface.
- 2. Click **QC Settings** to enter the **QC Settings** interface. See Figure 9-1.

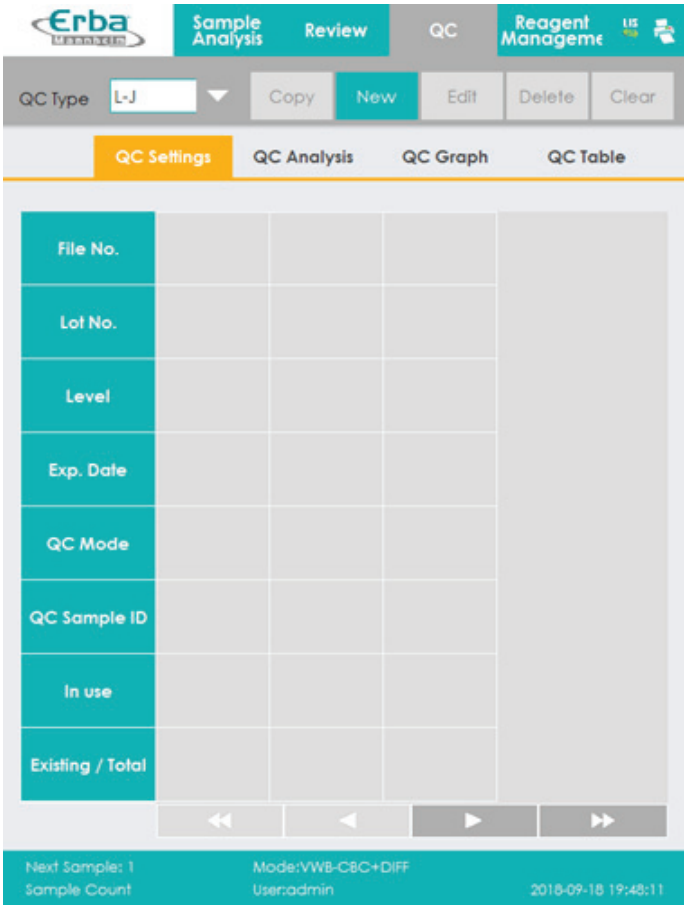


Figure 9-1 L-J Quality Control

- 3. Click the **New** button, or select a QC file (**Existing/Total** is **0/500**) without QC counting results and click the **Edit** button.

The interface as shown in Figure 9-2 will pop up on the screen.

New

File No. Level Normal QC Mode Whole Blood-CBC+DIFF

Lot No. Exp. Date 2020/10/26 QC Sample ID

Para.	Target	Limits (#)	Para.	Target	Limits (#)
WBC			MCH		
Neu%			MCHC		
Lym%			RDW-CV		
Mon%			RDW-SD		
Eos%			PLT		
Bas%			MPV		
Neu#			PDW-SD		
Lym#			PDW-CV		
Mon#			PCT		
Eos#			P-LCR		
Bas#			P-LCC		
RBC			GRAN-X		
HGB			GRAN-Y		
HCT			GRAN-Z		
MCV					

Set Limits Get Preset Values Import Save Close

Figure 9-2 Entering QC Information

You can also select the QC file of which data has been set and then click **Copy**, and edit the content based on the original data.

- Set related information of the controls with reference to Table 9-1.

Table 9-1 QC File Information

Parameter	Parameter Description	Operation Description
File No.	QC file No.. The system provides 60 QC files in total for users to set the parameters.	Read only.
Lot No.	Lot number of controls.	Enter into the textbox directly. The lot No. cannot be empty and up to 16 digits can be entered. You can enter characters, numbers, letters and special characters, but no Chinese characters are allowed. NOTE
Level	Level of the controls, including 3 levels, i.e. High, Normal and Low.	Select from the dropdown list.
Exp. Date	Exp. date of the controls.	The default Exp. Date is the current system date and needs to be changed to the actual Exp. date of the controls.

Parameter	Parameter Description	Operation Description
QC Mode	QC mode of the controls, including Whole Blood-DIFF and Predilute-DIFF .	Select from the dropdown list.
QC Sample ID	Number of the QC sample - Users need to set the number of the controls here if he/she is used to performing the analysis with the controls placed among the daily samples. See section 9.2.3.2 Completing QC Analysis in the Sample Analysis Interface. - If the user performs the analysis in the QC Analysis interface, the ID cannot be entered.	Enter into the textbox directly. NOTE Letters, numbers and all characters that can be entered through the keyboard (including special characters) are allowed for the Sample ID. Chinese and other languages (such as Japanese, Korean, etc.) are not supported. The length of the entries ranges from 1 to 25 and the entries shall not be empty. The last character of a sample ID must be numeric, but a string of "0" only is not an acceptable sample ID.
Target	Target of the QC parameter.	Enter the targets in the cell corresponding to the expected QC parameter according to the control target list with the corresponding lot No. NOTE You can also click Get Preset Values button to read the saved preset target and limits (corresponding to the current level) into the current QC file. Please refer to 9.2.4.1 QC Graph for calculation and saving of the preset values.
Limits (#)	Limits (#) of the QC parameter.	Enter the limits in the cell corresponding to the expected QC parameter according to the control target list with the corresponding lot No. NOTE You can click Set Limits to set the display form of the limits or the calculation method of the limits among the pre-set values. By SD: the limits displays in form of absolute value. Click 2SD or 3SD to select either double or triple standard deviation to be the limits. By CV: the limits displays in form of percentage. Click 2CV or 3CV to select either double or triple coefficient of variation to be the limits.

Parameter	Parameter Description	Operation Description
In use	Set if you want to specify the QC sample ID in the selected file so that you can run the QC sample in the interface other than the QC interface. • If it's checked, you can run the sample with the corresponding sample ID in any interface and the system will run the QC analysis for this sample. • If it's not checked, you can only run the QC sample in the QC interface.	It's unchecked by default. Set the parameter according to the actual situation.
Existing/Total	The existing data and total QC results in the current QC file. Up to 500 QC results can be saved for each QC file.	Read only.

- According to the value sheet of the corresponding LOT No., enter the Target and Limits into the textboxes and choose the Level, QC Mode and Exp. date.
Another option is to use button **Import** to upload the .qcs file.
- Click the **Save** button to save all the settings of the QC.

9.2.2.2 Deleting QC Files

If you want to delete the QC files which will not be used any more, please take the following steps:

- Click **QC** to access the **QC** interface.
- Click **QC Settings** to enter the **QC Settings** interface.
- Select the QC file to be deleted, and click **Delete**.
- Click **Yes**.

All selected QC files together with their QC results will be completely deleted.

9.2.2.3 Clearing QC results

If you want to delete QC results of a specified file, please take the following steps:

- Click **QC** to access the **QC** interface.
- Click **QC Settings** to enter the **QC Settings** interface.
- Select the QC file in which the QC results are expected to be cleared, and click **Clear**.
- Click **Yes**.

QC results in the selected QC file will be deleted. See the Figure 9-3. The value in the **Existing/Total** line will be restored to the initial value.



Sample Analysis

Review

QC

QC Type

L-J

Copy

New

Edit

QC Settings

QC Analysis

QC Graph

File No.	59		
Lot No.	E5112-02		
Level	Low		
Exp. Date	2017-01-10		
QC Mode	Whole Blood- CBC+DIFF		
QC Sample ID			
In use			
Existing / Total	0/500		

Figure 9-3 Clearing QC results

9.2.2.4 Details of QC file

You can click it from the drop-down list for scanning QC information.

9.2.3 Quality Control Analysis

After completing the QC settings, you can choose one of the following two modes according to the selected QC mode to run the quality control samples.

- Completing QC analysis in the **QC Analysis** interface
- Completing QC analysis in the **Sample Analysis** interface

9.2.3.1 Completing QC Analysis in the QC Analysis Interface



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.



WARNING

The sample probe is sharp and potentially biohazardous. Exercise caution to avoid contact with the probe when working around it.

The sample may spill from the unclosed collection tubes and cause biohazard. Exercise caution to the unclosed collection tubes.

Collection tubes broken may cause personal injury and/or biohazard. Be sure to



place the collection tubes in the right adapter before running, otherwise, the collection tubes may be broken and cause biohazard.

Keep your clothes, hairs and hands away from the moving parts to avoid injury.

Reagents can be irritating to the eyes, skin, and mucosa. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them in the laboratory.

If the reagent accidentally comes in contact with your skin, wash it off immediately with plenty of water and see a doctor if necessary. Do the same if you accidentally get any of the reagent in your eyes.

CAUTION



Running quality controls in presence of errors may lead to incorrect analysis results. If you see the error alarms when running the quality controls, please stop and resume the analysis until the errors are removed.

Do not re-use such disposable products as collection tubes, test tubes, capillary tubes, etc.

Sample clump may lead to incorrect analysis results. Check if clump exists before running the controls; if it does, handle it as per the related laboratory procedures.

You should only use the Erba Lachema-specified controls and reagents. Store and use the controls and reagents as instructed by instructions for use of the controls and reagents. Using other controls may lead to incorrect QC results.

NOTE

Before being used for analysis shake well the controls that have been settled for a while. Be sure to use the Erba Lachema -specified disposable products including vacutainer blood collection tube, vacutainer blood collection tubes with anticoagulant and capillary tubes etc.

After completing the QC settings, users can perform the QC analysis in the **QC Analysis** interface. Detailed steps are shown below:

1. Click **QC** to access the **QC** interface.
2. Click **QC Analysis** and enter the QC analysis interface as shown in Figure 9-4.

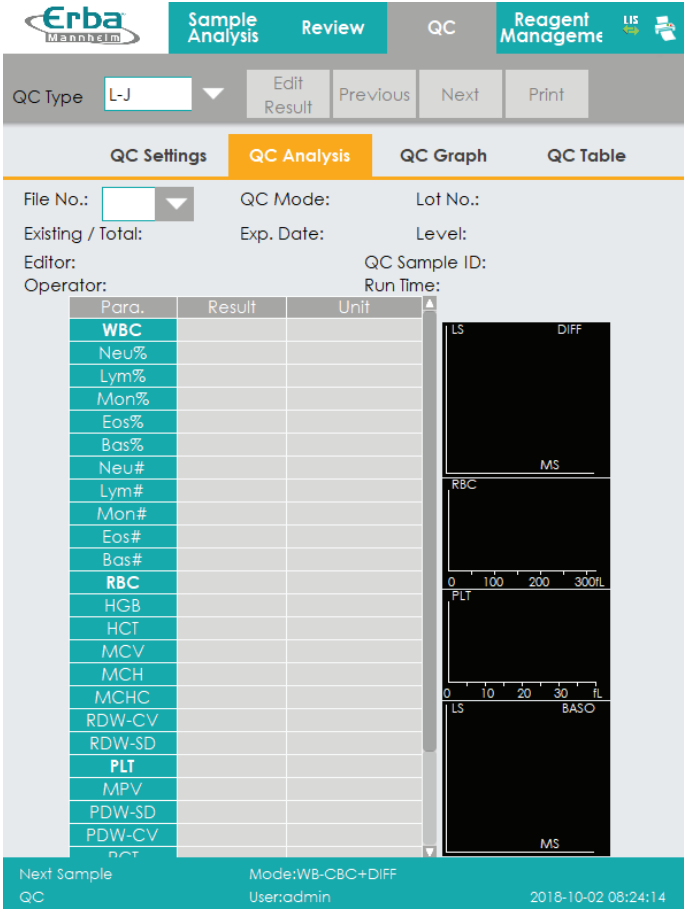


Figure 9-4 QC Analysis

- 3. Select the QC file No. to be run.
The screen will display the corresponding information and QC parameters.
- 4. Be sure that the level of the control to be run is the same with the current QC file, and the control to be run is not expired.
- 5. Prepare the controls according to the set control mode and control instructions.
Predilute the controls with reference to **6.5 Sample Collection and Handling** and get diluted QC samples if the QC mode is **Predilute-DIFF**.

NOTE Be sure to evaluate predilute stability based on your laboratory’s sample population and sample collection techniques or methods.

- 6. Shake the prepared control as shown in Figure 9-5 to mix it well.

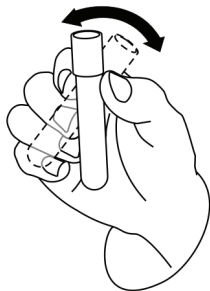


Figure 9-5 Mixing the Controls

7. In the ready for counting state (namely, the indicator light of the main unit is green), place the controls under the sample probe where the probe can aspirate the well-mixed controls.
8. Press the aspirate key and start running the controls.
9. Upon the completion of the aspiration, you'll hear a beep and you can remove the controls.
10. When the running of QC analysis is complete, the QC results will be displayed in the current screen (as shown in Figure 9-6) and saved in the QC file automatically.

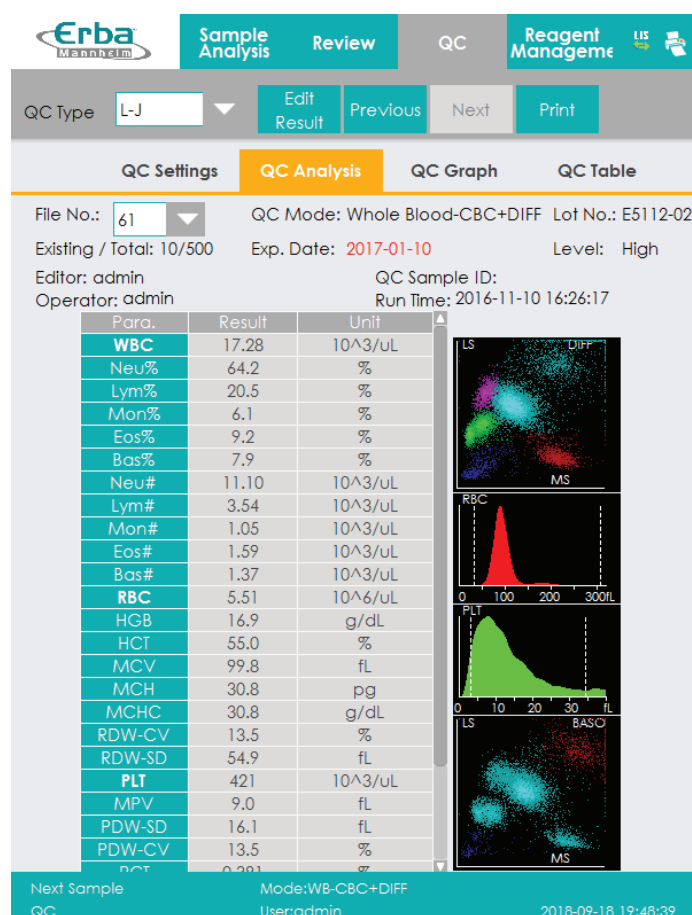


Figure 9-6 QC Analysis Results

11. Perform the above procedures to continue running the controls if necessary.

If the QC file is outdated, its valid period will be displayed in red.

NOTE “↑” or “↓” alarm symbol will be displayed next to the results with deviations exceeding the set limits.

9.2.3.2 Completing QC Analysis in the Sample Analysis Interface



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.

**WARNING**

The sample probe is sharp and potentially biohazardous. Exercise caution to avoid contact with the probe when working around it.

The sample may spill from the unclosed collection tubes and cause biohazard. Exercise caution to the unclosed collection tubes.

Collection tubes broken may cause personal injury and/or biohazard. Be sure to place the collection tubes in the right adapter before running, otherwise, the collection tubes may be broken and cause biohazard.

Keep your clothes, hairs and hands away from the moving parts to avoid injury.

Reagents can be irritating to the eyes, skin, and mucosa. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them in the laboratory.

If the reagent accidentally comes in contact with your skin, wash it off immediately with plenty of water and see a doctor if necessary. Do the same if you accidentally get any of the reagent in your eyes.

**CAUTION**

Running quality controls in presence of errors may lead to incorrect analysis results. If you see the error alarms when running the quality controls, please stop and resume the analysis until the errors are removed.

Do not re-use such disposable products as collection tubes, test tubes, capillary tubes, etc.

Sample clump may lead to incorrect analysis results. Check if clump exists before running the controls; if it does, handle it as per the related laboratory procedures.

NOTE

You should only use the Erba Lachema-specified controls and reagents. Store and use the controls and reagents as instructed by instructions for use of the controls and reagents. Using other controls may lead to incorrect QC results.

Before being used for analysis shake well the controls that have been settled for a while. Be sure to use the Erba Lachema -specified disposable products including vacutainer blood collection tube, vacutainer blood collection tubes with anticoagulant and capillary tubes etc.

After completing the QC settings, you can place the controls among the daily samples and perform analysis together in the **Sample Analysis** interface. After the analysis is completed, the system will store the results to the QC file with the corresponding ID.

Specific steps for performing QC analysis in the Sample Analysis interface are as follows:

1. Prepare the controls according to the set control mode and control instructions.
Predilute the controls with reference to **6.5 Sample Collection and Handling** and get diluted QC samples if the QC mode is **Predilute**.
2. Click **Mode & ID** in the Sample Analysis screen.

The interface pops up a dialog box as in Figure 9-7.

Mode & ID

☒ Venous Whole Blood (VWB)
 ☐ CBC

☐ Capillary Whole Blood (CWB)
 ☒ CBC+DIFF

☐ Predilute (PD)

Sample ID

☐ Bidirectional LIS/HIS Communication

OK Cancel

Figure 9-7 Set up mode and ID of the sample

3. Enter the set QC Sample ID in the **Sample ID** edit box (other options can be ignored). Refer to **9.2.2.1 Entering QC Information** for the setting of the QC Sample ID.
4. Well mix the prepared controls.
5. In the ready for counting state (namely, the indicator light of the main unit is green), place the controls under the sample probe where the probe can aspirate the well-mixed controls.
6. Press the aspirate key and start running the controls.
7. Upon the completion of the aspiration, you'll hear a beep and you can remove the controls. When the running of the controls is complete, the QC results will be saved in the QC file automatically.
8. Perform the above procedures to continue running the controls if necessary.

If the QC file is outdated, its valid period will be displayed in red.

NOTE “↑” or “↓” alarm symbol will be displayed next to the results with deviations exceeding the set limits.

9.2.3.3 Edit Result

Clicking **Edit** will allow you to edit the QC analysis result after the QC analysis is performed. See Figure 9-8.

Edit Result

WBC	17.28	10^3/uL	Bas#	1.37	10^3/uL	RDW-SD	54.9	fL
Neu%	64.2	%	NRBC#	0.420	10^9/L	PLT	421	10^3/uL
Lym%	20.5	%	NRBC%	8.70	%	MPV	9.0	fL
Mon%	6.1	%	RBC	5.51	10^6/uL	PDW-SD	16.1	fL
Eos%	9.2	%	HGB	16.9	g/dL	PDW-CV	13.5	%
Bas%	7.9	%	HCT	55.0	%	PCT	0.381	%
Neu#	11.10	10^3/uL	MCV	99.8	fL	P-LCR	30.8	%
Lym#	3.54	10^3/uL	MCH	30.8	pg	P-LCC	55	10^3/uL
Mon#	1.05	10^3/uL	MCHC	30.8	g/dL			
Eos#	1.59	10^3/uL	RDW-CV	13.5	%			

OK

Cancel

Figure 9-8 Editing QC Results

The edited data will be marked with an **E** by the result.

9.2.4 QC Result Review

After running controls, you can review the QC results in the following two forms:

- QC Graph
- QC Table

9.2.4.1 Graph



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.

You can review the result of L-J QC graph as per the following steps.

1. Click **QC** to access the QC interface.
2. Click **QC Graph** to enter the interface as shown in Figure 9-9.

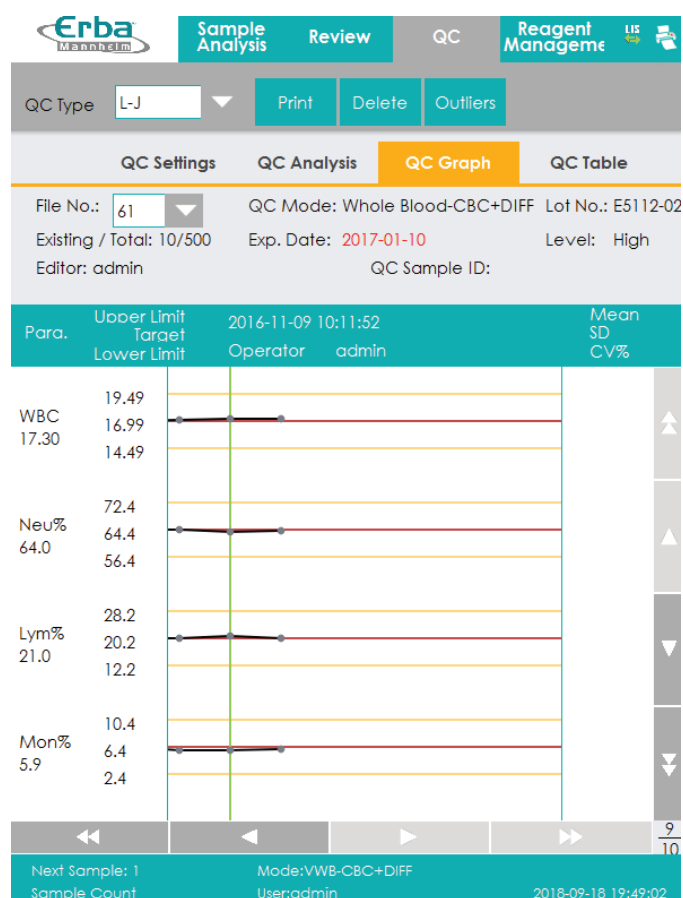


Figure 9-9 L-J QC Graph Interface

3. Select the QC **File No.** you want to review.

The screen will display the corresponding information and the graph.

4. Click the buttons at the right side of the QC graph, then you can browse QC graphs of different parameters; click the buttons at the bottom of the QC graph, then you can browse all QC results. Introduction to the Graph Interface in Figure 9-10.

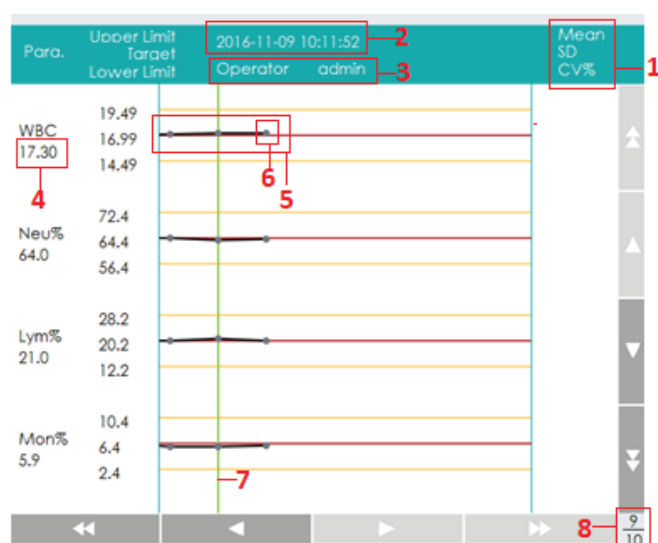


Figure 9-10 L-J QC Graph Interface

Interface Description:

1. The Mean, SD and CV% of all the QC results of each parameter in the current graph.
2. The saving date and time of the QC points located on the green line.
3. The operator who run the QC analysis and obtained the QC points located on the green line.
4. The QC results of the parameters that correspond to the QC points located on the green line.
5. The QC points in each graph are displayed from left to right according to the sequence from the earliest to the latest. The QC points are connected by a line to illustrate the distribution trend.
6. The QC point corresponds to each QC result. Only the selected QC point displays its value under the parameter. The black QC point indicates the value is within the limit; the red QC point indicates the value is out of the limit.
7. When you clicking a QC point in the graph, the QC points of other parameters saved together with this one will be marked by a green line.
8. The relative position of the QC point located on the grey line and the total QC points saved currently.

NOTE The outliers are excluded from the calculation of Mean, SD and CV%.

Calculate Preset Values/Save Preset Values

If the QC file of the QC graph has 3 groups or more of valid data, the administrator can take the followings steps to calculate and save the preset values of the QC parameters.

NOTE If the valid data (without “*”) in the QC file are less than 3 groups, the default value cannot be calculated.

1. Click **Cal Preset Values**.
The screen will display two black lines for you to select the range for calculating the preset values.
2. Click and drag the two lines respectively to place them at the beginning and the ending of the range for calculating the preset values.
The Mean, SD and CV% (on the right of the graph) will change into the new results obtained by calculating the selected range.
3. Click **Save Preset Values** to save the current Mean, SD and CV% as the preset values for the corresponding level (high/normal/low).
Then, the two selecting lines will disappear and the Mean, SD and CV% will return to the calculated results of all QC results.

- **Delete**

The administrator can delete the QC results by the following steps:

- Delete a single QC result
 1. Move the green line to the desired QC result, and click **Delete**.
 2. Select **Current Data** in the pop-up dialog box.
 3. Click **OK**.
- Deleting all the QC results in the current QC file
 1. Click **Delete**.
 2. Select **All Data** in the pop-up dialog box.
 3. Click **OK**.

- Entering the Reasons for the **Outliers**

Do as follows to enter the reasons for the outliers:

1. Move the green line to the desired QC point, and then click Outliers.

The pop-up window displays the QC results, reference values and deviation limits of all parameters corresponding to the green line as shown in Figure 9-11 .

	HCT	MCV	MCH	MCHC	RDW-CV
Target	53.7				
Limits (#)	2.4				
Outliers Data	57.1				

Cause of Outliers

☐ Reagent Expired ☐ Control Ineffective ☐ Control Expired
☐ Reagent Contamina ☐ Control not Mixed W
☐ Others

OK Cancel

Figure 9-11 Enter Cause of Outliers

The QC results exceeding the limit will be displayed in red.

2. You can select the reason from the given ones or manually enter the reasons (up to 200 characters) into the textbox after selecting **Others**.
3. Click **OK** to save the reasons for the outliers and exit.

NOTE

If you enter the reason for the group of QC points whose results are actually within the limits, then their corresponding QC data both in the QC Graph and QC Table will be displayed in red. And the data will return in black if you cancel the reason and then save the changes.

- **Print**

You can have the QC data of the current page or all QC data in the QC file printed by clicking the **Print** button.

NOTE

The printed QC graph will not show any parameters which are not involved in the quality control.

9.2.4.2 Table



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.

1. Click QC to access the QC interface.
2. Click QC Table to access the interface as shown in Figure 9-12.

Erba Mannheim

Sample Analysis Review QC Reagent Management LIS

QC Type: L-J Edit Result Delete Print Comm. Export

QC Settings QC Analysis QC Graph **QC Table**

File No.: 61 QC Mode: Whole Blood-CBC+DIFF Lot No.: E5112-02
 Existing / Total: 10/500 Exp. Date: 2017-01-10 Level: High
 Editor: admin QC Sample ID:

	Target	Limits (#)	10	9	8	
Date	/	/	2016-11-10	2016-11-09	2016-11-	▲
Time	/	/	16:26:17	10:11:52	09:16:0	
WBC	16.99	2.50	17.28	17.30	17.19	
Neu%	64.4	8.0	64.2	64.0	64.5	▲
Lym%	20.2	8.0	20.5	21.0	20.3	
Mon%	6.4	4.0	6.1	5.9	6.0	
Eos%	9.0	8.0	9.2	9.1	9.2	▼
Bas%	7.4	10.0	7.9	7.4	7.2	
Neu#	10.95	1.40	11.10	11.08	11.09	
Lym#	3.43	1.40	3.54	3.63	3.49	▼
Mon#	1.08	0.70	1.05	1.02	1.03	

Next Sample: 1 Sample Count Mode: VWB-CBC+DIFF User: admin 2018-09-18 19:50:03

Figure 9-12 L-J QC Graph Interface

3. Select the QC **File No.** you want to review.

- **Editing**

Choose a row in the QC table and click **Edit**, then you can edit the selected QC data. The edited data will be marked with an **E**.

- **Delete**

With the administrator-level access, users can delete the selected QC data, QC data on the current page and all QC data.

- Delete a selected QC result

1. Click the column containing the desired QC result, and then click **Delete**.
2. Select **Current Data** in the pop-up dialog box.
3. Click **OK**.

- Delete QC data on the current page

1. Click **Delete** on the page which contains the QC results expected to be deleted.
2. Select **Current Page Data** in the pop-up dialog box.
3. Click **OK**.

- Delete all QC results

1. Click **Delete**.
2. Select **All Data** in the pop-up dialog box.
3. Click **OK**.
4. Click **Yes** to delete all the QC results in the current QC file.

- **Print**

You can print all the QC data or the data within the specified date range of the selected QC file.

Detailed steps are shown below:

1. Select a QC File No. to be printed.
2. Click **Print**.
3. Select the QC data to be printed: all data or specified data.

When **All Data** is selected, all the QC data of the table will be printed.

When **Specified Data** is selected, and the date range is set in the date controls, the QC data within the specified date range will be printed.

4. Click **OK** to print the data.

- **Communication**

The current QC data, the data within the specified date range or all the QC data can be transmitted to LIS/HIS.

- Communication for current data
 1. Select a QC record to be transmitted, and click **Comm**.
 2. The default option is **Current Data**.
 3. Click **OK**.
 4. Click **OK** to close the message box.
- Communication for all data
 1. Click **Comm..**
 2. Select **All Data**.
 3. Click **OK**.
 4. Click **OK** to close the message box.
- Transmitting the data within specified date range
 1. Click **Comm..**
 2. Select **Records of the Specified Dates**, and set the starting and ending dates for the data to be communicated.
 3. Click **OK**.
 4. Click **OK** to close the message box.

- **Export**

If you wish to export the information and the result of the current QC file, do as follows:

1. Insert a USB flash disk in the USB interface on the analyser.
2. Click **Export**.
3. Select an export path for the data and enter the file name.

The file will be exported to the root directory of the USB flash disk (**/udisk/sda1**) and named in the format of **SampleInfo_YYYYMMdd_hhmmss.csv**. Among which, **YYYYMMdd_hhmmss** means data export year, month, date, hour, minute, and second.

4. Click **Save**. When the export is finished, a message box will pop up.
5. Click **OK** to close the message box.

9.3 X-B Quality Control

9.3.1 QC Principle

The X-B analysis is a weighted moving average analysis that uses values obtained from patient samples. It uses the 3 red cell indices, MCV, MCH and MCHC to indicate the haematology instrument performance. This is QC with no controls, which is a method of performance control like QC with controls. Both methods reflect the analysis performance of the analyser from different perspective. Thus, one method should not be replaced with the other.

It is recommended the X-B analysis be activated when the sample volume of your laboratory is greater than 100 samples per day. Effective use of X-B requires randomization of samples and a normal cross section of patients to prevent skewing of indices. A reference range is established by the given reference values as well as lower and upper limits for the purpose of observing the variation of QC results within the reference range.

The analyser performs X-B QC for three parameters, MCV, MCH, and MCHC. Twenty to two hundred samples can be grouped together for X-B numerical analysis. The samples are derived from the results of normal analyser counting, with no distinction of whole-blood or predilute mode. The analyser can save maximum 500 X-B QC results. When the saved QC results have reached the maximum number, the newest result will overwrite the oldest.

9.3.2 QC Settings



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.

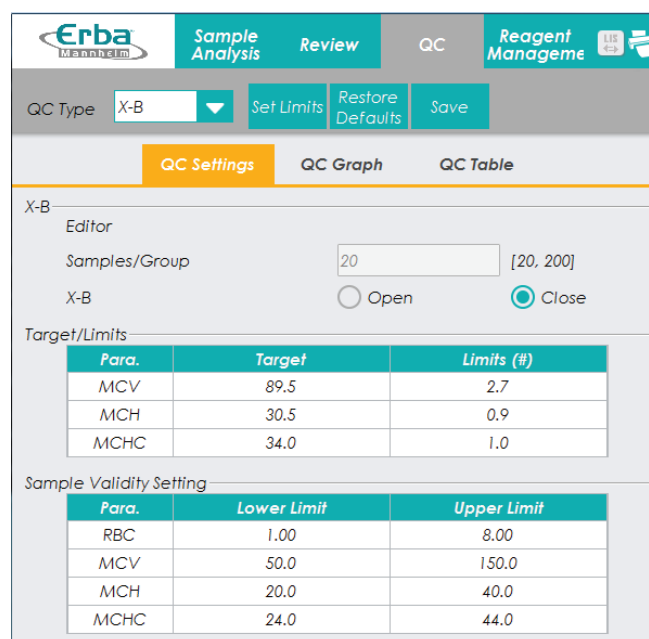
NOTE	Only users with administrator-level access can edit the L-J settings.
-------------	---

Perform the QC Settings before running the controls. You can complete the QC settings by entering the QC information.

9.3.2.1 Entering QC Information

You can complete the X-B QC settings as per the following steps:

1. Click **QC** to access the **QC** interface.
2. Select **X-B** from the dropdown list of the **QC Type**.
3. Click **QC Settings**. You'll enter the **QC Settings** interface as shown in Figure 9-13.



Erba H560

Sample Analysis Review QC Reagent Management

QC Type: X-B Set Limits Restore Defaults Save

QC Settings QC Graph QC Table

X-B Editor

Samples/Group: 20 [20, 200]

X-B: ☐ Open ☒ Close

Target/Limits

Para.	Target	Limits (#)
MCV	89.5	2.7
MCH	30.5	0.9
MCHC	34.0	1.0

Sample Validity Setting

Para.	Lower Limit	Upper Limit
RBC	1.00	8.00
MCV	50.0	150.0
MCH	20.0	40.0
MCHC	24.0	44.0

Figure 9-13 X-B QC Settings

- In the **Samples/Group** edit box, enter the amount of samples to be included in calculating for an X-B QC point. The range is between 20 and 200 and the recommended value is 20.

NOTE Once the **Samples/Group** is changed, the number of valid sample results will be re-calculated. For example, if 20 valid samples are needed for the X-B QC calculation, when you change the value of **Samples/Group** after 10 group of valid sample results have been acquired, these 10 group of results will be discarded, and only valid sample results generated afterwards will be used in the QC calculation.

- Click the **Open** button of **X-B** to open the X-B quality control. The samples results will be included to calculate the X-B.
- Enter the targets and limits for the QC parameters.

NOTE All the targets and limits for the QC parameters must be entered. When first use, the default setting will provide the Initial values for the targets and limits of the three QC parameters. If the QC data have existed in the QC file, you are not allowed to edit the target and limits.

You can set the display form of the limits or the calculation method of the limits among the pre-set values. See section **9.3.2.2 Setting Limits**.

- Set the valid upper and lower limits for the QC parameter in **Sample Validity Setting** field. Setting sample validity is to set the valid range of four QC parameters, RBC, MCV, MCH and MCHC. To be incorporated into X-B QC calculation, the sample results should satisfy the validity ranges of all these four parameters.
- Click the **Save** button to save all the settings of the QC. If the entered value exceeds the acceptable range or the upper limit is lower than the lower limit, a reminder message will pop up and you will be prompted to re-enter the correct data and save the entry again.

9.3.2.2 Setting Limits

You can take the following steps to adjust the display format of the limits and the calculation method of the pre-set limits.

1. Click **Set Limits**. The interface pops up a dialog box as in Figure 9-14 .

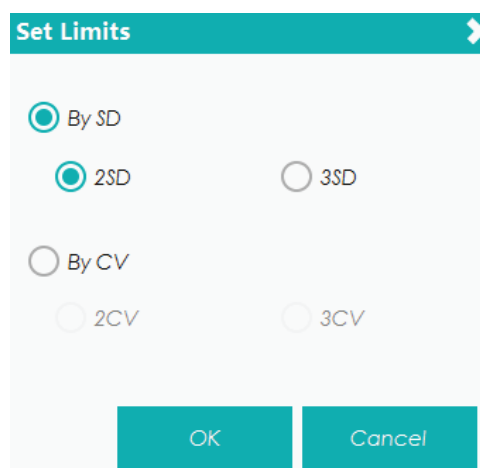


Figure 9-14 Set limits for X-B QC

2. Select **By SD** or **By CV** according to the actual needs.
 - If **By SD** is selected, the limits will be displayed in form of absolute value.
Click **2SD** or **3SD** to select either double or triple standard deviation to be the limits.
 - If **By CV** is selected, the limits will be displayed in form of percentage.
Click **2CV** or **3CV** to select either double or triple coefficient of variation to be the limits.
3. Click **OK** to save all the settings for the limits.

9.3.2.3 Restoring Defaults

In QC setting, click **Restores Defaults** button to restore the parameter reference values, limits and sample validity to the default settings.

NOTE If the QC data have existed in the QC file, you are not allowed to restore the parameters. Clicking Restores Defaults can only store the default settings of Target, Limits and Sample Validity Setting, while Samples/Group, X-B QC switch and limit settings cannot be restored.

9.3.3 Quality Control Analysis



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.

After the QC settings, the analyser will automatically start the X-B QC analysis.

After every 20~200 results (determined by the setting) are obtained, the system will perform the X-B calculation once automatically. You can review the result in X-B graph or X-B table.

In X-B QC, sample results conforming to any of the following conditions will be considered as invalid and cannot be used in the QC calculation.

Sample results exceeding the linearity range

- Background results
- Sample results not conforming to the **Sample Validity Setting**
- QC data for other QC programs (such as L-J QC)
- Calibration data
- Results generated while there are errors which could affect the accuracy of the results (insufficient aspiration volume or clogging for example).

9.3.4 QC Result Review

After running controls, you can review the QC results in the following two forms:

- QC Graph
- QC Table

9.3.4.1 QC Graph



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.

Access the X-B QC Graph interface by taking the following steps:

1. Click **QC** to access the **QC** interface.
2. Select **X-B** from the dropdown list of the **QC Type**.
3. Click **Graph**.

The X-B QC Graph interface will be displayed. See Figure 9-15.

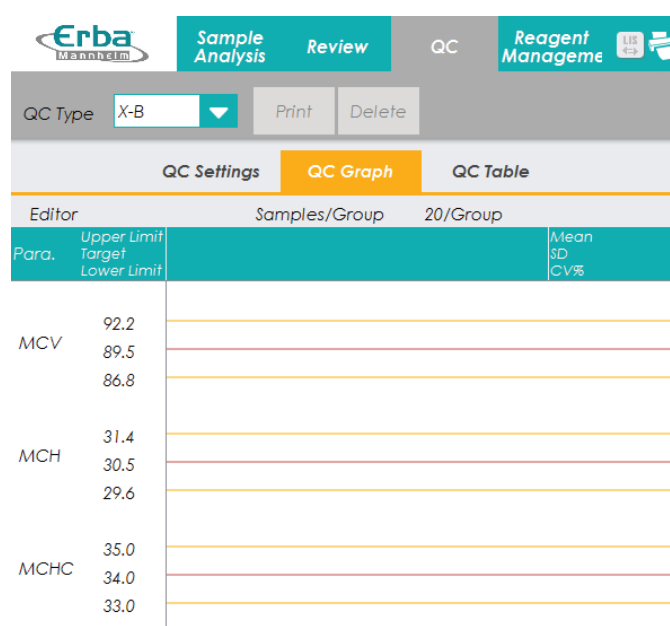


Figure 9-15 QC Graph

4. You can also drag the scroll bar down to the graph horizontally to browse all the QC results. Introduction to the Graph Interface in Figure 9-16.

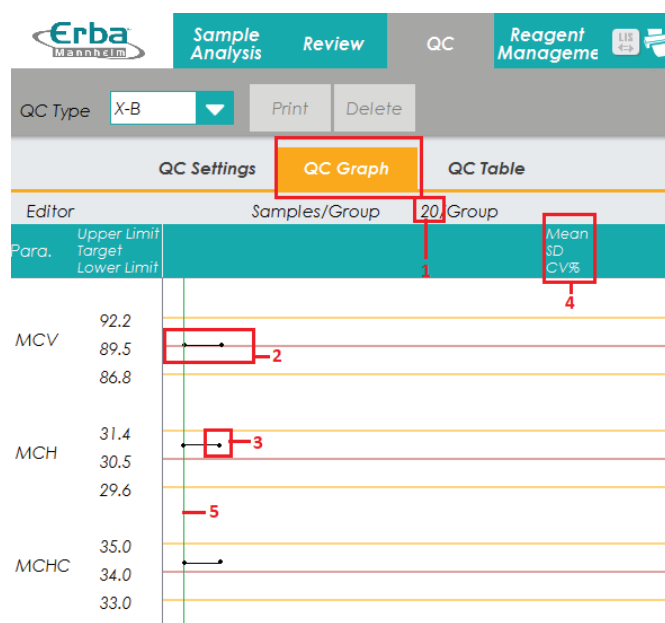


Figure 9-16 X-B QC Graph

1. The amount of samples included in calculating for each QC point.
2. The QC points in each graph are displayed from left to right according to the sequence from the earliest to the latest. The QC points are connected by a line to illustrate the distribution trend.
3. The QC point corresponds to each QC result. Only the selected QC point displays its value under the parameter. The black QC point indicates the value is within the limit; the red QC point indicates the value is out of the limit.
4. The Mean, SD and CV% of all the QC results of each parameter in the current graph.
5. When you clicking a QC point in the graph, the QC points of other parameters saved together with this one will be marked by a green line.

- **Delete**

The administrator can delete the QC results by the following steps:

1. Delete a single QC result
 2. Move the green line to the desired QC result, and click **Delete**.
 3. Select **Current Data** in the pop-up dialog box.
 4. Click **OK**.
- Deleting all the QC results in the current QC file
 1. Click **Delete**
 2. Select **All Data** in the pop-up dialog box
 3. Click **OK**.

- **Print**

Click the **Print** button to print the QC graph.

9.3.4.2 Table



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.

Access the X-B QC Graph interface by taking the following steps:

1. Click **QC** to access the **QC** interface.
2. Select **X-B** from the dropdown list of the **QC Type**.
3. Click **QC Table**.

The X-B QC table interface will be displayed. See Figure 9-17.

Editor	Target	Limits (#)	20/Group
Date	/	/	
Time	/	/	
MCV	89.5	2.7	
MCH	30.5	0.9	
MCHC	34.0	1.0	

Figure 9-17 QC Table

- **Delete**

With the administrator-level access, users can delete the selected QC data, QC data on the current page and all QC data.

- Delete a selected QC result
 1. Click the column containing the desired QC result, and then click **Delete**.
 2. Select **Current Data** in the pop-up dialog box.
 3. Click **OK**.
- Delete QC data on the current page
 1. Click **Delete** on the page which contains the QC results expected to be deleted.
 2. Select **Current Page Data** in the pop-up dialog box.
 3. Click **OK**.

- Delete all QC results
 1. Click **Delete**.
 2. Select **All Data** in the pop-up dialog box.
 3. Click **OK**.
 4. Click **Yes** to delete all the QC results in the current QC file.

- **Print**

You can print all the QC data or the data within the specified date range of the selected QC file. Detailed steps are shown below:

1. Select a QC file No. to be printed.
 2. Click **Print**.
 3. Select the QC data to be printed: all data or specified data.
 - When **All Date** is selected, all the QC data of the table will be printed.
 - When **Specified Data** is selected, and the date range is set in the date controls, the QC data within the specified date range will be printed.
 4. Click **OK** to print the data.
- **Communication**

The current QC data, the data within the specified date range or all the QC data can be transmitted to LIS/HIS.

- Communication for current data
 1. Select a QC record to be transmitted, and click **Comm**.
 2. The default option is **Current Data**.
 3. Click **OK**.
 4. Click **OK** to close the message box.
- Communication for all data
 1. Click **Comm**.
 2. Select **All Data**.
 3. Click **OK**. After the data is transmitted to LIS/HIS, a message box will pop up.
 4. Click **OK** to close the message box.
- Transmitting the data within specified date range
 1. Click **Comm**.
 2. Select **Records of the Specified Dates**, and set the starting and ending dates for the data to be communicated.
 3. Click **OK**. After the data is transmitted to LIS/HIS, a message box will pop up.
 4. Click **OK** to close the message box.

- **Export**

If you wish to export the information and the result of the current QC file, do as follows:

1. Insert a USB flash disk in the USB interface on the analyser.
2. Click **Export**.
3. Select an export path for the data and enter the file name.

The file will be exported to the root directory of the USB flash disk (**/udisk/sda1**) and named in the format of **SampleInfo_yyyyMMdd_hhmmss.csv**. Among which, **yyyymmdd_hhmmss** means data export year, month, date, hour, minute, and second.

4. Click **Save**. When the export is finished, a message box will pop up.
5. Click **OK** to close the message box.

10 Calibration

10.1 Introduction

Calibration is a procedure to standardize the analyser by determining its deviation, if any, from calibration references and to apply any necessary correction factors. To get accurate blood analysis results, perform calibration of the analyser following the procedures given in this chapter when it's needed.

Calibration procedures can only be performed by users with the administrator-level access. The login users with the access level of general users cannot perform the calibration procedures but only browse the calibration coefficients.

NOTE

You should only use the Erba Lachema-specified calibrators and reagents. Store and use the calibrator and reagents following the instructions for use of the calibrations and reagents.

The analyser identifies a sample as a calibration sample only if the analysis is started from the Cal interface.

The calculation of repeatability is included in the calibration procedure.

10.2 When to Calibrate

This analyser is calibrated at the factory just before shipment. It is electronically stable and does not require frequent recalibration if you operate and maintain it as instructed by this manual. You need to recalibrate this analyser if:

It is the first time this analyser has been used (usually done by an Erba Lachema-authorized representative when installing the analyser).

An analytical component has been changed.

The quality control results indicate that there may be a problem.

The operating environment (such as the temperature) has changed significantly.

All of the measured parameters must be calibrated before readings of this analyser can be used as valid analysis results.

NOTE

For laboratories conducting routine tests, the calibration should be applied at least once every six months.

10.3 How to Calibrate

There are three calibration programs available on this analyser: manual calibration, auto calibration using calibrators and auto calibration using fresh blood samples.

All or part of the parameters of WBC, RBC, HGB, MCV and PLT can be calibrated by the calibration procedure.

10.3.1 Preparation



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.



WARNING

The sample probe tip is sharp and may contain biohazardous materials. Exercise caution to avoid contact with the probe when working around it.

Reagents can be irritating to the eyes, skin, and mucosa. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them in the laboratory.

If the reagent accidentally comes in contact with your skin, wash it off immediately with plenty of water and see a doctor if necessary. Do the same if you accidentally get any of the reagent in your eyes.

Keep your clothes, hairs and hands away from the moving parts to avoid injury.

Be sure to dispose of reagents, waste, samples, consumables, etc. according to local legislations and regulations.



CAUTION

Do not re-use such disposable products as collection tubes, test tubes, capillary tubes, etc.

NOTE

You should only use the Erba Lachema-specified controls and reagents. Store and use the controls and reagents by following the instructions for use of the controls and reagents.

Be sure to use the Erba Lachema -specified disposable products including vacutainer blood collection tube, vacutainer blood collection tubes with anticoagulant and capillary tubes etc.

Carry out the calibration only when the background range, repeatability and carryover are within the specified limits given in the manual, otherwise, the problems must be identified and solved before you determine if calibration is needed. If you cannot solve the problems, please contact Erba Lachema Service Department.

1. Check and make sure enough reagents have been prepared for the calibration. You need to start over the calibration if the reagents run out during the process.

2. Do the background check.

If the analyser alarms are activated for abnormal background results, see **13 Troubleshooting** for solutions. (Refer to **15.4.2. Normal Background** for background range.)

3. Do the repeatability test.

Run the median controls in whole blood-CBC+DIFF mode consecutively for 11 times.

Take and view repeatability of the counting results from the 2nd run through the 11th run in the **Review** interface and make sure they are within the range specified in **15.4.4 Repeatability**.

4. Do the carryover test.

Run the corresponding diluent for 3 times immediately after running the high-level controls for 3 times. Calculate the carryover by the following formulae:

$$\text{Carryover (\%)} = \frac{\text{First low-value sample result} - \text{Third low-level sample result}}{\text{Third high-value sample result} - \text{Third low-level sample result}} \times 100\%$$

The calculated carryovers shall meet the requirements in **15.4.5 Carryover**.

5. It is recommended that you create a log table for your analyser. The suggested items that you may want to include in the log table are: calibration date, supplier of calibrator, lot number, expected results and limits, and result of background check.

10.3.2 Manual Calibration

Complete the manual calibration as per the following procedure:

1. Click **Cal** in the menu page to access the calibration interface.
2. Click **Manual** to access the manual calibration interface. See Figure 10-1. The calibration coefficients of whole blood mode and predilute mode are displayed on the Manual interface.

Manual		
Whole Blood		
Para.	Cal. Coefficient (%)	Cal. Date
WBC	100.00	2018/08/29
RBC	100.00	2018/08/29
HGB	100.00	2018/08/29
MCV	100.00	2018/08/29
PLT	100.00	2018/08/29
Predilute		
Para.	Cal. Coefficient (%)	Cal. Date
WBC	100.00	2018/08/29
RBC	100.00	2018/08/29
HGB	100.00	2018/08/29
MCV	100.00	2018/08/29
PLT	100.00	2018/08/29
Save Print Exit		

Figure 10-1 Manual Calibration

3. Check the calibration coefficient and calculate the new coefficient using the following equation.

$$\text{New calibration factor} = \frac{\text{Current calibration factor} \times \text{Reference value}}{\text{Mean}}$$

For example, the WBC reference value of a calibrator is 8.3, and the current calibration coefficient of the whole blood mode is 99.00%.

Run the calibrator in whole blood mode for 11 consecutive times and calculate the WBC results of the 2nd to 11th runs (n=10): The obtained CV is 1.1% and the Mean is 8.22, which meet the requirements.

The new calibration coefficient is obtained:

$$\text{New calibration factor} = \frac{99.00\% \times 8.3}{8.22} = 99.96\%$$

The calculated calibration coefficients shall be between 75%~125%. In case of an invalid calibration coefficient, try to find out the reason (e.g. calibration material not thoroughly mixed, incorrect operation, etc.). Then recalibrate the analyser and recalculate the calibration coefficients.

4. Enter the new calibration coefficients into the factor cell of the parameter that requires calibration.

NOTE The entered calibration coefficients shall be between 75.0%~125.0% (calculation results rounded to two decimal places).

5. Click **Save**.

On the screen, the calibration coefficient is refreshed to be the new one and the calibration date is refreshed to be the current system date.

- If the new calibration coefficients are invalid, the message box will pop up. Click **OK** to close the message box and enter a valid factor.
6. (Optional) Click **Print** to print the current calibration coefficient.
 7. Click **Exit** to close the Manual interface.

10.3.3 Auto Calibration Using Calibrators



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.

NOTE

Only Erba Lachema-specified calibrators shall be used. Erba Lachema will not be responsible for any erroneous result caused by using other calibrators. See the instructions for use of the calibrators for the lot No., Exp. Date and the target.

Complete the calibration with calibrators as per the following procedure:

1. Click **Cal** in the menu page to access the calibration interface.
2. Click **Calibrator**. The **Calibrator** interface pops up as shown in Figure 10-2.

Calibrator

Lot No. Exp. Date 2018-09-19 Mode ☒ Whole Blood

Para.	WBC	RBC	HGB	MCV	PLT
Target					
☐ 1					
☐ 2					
☐ 3					
☐ 4					
☐ 5					
☐ 6					
☐ 7					
☐ 8					
☐ 9					
☐ 10					
Mean					
CV(%)					
New Calibration Coefficient (%)					
Original Calibration Coefficient (%)	100.00	100.00	100.00	100.00	100.00

Clear Print Save

Figure 10-2 Auto Calibration Using Calibrators

- Enter the lot No. of the calibrator into the **Lot No.** box.
- Click the **Exp. Date** box, and then edit the **Exp. Date**.

The **Exp. Date** can be no earlier than the current system date.

NOTE The entered **Exp. Date** should be either the **Exp. Date** printed on the labelling or the open-container expiration date, whichever is earlier. The open-container expiration date is calculated as follows: the date on which the container is opened + the open-container stability days.

- Input the target values of the parameters in the corresponding cell of the Target.
- Prepare the calibrators following their instructions for use and place the calibrators under the sampling probe.
- Press the aspirate key to start the calibration counting.
After every calibration run, the progress bar will close automatically and the analyser will have different responses according to different analysis results.
The valid results within the linearity range will be displayed directly.
If the calibration counting data of any parameter in the current counting are out of the display range or linearity range of the parameter, a message box will pop up on the screen prompting that the calibration data is invalid.
- Click **OK** to close the message box and delete the data from the table without saving.
If any of the parameter's value in the calibration counting differs from the Target value by more than 50%, the system will prompt you with a message box asking if the calibration counting results should be kept. To keep the results, click **Yes**; to remove the results, click **No**.

NOTE

After the valid calibration result is obtained, the parameters with corresponding checkboxes ticked off will be involved in the calculation of the calibration coefficients by default.

If you switch to other interfaces before the new calibration coefficients are obtained, the system will discard the current calibration data and keep the original calibration coefficients.

9. To get 10 valid counting results, repeat steps Prepare the calibrators following their instructions for use and place the calibrators under the sampling probe. Press the aspirate key to start the calibration counting. ten times.

The analyser will, by default, calculate the Mean, CV% and the new calibration coefficients based on all the ticked-off calibration data according to the formulae.

10. Select at least 5 groups of data for the calculation of the calibration coefficients.

When the amount of the valid calibration data in the list reaches 10, a message box of **Calibrator calibration done!** will pop up. Click **OK** to close the message box.

If the calibration coefficients are invalid, click **Yes** to close the dialog box. Then click **Clear** to delete the current data and redo the calibration.

NOTE

The out-of-range CV% does not influence the display of the calibration coefficients.

11. Click **Save**.

If the calculated calibration coefficients of all parameter are within the range of 75%~125% and the CV% of all parameter are also within the repeatability, then a dialog box prompting the successful calibration setting will pop up. Click **OK** to close the message box.

If the obtained calibration coefficient of any parameter is not within the range of 75%~125% or the CV% of any calibrated parameter does not meet the repeatability, the calibration coefficient will not be saved and a dialog box indicating invalid new calibration coefficient will pop up. Click **Yes** to close the dialog box and repeat the calibration operations.

12. (Optional) Click **Print** to print the calibration results.

10.3.4 Auto Calibration Using Fresh Blood Samples



All the samples, controls, calibrators, reagents, wastes and areas in contact with them are potentially biohazardous. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them and the relevant areas in the laboratory.

Complete the calibration using fresh blood samples as per the following procedure:

1. Click **Cal** in the menu page to access the calibration interface.
2. Click **Fresh Blood**. The fresh blood sample calibration interface pop up, as shown in Figure 10-3.

Fresh Blood
✕

Para.	WBC	RBC	HGB	MCV	PLT
Target					
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
Mean					
CV(%)					
Calibration Coefficient 1 (%)					

Mode

☒ Whole Blood

Blood Sample

☒ Blood Sample 1

☐ Blood Sample 2

☐ Blood Sample 3

☐ Blood Sample 4

☐ Blood Sample 5

Calculate

Clear

Print

Figure 10-3 Auto Calibration Using Fresh Blood Samples

3. Prepare 3 to 5 normal fresh blood samples as instructed by **6.5 Sample Collection and Handling**.
4. Run each of the prepared samples on the reference instrument three times at least. Average the results for your reference values.

NOTE The reference instrument must be a properly running standard analyser so as to ensure the accuracy of the reference values.

5. Enter the reference values for the parameters to be calibrated in the corresponding **Target** textbox.
6. Place the blood sample under the sampling probe, press the aspirate key on the analyser to run the samples.
7. The system will calculate the values for WBC, RBC, HGB, MCV and PLT of the sample.
8. Repeat step Place the blood sample under the sampling probe, press the aspirate key on the analyser to run the samples for 10 times and calculate the counting results for sample No. 1 in the 10 runs.

The system will calculate the Mean, CV and Calibration coefficient for each parameter of the sample. If the obtained calibration coefficient for any sample is not within the valid range or CV% or any calibrated parameters does not meet the repeatability, a dialog box indicating invalid new calibration coefficient will pop up. Click **Clear** to delete the calibration data of the sample. Redo the calibration or redo the calibration with another sample meeting all criteria.

9. Refer to steps f. and g. and perform the counting operations for the remaining four blood samples. The system will calculate the Mean, CV and Calibration Coefficient for each parameter of the remaining 4 blood samples.
10. Click **Calculate**.
The system will calculate the average of the calibration coefficients, namely, the mean calibration coefficient (%), as the new calibration coefficient based on the five blood samples.
You can also check at least three accurate calibration coefficients and the system will re-calculate the mean calibration coefficient (%).

NOTE The mean calibration coefficient is invalid if its absolute value of deviation from the original calibration coefficient is greater than or equal to 5%.

11. Click **Save**.
If the mean calibration coefficient is within the valid range (the absolute value of deviation from the original calibration coefficient is less than 5%), you'll be prompted that the mean calibration coefficient is saved successfully.
If the mean calibration coefficient is not within the valid range (the absolute value of deviation from the original calibration coefficient is greater than or equal to 5%), you'll be prompted that the mean calibration coefficient is invalid.

NOTE CV% out of standard will not affect the display of calibration coefficient.

12. (Optional) Click **Print** to print the calibration results.

10.4 Verifying Calibration Coefficients

It is recommended that you take the following steps to verify the calibration coefficients:

1. Run the calibrator at least three times and check whether the means of the obtained results are within the expected ranges.
2. Run the low-, normal- and high-level controls each for three times at least, and check whether the means of the obtained results are within the expected ranges.
3. Run at least three fresh blood samples with known reference values, each for six times at least, and check whether the means of the obtained results are within the expected ranges.

10.5 Calibration History

Click **Cal > History** to enter the calibration history screen. You can view the calibration history list and the detailed calibration data.

- **Calibration History List**

The list shows the latest 100 calibration history records, including the following items:

Date: The operating system date when the calibration coefficient is saved.

Cal. Operator: The one who performs the calibration operations, such as the Admin.

Calibration Method: Including **Manual**, **Calibrator** and **Fresh Blood**.

Calibration Mode: The mode adopted for the calibration, including **Whole Blood** and **Predilute**.

Description: Supplementary description of the calibration information about the corresponding entries.

- **Detailed Calibration Data**

Select any row of record in the Calibration History List to view the detailed calibration data of that record.

If the calibration method of the selected record is Fresh Blood, you can click **Details...** next to each intermediate calibration record and view the detailed calibration data of each intermediate calibration record.

- **Print**

You can click the **Print** button to print the calibration record.

11 Reagent Management

Once the new reagent is connected to the analyser, you can set the reagent configurations, including validity period, residue volume and reagent barcode on the Reagent Management interface. Upon the completion of reagent configuration, you can perform the procedures for reagent replacement.

WARNING



Reagents can be irritating to the eyes, skin, and mucosa. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them in the laboratory.

If the reagent accidentally comes in contact with your skin, wash it off immediately with plenty of water and see a doctor if necessary. Do the same if you accidentally get any of the reagent in your eyes.

After long-distance transportation, the reagent must be allowed to settle for more than one day before use.

NOTE

When you have changed the diluents, cleansers or lyses, run a background check to see if the results meet the requirement.

11.1 Accessing the Interface

Click **Reagent Management** in the menu navigation area, to access the reagent management setting interface. See Figure 11-1.

Reagent Name	Erba H560 DII	Erba H560 Lyse1	Erba H560 Lyse2
Unopened expiry date			
Period After Opening			
Open-container Date			
Reagent Expiry Date			
Residue Volume			

Next Sample: 4 Mode: VWB-CBC+DIFF **Background abnormal.**
Sample Count User: admin 2018/12/11 10:31:50

Figure 11-1 Reagent Management

Refer to Table 11-1 for related parameter descriptions.

Table 11-1 Parameter Description for Reagent Management

Parameter	NOTE
Reagent Name	Name of the reagent.
Exp. Date	Exp. Date of the unopened reagent will be shown upon the completion of the reagent settings. Any reagent, regardless of its container being opened or not, should not be used beyond this date.
Open-container Date	The date on which the reagent container is opened. The default open-container date is the date on which the reagent settings are completed.
Period after opening (PAO)	The validity period (days) after the reagent container is opened. It will be shown upon the completion of the reagent settings.
Open-container Exp. Date	Expiration date of the opened reagent, and it will be shown upon the completion of the reagent settings.
Residue Volume	The current residue volume of the reagent, and it will be shown in ml upon the completion of the reagent settings. The unit is ml.

11.2 Setting Reagent Information

Once the new reagent is connected to the analyser, you should set the reagent configurations, including validity period, residue volume and reagent barcode on the **Reagent Management** interface. Upon the completion of reagent configuration, you can perform the procedures for reagent replacement.

Taking **Insufficient Erba H560 Lyse 1** as an example, this section introduces the setting procedures for the closed reagent:

1. When the **Insufficient Erba H560 Lyse 1** is prompted on the lower right of the screen, double click the message.
2. Select the error name in the popup dialog box, and click **Remove Error**. A dialog box pops up.
3. Attach the RF card attached to the reagent packing on the RF card reader in front part of the analyzer and keep it for 3-4 seconds.
4. The beeping of the card reader and a pop-up dialog box indicate the successful reagent settings.
5. Click **OK**.

The RF card is intended for single use only.

If the verification fails, try to attach the card by front side back on the RF reader.

If the verification fails again, please follow the system prompts and use a valid RF card for re-reading.

NOTE Once the reagent settings are successfully completed, the system prompt at the lower right corner of the screen will show that the reagent has not been replaced. To remove this error, click the error message and then click **Remove Error** in the pop-up dialog box. The analyser will complete the replacement of the reagent and remove the error.

11.3 Replacing Reagents

After completing the reagent settings, you should perform the reagent replacement operations. You can select to replace one type of reagent at a time or all reagents. The method is applied as follows:

1. Select a type of reagent to be replaced, and click **Replace**; or click **Replace All** to replace all the reagents.
2. After the replacement is completed, a message box will pop up on the screen. Click **OK** to close the message box.

NOTE When you have changed the reagents, run a background check to see if the results meet the requirement.

12 Service

12.1 Introduction

This analyser provides multiple maintenance functions for this purpose. This chapter introduces how to use the provided functions to maintain and troubleshoot your analyser. Preventive and corrective maintenance procedures are required to keep the analyser in a good operating condition.



All the analyser components and surfaces are potentially infectious, take proper protective measures for operation or maintenance.



CAUTION

Performing unauthorized maintenance procedures can damage your analyser. Do not perform any maintenance procedures that are not described in this chapter.

In case of problems not specified in this manual, contact Erba Lachema customer service department or your local agent for assistance.

Only Erba Lachema -supplied parts can be used for maintenance. For any question, contact Erba Lachema customer service department or your local agent.

Exercise caution to avoid contact with the sharp sample probe when performing maintenance.

12.2 Maintenance

The analyser provides multiple service functions helping users to perform daily maintenance.

12.2.1 Reagent Replacement



WARNING

Reagents can be irritating to the eyes, skin, and mucosa. Wear proper personal protective equipment (e.g. gloves, lab uniforms, etc.) and follow laboratory safety procedures when handling them in the laboratory.

If the reagent accidentally comes in contact with your skin, wash it off immediately with plenty of water and see a doctor if necessary. Do the same if you accidentally get any of the reagent in your eyes.

NOTE

After long-distance transportation, the reagent must be allowed to settle for more than one day before use.

When you have changed the diluents, cleansers or lyses, run a background check to see if the results meet the requirement.

You should replace the reagents when:

- The system indicates that the reagent is used up
- The suspicious flag indicates that the reagent in the pipeline is contaminated
- The reagent is contaminated or expired
- WBC or RBC bubbles are identified.

You can replace any of the following reagents:

- Erba H560 Dil
- Erba H560 Lyse1
- Erba H560 Lyse2

Do as follows to replace the reagents:

1. Refer to Figure 2-2 in **2.6.1 Electrical Connections** for reagent connections.
2. Click the **Service** icon in the menu page to access the Service interface as shown in Figure 12-1.

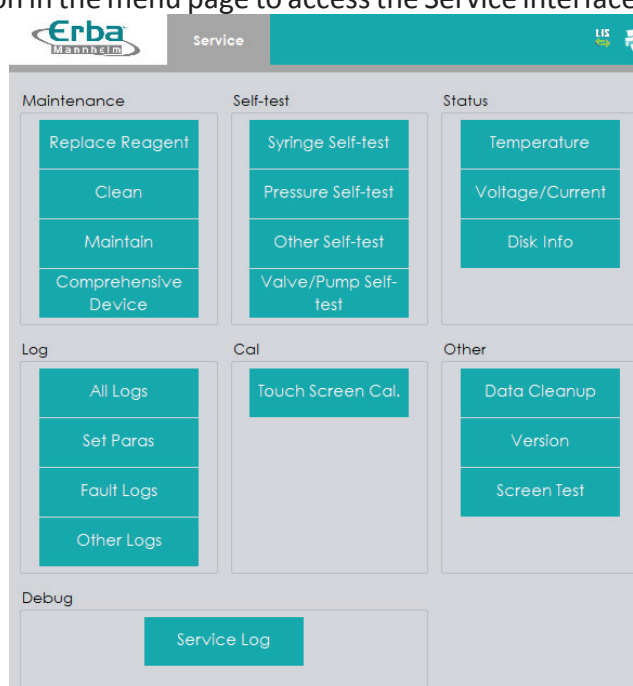


Figure 12-1 Service

3. Click **Replace Reagent** in the Maintenance selection. The interface as shown in Figure 12-2 will pop up on the screen.

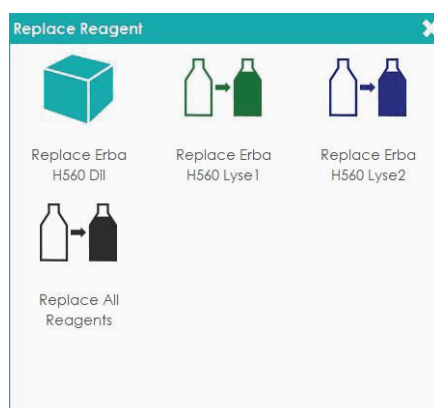


Figure 12-2 Reagent Replacement

4. Click the name of the reagent that needs to be replaced, such as **Replace All Reagents**.
5. Click **OK** to close the message box.
6. Perform the above procedures to replace other reagents if necessary.

12.2.2 Cleaning

Clean corresponding parts according to the actual situation:

- Cleaning of WBC bath:
 - When the background of the scattergram has abnormal excessive cells
 - When the background of WBC- and/or HGB-specific parameters exceeds the reference range
- Cleaning of RBC bath:
 - When the background of RBC- and (or) PLT-specific parameters exceeds the reference range
- Cleaning of flow chamber:
 - When the background of the scattergram has abnormal excessive cells, or bad differential of WBC
- Cleaning of sample probe
 - When the sample probe is dirty

The cleaning procedures are as follows:

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Clean** in the **Maintenance** selection, an interface as shown in Figure 12-3 will pop up on the screen.

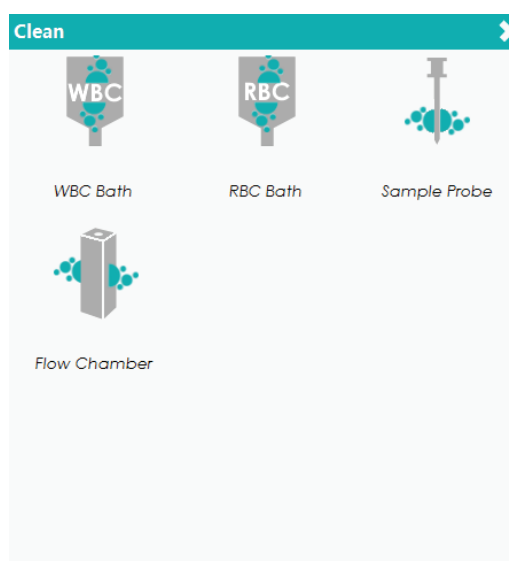


Figure 12-3 Cleaning

3. Click the icon of the part that needs to be cleaned, such as **Clean Sample Probe**.
4. When the system cleaning is complete, the message box will pop up to show that the cleaning is done.
5. Click **OK** to close the message box.
6. Perform the above procedures to clean other components if necessary.

12.3.3 Maintenance

Maintenance of the analyser includes: unclogging, cleanser soak, cleanser soak for WBC channel, and cleanser soak for RBC channel.

12.2.3.1 Unclogging

If clogging is found, or it is suspected that the counting results are not accurate due to aperture clogging, you can perform the unclogging operations. The unclogging procedures are shown as follows:

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Maintain** in the **Maintenance** selection. The interface as shown in Figure 12-4 will pop up on the screen.



Figure 12-4 Maintenance

3. Click the **Unclog** icon.
4. Click **OK** to close the message box.
5. Perform the above procedures to continue unclogging if necessary.

12.2.3.2 Cleanser Soak

The cleanser soak should be performed under the following circumstances:

- When the problems including the background results exceed the Ref. Range, bad differential of scattergram and clogging still exist after other maintenance procedures have been adopted.
- Analyser has been running for more than 24 hours.

The cleanser soak procedures are shown as follows:

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Maintain** in the **Maintenance** selection.
3. Click the icon of **Cleanser Soak**.
4. Click **Yes**.
5. Present the cleanser to the sample probe as per the prompt, and press the aspirate key or click the **Aspirate** button.
6. After one minute of soaking, you can stop it manually. Click the **Stop soaking** button, or wait for 19 minutes until the automatic soaking is completed. After the soaking is completed, a prompt **Cleanser Maintenance done!** appears.

7. Click **Close**.
8. Perform the above procedures to perform the cleanser soak again if necessary.

12.2.3.3 Cleanser Soak for WBC Channel

Probe cleanser soaking for WBC channel can be used to remove the errors for aperture clogging or abnormal scattergram. Please refer to **12.2.3.2 Cleanser Soak** for performing the operations for cleanser soaking for WBC channel.

12.2.3.4 Cleanser Soak for RBC Channel

In case the RBC distribution histogram is abnormal or the clogging is believed to exist in the flow chamber, cleanser soak for RBC channel feature can be used as a means for troubleshooting. Please refer to **12.2.3.2 Cleanser Soak** for performing the operations for cleanser soaking for RBC channel.

12.2.4 Comprehensive Device Maintenance

The comprehensive device maintenance feature includes fluidics initialization, comprehensive device cleaning, emptying fluidics and preparing to ship.

12.2.4.1 Fluidics Initialization

After maintaining the fluidic system or replacing a main part of the analyser, you should perform this procedure to initialize the fluidic system. Do as follows to perform the fluidics initialization:

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Comprehensive Device** in the **Maintenance** selection. The interface as shown in Figure 12-5 will pop up on the screen

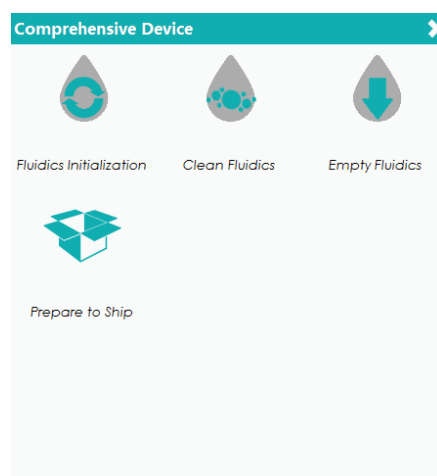


Figure 12-5 Comprehensive Device Maintenance

3. Click the icon of **Fluidics Initialization**.
4. The analyser starts to perform the fluidics initialization procedure. After the initialization is complete, a message box will pop up. Click **OK**.

12.2.4.2 Clean Fluidics

If the background results of parameters are out of the background range, the comprehensive device cleaning should be cleansed. Procedures for comprehensive device cleaning are shown as below:

1. Click the **Service** icon in the menu page to access the Service interface.
2. Click **Comprehensive Device** in the **Maintenance** selection.
3. The analyser starts to perform the fluidics cleaning procedure. After the cleaning is completed, the message box will pop up. Click **OK**.

12.2.4.3 Empty Fluidics

This function enables the device to empty fluidics to prevent crystallization and maintain device performance when the device has not been used for more than one week. Procedures for emptying fluidics are shown as below:

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Comprehensive Device** in the **Maintenance** selection.
3. Click the icon of **Empty Fluidics**.
4. Click **Yes**.
5. Remove all reagent pickup tube assemblies according to the prompt, and then click **OK** to start emptying the fluidic system. After the emptying is complete, a message box will pop up:

Empty Fluidics done. Please power off the analyzer!

6. Place the [O/I] switch at the left side of the main unit in the [O] position.
7. After shutdown, empty the waste in the waste container, and dispose of it.



Be sure to dispose of reagents, waste, samples, consumables, etc. according to local legislations and regulations.

12.2.4.4 Prepare to Ship

If the analyser is not to be used for over two weeks or needs be transported over a long distance (transporting time>2h), you should perform this procedure:

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Comprehensive Device** in the **Maintenance** selection.
3. Click the icon of **Prepare to Ship**.
4. Click **Yes**.
5. Remove all reagent pickup tube assemblies according to the prompt, and then click **OK** to start emptying the fluidic system.
6. Place all reagent pickup tube assemblies into the distilled water, and then click **OK** to start priming.
7. Take out the diluent and lyse pipes from the distilled water as per the prompt, then click **OK**.
8. Place the [O/I] switch at the left side of the main unit in the [O] position.
9. After shutdown, empty the waste in the waste container, and dispose of it.



Be sure to dispose of reagents, waste, samples, consumables, etc. according to local legislations and regulations.

NOTE

Be sure to use distilled water in order to ensure the normal use of the device in the future. In addition, the beaker holding the distilled water needs to be cleaned thoroughly.

The diluent pipe and lyse pipes should be stored separately in two beakers.

12.2.5 Auto Clean

There will be a certain amount of contamination accumulated after running a certain amount of samples without shutting down the analyser. When the sample count amounts to over 100, the analyser will perform the cleaning procedure automatically once, and a prompt will be displayed on the screen.

In addition, the analyser will perform the auto clean procedures if there has been no fluidics sequential operation for more than one hour.

NOTE

Once the auto clean is performed or the analyser is shut down, the statistical data of auto clean will be cleared automatically.

12.2.6 Auto Prompt for Cleanser Soak

If the analyser has been running for more than 24 hours but hasn't performed cleanser maintenance when the auto maintenance time is reached, the system will prompt to perform cleanser soak immediately, so as to prevent the accumulation of contamination.

- Click **Yes**, then you can perform the cleanser maintenance as per the prompt and the description in **12.2.3.2 Cleanser Soak**.
- Click **No**, the cleanser maintenance will be cancelled temporarily. You will be reminded after set waiting time and you can cancel 3 times at most. When the system reminds the forth time, you must perform the maintenance, otherwise the normal operation of the analyzer may be affected.

At the **Self-test** or **Status** interface, the analyser does not ask for confirmation to perform the cleanser soak.

NOTE

If the analyser is running or has problems when the conditions of auto prompt for cleanser soak is satisfied, the analyser will prompt again after the current operation is completed or the problems are resolved.

After cleanser soak is completed, the accumulative count values will be cleared automatically.

Cleanser soak is an important step in comprehensive device maintenance. It is recommended not to stop soaking halfway.

12.2.7 Auto Sleep

When the fluidics system stops working for a specified waiting time for auto sleeping (30 minutes by default), the analyser will enter the sleeping status automatically. You can change the waiting time for auto sleeping as needed.

When the analyser is in the sleep mode, a prompt will be displayed on the screen. Touch the screen or press the aspirate key on the analyser to wake it up.

If it is the time to auto sleep but the analyser is error status, then only after the error is removed will auto sleep start accordingly.

NOTE Different maintenances will be performed by the analyser automatically when exiting the sleep mode, and the exiting time depends on how long the analyser was in the sleep mode.

If errors occur when you are trying to cancel the auto sleep of the analyser, please refer to **13 Troubleshooting** for solving the problems.

12.3 Self-test

This feature is to test if some important components of the device can function properly or not, including syringe and sampling assembly self-test, pressure and vacuum self-test, valve self-test and other self-test.

NOTE If the testing result is abnormal, you should try again for several times; if the abnormalities persist, please contact Erba Lachema customer service department or your local agent.

12.3.1 Syringe and Sampling Mechanism

You can test the performance of all syringes and sampling mechanisms. The self-inspection procedures are shown as below:

1. Click the **Service** icon in the menu page to access the Service interface.
2. Click **Syringe Self-test** in the **Self-test** selection. The interface as shown in Figure 12-6 will pop up on the screen.

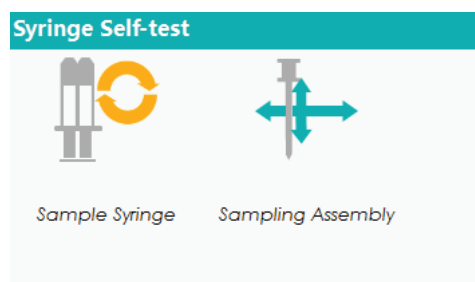


Figure 12-6 Syringe

3. Click the part that needs to be tested, e.g. **Sample Syringe**, and wait for the self-inspection results.
4. Click **OK** to close the message box.

12.3.2 Pressure and Vacuum

This feature is to test the pressure and vacuum inside the device.

Procedures for pressure (or vacuum) self-inspection are shown as below:

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Pressure Self-test** in the **Self-test** selection. The interface as shown in Figure 12-7 will pop up on the screen.

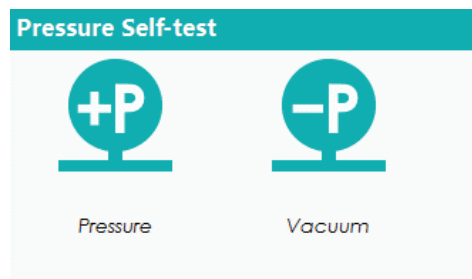


Figure 12-7 Pressure and Vacuum Self-inspection

3. Click **Pressure** (or **Vacuum**). The system will perform the corresponding self-test operations. After the self-test is completed, a dialog box will pop up to show the self-test results.
4. Click **OK** to close the message box.

12.3.3 Valve & Pump

When controlling the switches of different valves (pumps), you can judge if the valves (pumps) are operating properly by the sound of opening, closing or manually touching the corresponding valves (pumps). The procedures for valve self-inspection are shown as follows:

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Valve/Pump Self-test** in the **Self-test** selection. The interface as shown in Figure 12-8 will pop up on the screen.

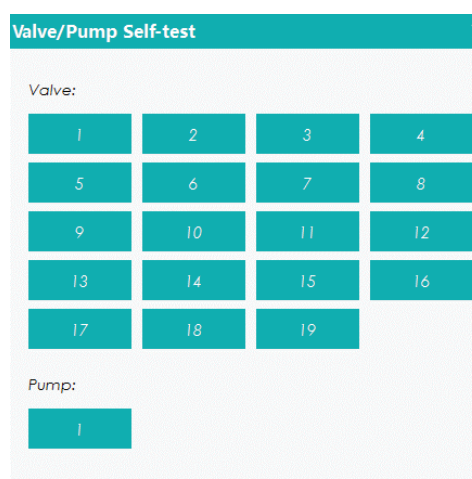


Figure 12-8 Valve Self-test

3. Click the desired valve number (e.g. 1), then confirm whether it works properly by the sound of its opening and closing.

12.3.4 Others

You can perform the self-test for RBC aperture voltage.

12.3.4.1 RBC Aperture Voltage

The self-test procedure of RBC aperture voltage is shown as below:

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Other Self-test** in the **Self-test** selection. The interface as shown in Figure 12-9 will pop up on the screen.

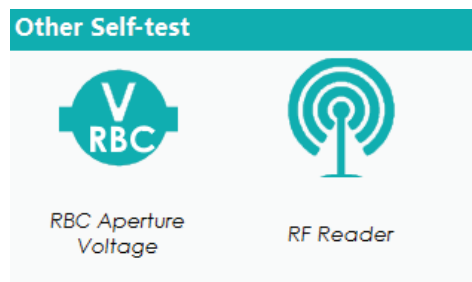


Figure 12-9 Other Self-test

3. Click **RBC Aperture Voltage** to start self-test.

The system will perform the corresponding self-test operations. After the self-inspection is completed, a dialog box will pop up to show the self-inspection results.

12.3.4.2 RF Card Reader

You can carry out a self-test on its built-in RF card reader. The operation procedures are as shown below:

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Other Self-test** in the **Self-test** selection.
3. Click the icon of **RF Reader** to start self-test.
4. According to the interface prompt, put the RF card on the card reader in front of the analyser. The system will perform the corresponding self-test operations. After the self-inspection is completed, a dialog box will pop up to show the self-inspection results.
5. Click **OK** to close the message box.

12.4 System Status

You can view the current status information of the analyser in the **Status** selection, including temperature, voltage and current, and version information.

12.4.1 Temperature

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Temperature** in the **Status** selection. The interface as shown in Figure 12-10 will pop up on the screen.

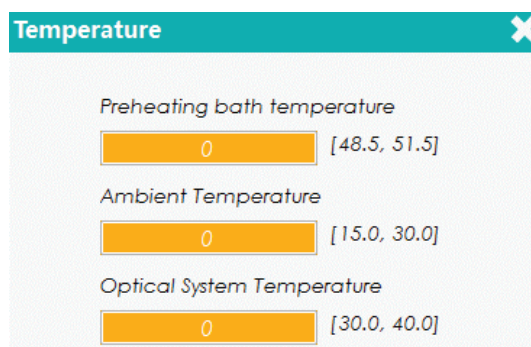


Figure 12-10 View Temperature Status

User can view the current temperature information of the analyser, including the temperature of preheating bath temperature, ambient temperature and the temperature of the optical system. If the results of the temperature testing exceed the normal range, they will be highlighted by the orange background.

12.4.2 Voltage and Current

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Voltage/Current** in the Status selection. The interface as shown in Figure 12-11 will pop up on the screen.

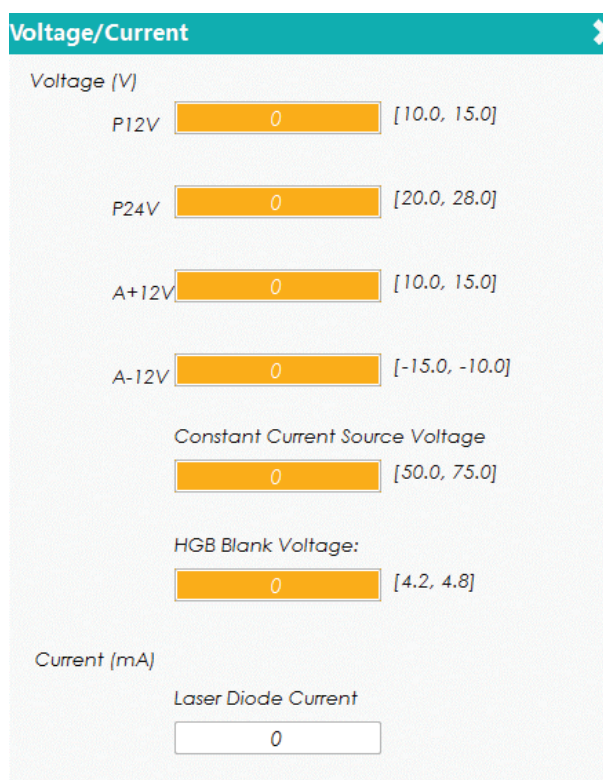


Figure 12-11 Voltage and Current

You can view the voltage and current information of the analyser. The voltage or current value that exceeds the normal range will be displayed in an orange background.

12.4.3 Disk Information

You can view the disk information of the analyser, including disk name, capacity and used space. Specific steps are shown below:

- 1. Click the **Service** icon in the menu page to access the **Service** interface.
- 2. Click **Disk Info** in the **Status** selection. The disk information interface displays. See Figure 12-12.

Disk Info

Item	Capacity	Used space
Flash	1013.5M	39%
SD card	3.7G	4%

Figure 12-12 Disk Information

12.5 Log

In the Log interface, you can view the records of **Set Paras**, **Other Logs**, **Fault Logs** and **All Logs**.

If a new record is added when the log is full, the newest record will overwrite the oldest one automatically.

NOTE

The administrator can view both his/her own operation logs and the general users' operation logs, while the general users can only review their own operation logs. The log can keep records of up to 5 years.

12.5.1 All Logs

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Log > All Logs** to access the **All Logs** interface. You can view all logs and reagent logs (available to the user of the current access level). See Figure 12-13. All logs interface is displayed by default, and you can click the drop-down list to select reagents logs.



No.	Time	Summary Information	Details	
1	2020/11/02 11:01:42	Login	develop(develop) Login	R&D Engineer
2	2020/11/02 11:01:13	Logout	admin(admin) Logout	Administrator admin (admin)
3	2020/11/02 11:00:32	Run	Background Count mode ...	Administrator admin (admin)
4	2020/11/02 11:00:31	Run Seque...	F014 sequence run succe...	Administrator admin (admin)
5	2020/11/02 10:59:03	Run Seque...	F079 sequence run succe...	Administrator admin (admin)
6	2020/11/02 10:56:12	Run Seque...	F016 sequence run succe...	Administrator admin (admin)
7	2020/11/02 10:55:40	Run Seque...	F083 sequence run succe...	Administrator admin (admin)
8	2020/11/02 10:55:13	Run Seque...	F084 sequence run succe...	Administrator admin (admin)

Date and Time:2020/11/02 11:01:42
 Operator:R&D Engineer
 Summary Information:Login
 Details:develop(develop) Login

Figure 12-13

All Logs

Select the dates in the two date textboxes, and then you can view all logs within the date range, which includes operation date and time, operation details and the operator.

Reagent Logs

- 1. Click the drop-down list of **All Logs**, and select **Reagent Logs**.
- 2. Select the dates in the two date textboxes, and then you can view all logs within the date range, which includes operation date and time, operation details, the operator and reagent expiration date.

12.5.2 Parameter Revision Logs

- 1. Click the **Service** icon in the menu page to access the **Service** interface.
- 2. Click **Set Paras** in the **Log** selection. You can view the parameter revision logs as in Figure 12-14 (which can be viewed by the user with the current level of access) within a specified date range.

Set Paras Logs

2018-09-19

2018-09-19

No.	Time	Summary Information	Details	Operator

Date and Time:

Operator:

Summary Information:

Details:

Figure 12-14 Parameter Revision Logs

- 3. Select the dates in the two date textboxes, and then you can view the parameter revision logs within the date range, including the revision date and time, revision summary and the operator.

12.5.3 Fault Logs

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Fault Logs** in the **Log** selection. You can view all logs as in Figure 12-15 (visible to the users of the current access level).

Fault Logs				
2018-09-19		---	2018-09-19	
No.	Time	Summary Information	Details	Operator
1	2018-09-19 13:49:05	Report Error	0xb2004001 : Backgroun...	Administrator admin (admin)
2	2018-09-19 13:43:28	Report Error	0xb2004001 : Backgroun...	Administrator admin (admin)

Date and Time:2018-09-19 13:49:05
 Operator:Administrator admin (admin)
 Summary Information:Report Error
 Details:0xb2004001 : Background abnormal.

Figure 12-15 Fault Logs

3. Select the dates in the two date textboxes, and then you can view the fault logs within the date range, including date and time when the faults occur, fault description and the operator.

12.5.4 Other Logs

- 1. Click the **Service** icon in the menu page to access the **Service** interface.
- 2. Click **Other Logs** in the **Log** selection. You can view other logs besides parameter revision logs and fault logs as in Figure 12-16.

Other Logs

2018-10-09

2018-10-09

No.	Time	Summary Information	Details	Operator
1	2018-10-09 09:45:01	Startup	Startup	Administrator admin (admin)
2	2018-10-09 09:44:54	Login	admin(admin) Login	Administrator admin (admin)
3	2018-10-09 08:11:19	Replace R...	Reagent Name:REAGEN...	Administrator admin (admin)
4	2018-10-09 08:11:19	Replace R...	Reagent Name:REAGEN...	Administrator admin (admin)
5	2018-10-09 08:11:19	Replace R...	Reagent Name:REAGEN...	Administrator admin (admin)
6	2018-10-09 08:10:39	Startup	Startup	Administrator admin (admin)
7	2018-10-09 08:10:30	Login	admin(admin) Login	Administrator admin (admin)

Date and Time:2018-10-09 09:45:01

Operator:Administrator admin (admin)

Summary Information:Startup

Details:Startup

Figure 12-16 Other Logs

- 3. Select the dates in the two date textboxes to view the logs within the date range, including operation date and time, operation records and the operator.

12.6 Data Clean-up

You can clean up the data stored in the analyser. Specific steps are shown below.

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Data Clean-up** in the **Other** selection. The data clean-up interface displays. See Figure 12-17.

Figure 12-17 Data Clean-up

3. Click the **End time** combo box, set the date range of the data to be cleaned up in the popup dialog box.
 - The input sequence of the controls is the same with the date format on the top right corner of the dialog box. For example, if the data format is **yyyy/MM/dd**, you should input the data in the sequence of year, month, and date.
 - Click or to select a date and time or enter the information in the textbox directly.
 - Click to clear the data and input again.

For example, If the **End time** is set to **2016/03/31**, the data generated from system installation date to 31 March 2016 will be cleared.

4. Click **OK** to save the settings and close the dialog box.
5. Select the data to be cleaned up.
 - You can clean up the following data:
 - Counting results
 - L-J QC results
 - X-B QC results
 - Log files
 - Scattergram
6. Click **Apply** or **OK**.

12.7 Version Information

You can view the current version information of all parts of the analyser, and export the version information to a USB flash disk. Detailed steps are shown below:

1. Click the **Service** icon in the menu page to access the **Service** interface.
2. Click **Version Info.** in the **Other** selection. Version information interface will pop up on the screen. See Figure 12-18.

The screenshot shows a 'Version' window with a teal header and a close button. It contains two columns of version information, each with a label and a text input field. The fields are as follows:

Software Full Version	Software Release Version
0.5.20.13902	5
Technical File Version	Machine Type
	1104
Application Software	Algorithm
	1104.1.20180109.718
Boot Software	MLO
0.11.9.0	0.11.9.0
MCU	FPGA
0.0.0.0	0.0.0.0
Fluidics Sequence	Operating System
0.1.17.3	3.2.0.0
LIBS	RF Reader MCU
	0.0.0.0

An 'Export' button is located at the bottom right of the window.

Figure 12-18 Version Information

3. Insert a USB flash disk in the USB interface on the analyser.
4. Click **Export**, and select the export path in the dialog box, and then enter the file name. The file will be exported to the root directory of the USB flash disk (**/udisk/sda1**) by default as shown below.
5. Click **Save** to start exporting.
6. Click **OK** to exit.

12.8 Touch Screen Calibration

When the touch screen has offset, it needs to be recalibrated. Detailed steps are shown below:

1. Click the Service icon in the menu page to access the Service interface.
2. Click Touch Screen Cal. in the Cal selection.
3. Click the calibration point “+” on the screen in order. When the calibration point disappears and the system return to the service screen, it indicates the completion of the calibration.

12.9 Downloading Service Logs

In the use of the analyser, when errors occur and cannot be removed, it's recommended that you export the service logs file to a USB flash disk and send the file to Erba Lachema customer service engineer. Specific steps are shown below.

1. Insert a USB flash disk into the USB interface on the analyser.
2. Click the **Service** icon in the menu page to access the **Service** interface.
3. Click **Service Log** in the **Debug** selection. Select the data range of the logs to be exported in the pop-up dialog box. See Figure 12-19.

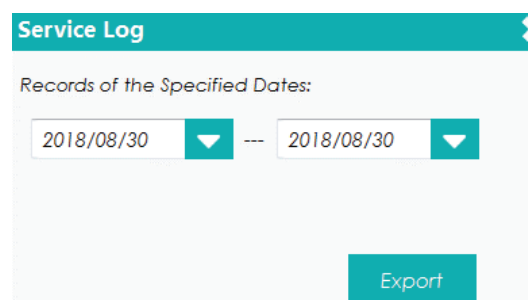


Figure 12-19 Downloading Service Logs

4. Click **Export**.
5. The **host_download.tar** file is exported to the root directory of the USB flash disk.
6. Send the **host_download.tar** file to our customer service engineer.

13 Troubleshooting

13.1 Introduction

This chapter contains information that is helpful in locating and resolving problems that may occur during the operation of your analyser.

NOTE This chapter is not a complete service manual and is limited to problems that are readily diagnosed and/or corrected by the user of the analyser. If the recommended solution fails to solve the problem, contact Erba Lachema customer service department or your local agent.

13.2 Dealing with Error Messages

In the use of the analyser, when the software detects abnormalities, an error message will be displayed on the upper right of the screen as shown in Figure 13-1 and the main unit will sound an alarm.



Figure 13-1 Error Messages

You can refer to the following steps to deal with the error messages.

1. Click the error message area.

As shown in Figure 13-2, the popup dialog box displays the error description and its help information. The error descriptions are displayed in the order of error occurrence.

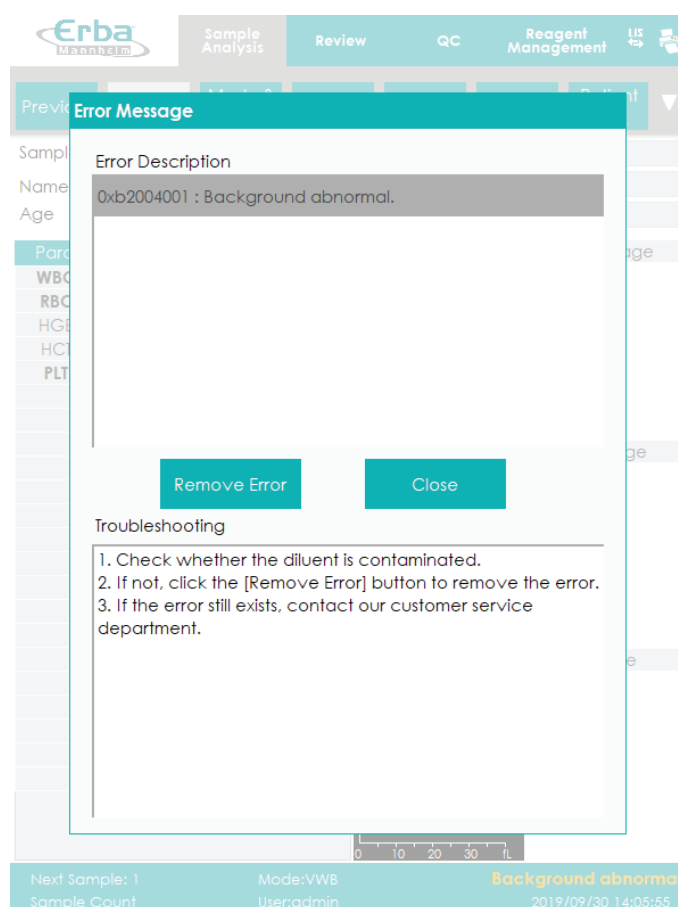


Figure 13-2 Error Message Dialog Box

2. Touch the screen to disable the beep.
3. Click **Remove Error**.

Normally, the system will automatically remove the errors. For errors which cannot be removed automatically, you can take appropriate actions by following the error help information or **Error Message Reference**.

13.3 Error Message Reference

Possible errors and the corresponding help information are shown in Table 13-1.

Table 13-1 Error Message Reference

Problem Name	Troubleshooting Information
-12V power is not working properly.	1. Please power off the analyser directly and restart later. 2. If the error still exists, contact our customer service department.
Optical assembly cover is open.	1. Close the optical assembly cover. 2. Click the Remove Error button to remove this error. 3. If the error still exists, contact our customer service department.

Problem Name	Troubleshooting Information
The CC source voltage is abnormal.	1. Please power off the analyser directly and restart later. 2. If the error still exists, contact our customer service department.
Abnormal laser current.	1. Please power off the analyser directly and restart later. 2. If the error still exists, contact our customer service department.
Start-up failure.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Start-up initialization is not executed.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
The right-side door is open.	1. Close the right side door. 2. Click the Remove Error button to remove this error. 3. If the error still exists, contact our customer service department.
+12V power is not working properly.	1. Please power off the analyser directly and restart later. 2. If the error still exists, contact our customer service department.
Erba H560 Dil expired.	1. Check if the Erba H560 Dil diluent expires. If so, replace it with a new container of Erba H560 Dil. 2. Click the Remove Error button, the Reagent Management screen will be displayed. 3. Set the reagent information by referring to 11 Reagent Management. 4. If the error still exists, contact our customer service department.
Erba H560 Lyse1 expired.	1. Check if the Erba H560 Lyse1 expires. If so, replace it with a new container of Erba H560 Lyse1. 2. Click the Remove Error button, the Reagent Management screen will be displayed. 3. Set the reagent information by referring to 11 Reagent Management. 4. If the error still exists, contact our customer service department.
Erba H560 Lyse2 expired.	1. Check if the Erba H560 Lyse2 expires. If so, replace it with a new container of Erba H560 Lyse2. 2. Click the Remove Error button, the Reagent Management screen will be displayed. 3. Set the reagent information by referring to 11 Reagent Management. 4. If the error still exists, contact our customer service department.
Preheating bath temperature out of working range.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Abnormal HGB background voltage.	1. Adjust the HGB background voltage within the specified range (4.2V~4.8V), preferably 4.5V. Refer to 5.5.1 Gain Settings. 2. If the error still exists, contact our customer service department.
Abnormal RBC aperture voltage.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.

Problem Name	Troubleshooting Information
Abnormal background.	1. Check whether the diluent is contaminated. 2. If not, click the Remove Error button to remove the error. 3. If the error still exists, contact our customer service department.
Failed to read sample syringe parameter.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Failed to configure sample syringe parameter.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Sample syringe timeout	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Sample syringe is busy.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Vertical motor instruction parameter error.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Failed to read vertical motor parameter.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Vertical motor timeout	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Failed to read the remaining steps of vertical motor.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
The vertical motor is busy.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Failed to read preheating bath temperature.	1. Make sure the temperature sensor is correctly installed. 2. Click the Remove Error button to remove the error. 3. If the error still exists, contact our customer service department.
Failed to read optical system temperature.	1. Make sure the temperature sensor is correctly installed. 2. Click the Remove Error button to remove the error. 3. If the error still exists, contact our customer service department.
Failed to read ambient temperature.	1. Make sure the temperature sensor is correctly installed. 2. Click the Remove Error button to remove the error. 3. If the error still exists, contact our customer service department.
Waste is full.	1. Empty the waste container or install a new waste container. 2. Click the Remove Error button to remove this error. 3. If the error still exists, contact our customer service department.
The setting temperature of optical system out of range.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.

Problem Name	Troubleshooting Information
Optical system temperature out of working range.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Flow cell clog.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Failed to read horizontal motor parameter.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Failed to configure Horizontal motor parameter.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Horizontal motor timeout	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
The optocoupler of the horizontal motor is not working properly.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
The horizontal motor is busy.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Erba H560 Dil running out.	1. Check whether the Erba H560 Dil container is empty. If so, install a new container of Erba H560 Dil. 2. Click the Remove Error button to remove the error. 3. If the error still exists, contact our customer service department.
Erba H560 Lyse1 running out or air bubbles in inlet tubing.	1. Check whether the Erba H560 Lyse1 is running out or there are air bubbles in the inlet tubing of Erba H560 Lyse1. If it is running out, install a new container of Erba H560 Lyse1; If there is still plenty of Erba H560 Lyse1 or there are bubbles, perform step 2. 2. Click the Remove Error button to remove the error. 3. If the error still exists, contact our customer service department.
Erba H560 Lyse2 running out or air bubbles in inlet tubing.	1. Check whether the Erba H560 Lyse2 is running out or there are air bubbles in the inlet tubing of Erba H560 Lyse2. If it is running out, install a new container of Erba H560 Lyse2; If there is still plenty of Erba H560 Lyse2 or there are bubbles, perform step 2. 2. Click the Remove Error button to remove the error. 3. If the error still exists, contact our customer service department.
Erba H560 Dil not replaced.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Erba H560 Lyse1 not replaced.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Erba H560 Lyse2 not replaced.	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.

Problem Name	Troubleshooting Information
DIFF probe clogging	1. Click the Remove Error button to remove this error. 2. If the error still exists, contact our customer service department.
Abnormal 12V driving power supply.	1. Please power off the analyser directly and restart later. 2. If the error still exists, contact our customer service department.
Abnormal 24V driving power supply.	1. Please power off the analyser directly and restart later. 2. If the error still exists, contact our customer service department.
Insufficient Erba H560 Dil.	1. Check whether the Erba H560 Dil container is empty. If so, install a new container of Erba H560 Dil. 2. Click the Remove Error button, the Reagent Management screen will be displayed. 3. Set the reagent information by referring to 11 Reagent Management. 4. If the error still exists, contact our customer service department.
Insufficient Erba H560 Lyse1.	1. Check whether the Erba H560 Lyse1 container is empty. If so, install a new container of Erba H560 Lyse1. 2. Click the Remove Error button, the Reagent Management screen will be displayed. 3. Set the reagent information by referring to 11 Reagent Management. 4. If the error still exists, contact our customer service department.
Insufficient Erba H560 Lyse2.	1. Check whether the Erba H560 Lyse2 container is empty. If so, install a new container of Erba H560 Lyse2. 2. Click the Remove Error button, the Reagent Management screen will be displayed. 3. Set the reagent information by referring to 11 Reagent Management. 4. If the error still exists, contact our customer service department.
The printer box is not connected.	1. The printer box is not turned on. 2. The connecting line of the printer box is abnormal. 3. The IP or port number of the printer box is misconfigured. 4. There exists an error in the printer box, try to restart the printer box. 5. Click Remove Error to automatically remove the error. 6. If the error still exists, please contact our company's customer service department.
Printing Error	1. The printer connected to the printer box is out of paper or has other errors. 2. The printer box is not connected to the designated printer. 3. If the above errors are excluded, try to restart the printer box. 4. Click Remove Error to automatically remove the error. 5. If the error still exists, please contact our company's customer service department.

14

Traceability and Reference Methods

14.1 Traceability

The assigned values of *in vitro* diagnostic medical calibrator ELite H CAL are traceable to internationally recognised reference methods for WBC, RBC, HGB, HCT and PLT, according to the recommendations of International Council for Standardization in Haematology (ICSH) and Clinical and Laboratory Standards Institute (CLSI).

The chain of traceability is based on ISO 17511 [1] (5.4 section) and is shown in Figure 14-1.

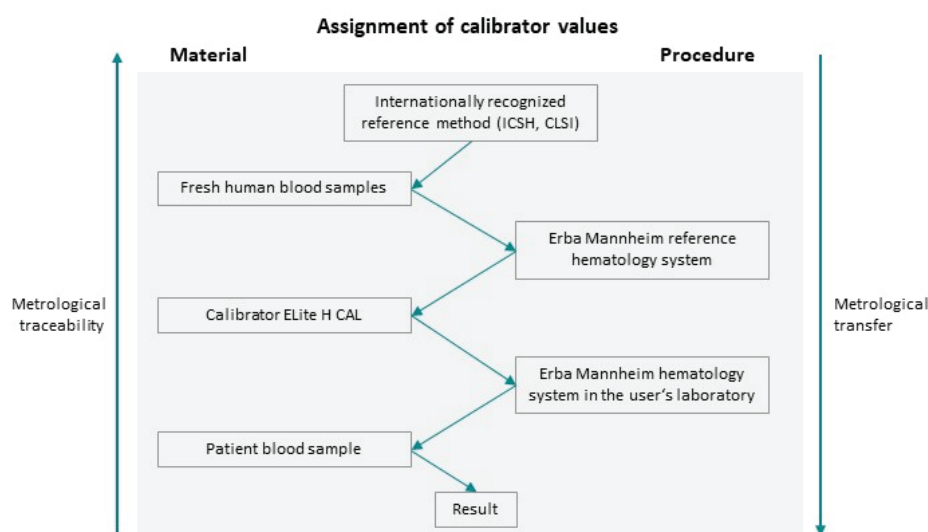


Figure 14-1 Value traceability chain of hematology analyser

14.2 Reference Methods

The hematology reference analyzers are regularly calibrated with fresh human blood samples directly against the reference methods recommended by ICSH and CLSI.

WBC: A 1:500 dilution is prepared using a 100 mL Class A volumetric flask filled with isotonic diluent. 1.2 mL of diluent is removed. Sample is added to the flask using a 200 µL calibrated positive displacement pipette, followed by 1.0 mL lysing agent. Counting is performed on a Coulter Counter Z series instrument. All counts are corrected for coincidence.

RBC: A 1:50,000 dilution is prepared using a 1000 mL Class A volumetric flask filled with isotonic diluent. Sample is added to the flask using a 20 µL T.C. micropipet. Counting is performed on a Coulter Counter Z series instrument. All counts are corrected for coincidence.

HGB: A 1:251 dilution is prepared using a 50 mL Class A volumetric flask filled with the CLSI recommended reagent for the hemoglobincyanide (cyanmethemoglobin) method [2]. Sample is added to the flask using a 200 µL calibrated positive displacement pipette. The sample is filtered with a 0.2 µm filter immediately before reading. Readings are made at 540 nm in a colorimeter/spectrophotometer calibrated according to CLSI H15-A3 and ICSH recommendations [2].

HCT: Plain glass microhematocrit tubes (not coated with anticoagulant) are filled with sample, sealed with sealing putty and centrifuged for 5 minutes in a microhematocrit centrifuge according to the CLSI H07-A3 document [3]. After centrifugation, the length of the whole column including the plasma, and the length of the red blood cell column, are viewed and measured using a microscope with graduated stage and an ocular micrometer. The hematocrit (packed cell volume) is calculated as the ratio of the two measurements. No correction is made for trapped plasma.

MCV: On some instruments MCV is the calibrated parameter instead of the HCT. The MCV is calculated from the HCT and RBC using the formula: $MCV = HCT \times 10/RBC$

PLT: A 1:126 dilution is prepared using a 50 mL Class A volumetric flask filled with filtered 1% ammonium oxalate. Sample is added to the flask using a 400 µL calibrated positive displacement pipette. The dilution is plated onto a clean, dry Neubauer ruled phase type hemocytometer. The hemocytometer is left for 10 minutes in a humidified chamber. Using phase contrast optics, the platelets in the entire central square millimeter on both sides of the hemocytometer are counted. The two counts are averaged and multiplied by 1260 (dilution factor 126 × volume factor 10 = 1260).

-
1. ISO 17511:2003 *In vitro* diagnostic medical devices – Measurement of quantities in biological samples – Metrological traceability of values assigned to calibrators and control materials.
 2. Clinical Laboratory Standards Institute. Reference and Selected Procedures for the Quantitative Determination of Hemoglobin in Blood: Approved Standard-Third Edition. CLSI document H15-A3.
 3. Clinical Laboratory Standards Institute. Procedure for Determining Packed Cell Volume by the Microhematocrit Method: Approved Standard-Third Edition. CLSI document H07-A3.

15 Specifications

15.1 Classification

According to the CE classification, the H560 Haematology Analyser belongs to *in vitro* diagnostic medical devices, rather than those covered by Annex II and devices for performance evaluation.

15.2 Reagents, controls and calibrator

Consumables	Type Name	Catalog number
Diluent	Erba H560 Dil	HEM0030
Lyse	Erba H560 Lyse1	HEM0031
	Erba H560 Lyse2	HEM0032
Medical cleanser	ELite H Clean	HEM0023
Low control	ELite H5 CON L	HEM0024
Normal control	ELite H5 CON N	HEM0025
High control	ELite H5 CON H	HEM0026
Calibrator	ELite H CAL	HEM0027

15.3 Parameters

Parameter	Abbreviation	Default Unit
White Blood Cell count	WBC	$10^3/\mu\text{L}$
Number of Neutrophils	Neu#	$10^3/\mu\text{L}$
Number of lymphocytes	Lym#	$10^3/\mu\text{L}$
Number of Monocytes	Mon#	$10^3/\mu\text{L}$
Number of Eosinophils	Eos#	$10^3/\mu\text{L}$
Number of Basophils	Bas#	$10^3/\mu\text{L}$
Number of Abnormal Lymphocytes	ALY# (RUO)	$10^3/\mu\text{L}$
Number of Large Immature Cells	LIC# (RUO)	$10^3/\mu\text{L}$
Number of Nucleated Red Blood Cells	NRBC# (RUO)	$10^3/\mu\text{L}$
Percentage of Neutrophils	Neu%	%
Percentage of Lymphocytes	Lym%	%
Percentage of Monocytes	Mon%	%
Percentage of Eosinophils	Eos%	%
Percentage of Basophils	Bas%	%
Percentage of Abnormal Lymphocytes	ALY% (RUO)	%

Parameter	Abbreviation	Default Unit
Percentage of Large Immature Cells	LIC% (RUO)	%
Percentage of Nucleated Red Blood Cells	NRBC% (RUO)	%
Corrected White Blood Cells	cWBC# (RUO)	$10^3/\mu\text{L}$
Neutrophil-to-Lymphocyte Ratio	NLR (RUO)	/
Red Blood Cell count	RBC	$10^6/\mu\text{L}$
Haemoglobin Concentration	HGB	g/dL
Haematocrit	HCT	%
Mean Corpuscular Volume	MCV	fL
Mean Corpuscular Haemoglobin	MCH	pg
Mean Corpuscular Haemoglobin Concentration	MCHC	g/dL
Red Blood Cell Distribution Width - Standard Deviation (RDW-SD)	RDW-SD	fL
Red Blood Cell Distribution Width - Coefficient of Variation (RDW-CV)	RDW-CV	%
Platelet count	PLT	$10^3/\mu\text{L}$
Mean Platelet Volume	MPV	fL
Platelet Distribution Width - Standard Deviation (PDW-SD)	PDW-SD	fL
Platelet Distribution Width - Coefficient of Variation (PDW-CV)	PDW-CV	%
Plateletcrit	PCT	%
Platelet-large cell count	P-LCC	$10^3/\mu\text{L}$
Platelet-large cell ratio	P-LCR	%
White Blood Cell Histogram	WBC Histogram	None
Red Blood Cell Histogram	RBC Histogram	None
Platelet Histogram	PLT Histogram	None
Basophils Scattergram	BASO Scattergram	None
DIFF Scattergram	DIFF Scattergram	None

15.4 Performance Specifications

15.4.1 Display Range

Parameter	Linearity Range	Display Range
WBC	$0\sim 300\times 10^3/\mu\text{L}$	$0\sim 999\times 10^3/\mu\text{L}$
RBC	$0.00\sim 8.50\times 10^6/\mu\text{L}$	$0\sim 18.00\times 10^6/\mu\text{L}$
HGB	$0\sim 25\text{ g/dL}$	$0\sim 30\text{ g/dL}$
PLT	$0\sim 3000\times 10^3/\mu\text{L}$	$0\sim 5000\times 10^3/\mu\text{L}$
HCT	$0\sim 67\%$	$0\%\sim 80\%$

15.4.2 Normal Background

Parameter	Normal Background
WBC	$\leq 0.2 \times 10^3/\mu\text{L}$
RBC	$\leq 0.02 \times 10^6/\mu\text{L}$
HGB	$\leq 0.1 \text{ g/dL}$
PLT	$\leq 10 \times 10^3/\mu\text{L}$
HCT	$\leq 0.5\%$

15.4.3 Linearity Range

Parameter	Linearity range	Deviation range (Whole blood mode)
WBC	$(0.00 \sim 100.00) \times 10^3/\mu\text{L}$	$\pm 0.30 \times 10^3/\mu\text{L}$ or $\pm 5\%$
	$(100.01 \sim 300.00) \times 10^3/\mu\text{L}$	$\pm 10\%$
RBC	$(0.00 \sim 8.50) \times 10^6/\mu\text{L}$	$\pm 0.05 \times 10^6/\mu\text{L}$ or $\pm 5\%$
HGB	$(0 \sim 25) \text{ g/dL}$	$\pm 0.2 \text{ g/dL}$ or $\pm 2\%$
PLT	$(0 \sim 1000) \times 10^3/\mu\text{L}$ (RBC ≤ 7.0)	$\pm 10 \times 10^3/\mu\text{L}$ or $\pm 8\%$
	$1001 \sim 3000 \times 10^3/\mu\text{L}$ (RBC ≤ 7.0)	$\pm 12\%$
HCT	$0 \sim 67\%$	$\pm 2\%$ (HCT value) or $\pm 3\%$ (deviation percent)

15.4.4 Repeatability

These repeatability requirements apply only to the situation in which a qualified sample has been run for 11 times and the results of the 2nd to 11th runs are used to calculate the repeatabilities.

Parameter	Condition	Whole Blood Repeatability (CV%/absolute deviation d*)
WBC	$(4.0 \sim 15.0) \times 10^3/\mu\text{L}$	$\leq 2.0\%$
Neu%	$50.0\% \sim 60.0\%$	± 4.0 (absolute deviation)
Lym%	$25.0\% \sim 35.0\%$	± 3.0 (absolute deviation)
Mon%	$5.0\% \sim 10.0\%$	± 2.0 (absolute deviation)
Eos%	$2.0\% \sim 5.0\%$	± 1.5 (absolute deviation)
Bas%	$0.5\% \sim 1.5\%$	± 0.8 (absolute deviation)
RBC	$(3.50 \sim 6.00) \times 10^6/\mu\text{L}$	$\leq 1.5\%$
HGB	$(11 \sim 18) \text{ g/dL}$	$\leq 1.5\%$
PLT	$(150 \sim 500) \times 10^3/\mu\text{L}$	$\leq 4.0\%$
MCV	$(70 \sim 120) \text{ fL}$	$\leq 1.0\%$
MPV	-	≤ 4.0

*: Absolute deviation d = analysis result - average of analysis results

15.4.5 Carryover

Parameter	Carryover
WBC	$\leq 0.5\%$
RBC	$\leq 0.5\%$
HGB	$\leq 0.5\%$
PLT	$\leq 1.0\%$
HCT	$\leq 0.5\%$

15.5 Sample Interference

If there is sample interference, the analysis results of the sample may be affected. See the table below.

Table 15-1 Possible sample interference

Parameter	Analysis Results	Interference Source
WBC	Low WBC count	Leukoagglutination
	High WBC count	- Possible Platelet agglutination - Cool insoluble protein - Cryoglobulins - Fibrin - Excessive numbers of giant platelets (platelets$>1000\times10^3/\mu\text{L}$) - Nucleated red blood cells
RBC	Low RBC count	- Agglutinated RBCs (Cold agglutinins) - Microcythemia - Schistocytes
	High RBC count	- Leukocytosis ($>100\times10^3/\mu\text{L}$) - Excessive numbers of giant platelets (platelets$>1000\times10^3/\mu\text{L}$)
HGB	High HGB count	- Leukocytosis ($>100\times10^3/\mu\text{L}$) - Chylaemia - Jaundice - Paraprotein
HCT	Low HCT value	- Agglutinated RBCs (Cold agglutinins) - Microcytes - Schistocytes
	High HCT value	- Leukocytosis ($>100\times10^3/\mu\text{L}$) - Severe diabetes - Uraemia - Spherocytes
PLT	Low PLT count	- Possible Platelet agglutination - pseudo thrombocytopenia - Giant platelets
	High PLT count	- Microcytes - Schistocytes - WBC fragments - Cool insoluble protein - Cryoglobulins

15.6 Input/output Device



WARNING

Accessory equipment connected to the analogue and digital interfaces must comply with the relevant Safety and EMC standards (e.g., IEC 60950 Safety of Information Technology Equipment Standard and CISPR 22 EMC of Information Technology Equipment Standard (CLASS B)). Anyone who connects additional equipment to the signal input or output ports and configures an IVD system is responsible for ensuring that the system works properly and complies with the safety and EMC requirements. If you have any problem, consult the technical services department of your local agent.

Be sure to use specified fuse only.

Analyser

- Touch screen: 10.4 inches embedded touch screen with a resolution of 800×600
- One LAN interface
- 4 USB interfaces

Power

- Voltage: A.C 100V~240V
- Input power: ≤200VA
- Frequency: 50/60 Hz

Fuse: T6.3AL 250V

Keyboard (Optional, USB)

Mouse (Optional, USB)

External barcode scanner (optional, USB)

External printer (optional, USB)

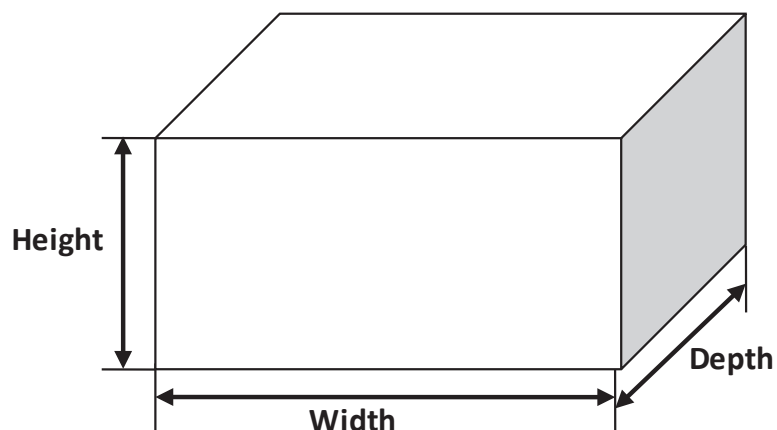
USB flash disk (optional, USB)

15.7 Environment Conditions

NOTE Be sure to use and store the analyser in the specified environment.

Environment Conditions	Operating Environment	Storage Environment	Running Environment
Ambient temperature	15°C~30°C	-10°C~40°C	5°C~40°C
Relative humidity	20%~85%	10%~90%	10%~90%
Atmospheric pressure	70kPa~106kPa	50kPa~106kPa	70kPa~106kPa

15.8 Dimensions and Weight



Analyser	Dimensions and Weight
Width (mm)	364
Height (mm)	498
Depth (mm)	431
Weight (kg)	26.5

15.9 Expected Service Life

8 years.

15.10 Contraindications

None

Terms and Abbreviations

CWB Capillary Whole Blood
PD Predilute
RUO Research use only
VWB Venous Whole Blood

17 Packing List

No.	Parameter	Quantity	Unit
1	H560 Haematology Analyser	1	PCS
2	Power Cable	1	PCS
3	Peripheral Grounding Cable	1	PCS
4	Operator's Manual	1	PCS
5	Quick Operation Guide Card	1	PCS
6	Diluent Adapter Tube	1	PCS
7	Waste Float Adapter Tube	1	PCS
8	Warranty Card	1	PCS
9	Waste container	1	PCS
10	Reagent Operation Guide	1	PCS
11	Inspection Record	1	PCS
12	Packing list	1	PCS

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